Remarks on Mixed Holomorphic-Topological Strings

Raeez Lorgat

root@raeez.com

http://math.raeez.com

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Abstract

This paper studies a local constrained mechanical system. The formal holomorphic normal surface has coordinates z_1, z_2 and symplectic form $dz_1 \wedge dz_2$. On a stack of N probes these coordinates become matrices $\phi_1, \phi_2 \in \mathfrak{gl}_N$. The open ghost-zero theory is the Dirac first-order system

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Here A is the multiplier for the first-class constraint $[\phi_1, \phi_2] = 0$, and the BV/Koszul variable $\psi = A^*$ records it by $Q\psi = [\phi_1, \phi_2]$. This is the open local model for a mixed holomorphic-topological BV sector on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$.

The infinitesimal canonical symmetries of the formal symplectic disk form the reduced Tate Lie algebra $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$. Its shifted cotangent semidirect extension $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1]$ is the dg Lie algebra of closed Hamiltonian BF theory. The classical local theorem is the umbrella of six interlocking statements. There is a universal cochain-level isomorphism of completed dg graded-commutative P_0-algebras

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\widehat{\mathfrak{h}})\big) \simeq Z^{P_0}_{\mathrm{fact}}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\widehat{\mathfrak{h}})\big),$$

sending the CE coordinate c^I dual to the Hamiltonian generator $H_I \in \widehat{\mathfrak{h}}$ to the constant polyvector $\theta^I = \partial/\partial O_I$, and the CE coordinate u_I dual to the shifted cotangent generator η_{∂_I} to the boundary Hamiltonian function $O_I = \overline{\mathrm{Tr}\, H_I(\phi_1, \phi_2)}$. The boundary trace observable is the image of the shifted cotangent CE coordinate, not of the unclosed CE coordinate dual to a Hamiltonian mode. The continuous dual $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ is canonically the principal-part residue module $\mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\, \rho_{a,b}$ with $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2$, paired by the Grothendieck residue. The polynomial Koszul descendants $\Psi_{a,b}$ are PBW special-fibre representatives carrying a raising action; the principal parts $\rho_{a,b}$ are deformation-correct coadjoint representatives carrying the lowering action; the Rees parameter $\hbar$ of the formal Moyal deformation interpolates them.

The cochain isomorphism upgrades to a quasi-isomorphism of locally constant P_0 factorization algebras on the brane line $\mathbb{R}$, in the connected Hamiltonian sub-sector, with strict factorization product on disjoint supports and equal-time delta-function bracket. It deforms to all orders in $\hbar$ on the degree-zero Hamiltonian sector: the Capelli/Weyl-renormalized stable trace $J_N(f)$ realizes the all-order Moyal target, the Harish-Chandra radial-parts identity gives the reduced Moyal commutator at all N, and the Costello first/third graph normalizations are fixed by Kontsevich uniqueness. The

scalar Hamiltonian/center-of-mass mode carries a single nontrivial class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ realised identically by the classical Poisson cocycle $\omega(z_1^k z_2^l, z_1^m z_2^n) = (kn - lm)\mathbf{1}[(k+m, l+n) = (1,1)]$ and the quantum Capelli shift $YX - XY + \hbar N I = 0$; the equivalence holds modulo $[\bar{c}]$.

The single dependency-closed residual is the unreduced Tate-coefficient lift of the cotangent direction. Two completion routes are named: a weighted graded-Mittag-Leffler completion compatible with the Costello heat-kernel calculus; or a nilpotent truncation in the Costello-Li $\mathfrak{gl}(N \mid N)$ tradition. The classical and quantum problems are pinned to the same obstruction, with the deformation adding no new conditionality. Cross-volume convention contracts to ~/calabi-yau-quantum-groups (Vol III) and ~/igusa-cusp-form (Δ_5/BKM) are local-only and explicitly non-transferable without separate matched-conventions proofs.

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0.1 Introduction

0.1.1 Local measurements, constraints, and brackets

The object of study is a local constrained mechanical system. Fix a defect world-line $\mathbb{R} \times p$ inside the $1+4$-real-dimensional space-time $\mathbb{R}^2 \times \mathbb{C}^2$. The transverse data is a single real line of *topological time* (the $\mathbb{R}^2$ factor), and a formal holomorphic symplectic disk

$$(z_1, z_2), \qquad \omega = dz_1 \wedge dz_2,$$

the formal neighbourhood of $p \in \mathbb{C}^2$. The defect is a *point in* $\mathbb{C}^2$*, dragged along* $\mathbb{R}$; nothing else has been postulated.

The two normal coordinates z_1, z_2 describe the position of the defect inside the symplectic disk. A stack of N identical probes at the defect represents each coordinate by an $N \times N$ matrix:

$$z_1 \rightsquigarrow \phi_1, \qquad z_2 \rightsquigarrow \phi_2, \qquad \phi_i \in \mathfrak{gl}_N.$$

The symplectic form pulls back to the matrix-valued two-form $\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2)$, and the symplectic potential along the world-line is

$$\theta_N = \mathrm{Tr}(\phi_1\, d\phi_2).$$

Acting on a probe stack by an internal change of basis is a gauge operation. The gauge generator is the $\mathfrak{gl}_N$-valued moment map

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N,$$

and demanding gauge-invariance of the world-line dynamics imposes the *first-class constraint*

$$[\phi_1, \phi_2] = 0.$$

The Lagrange multiplier for this constraint is a connection $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$. The boundary action is therefore *not chosen*; it is the unique first-order action carrying the symplectic potential and enforcing the first-class constraint:

$$S_\partial = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big). \tag{1}$$

This is the Dirac standard. The "open string field theory on $\mathbb{R} \times p$" is the canonical mechanics of an N-probe stack on the formal symplectic normal disk, with the matrix moment-map equation as its first-class constraint. Lemma 0.2.6 records the precise constrained-mechanics statement and identifies A as the multiplier.

The BV reduction of this constrained system is an algebraic step. The constraint is recorded as a Koszul differential by a single odd matrix antifield $\psi = A^* \in \mathfrak{gl}_N[1]$ with

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0.$$

The reduced local polynomial algebra at a brane point is

$$\mathcal{R}_N = \mathbb{C}[{\phi_1}^i{}_j, {\phi_2}^i{}_j] \otimes \Lambda({\psi}^i{}_j),$$

and $H^0(\mathcal{R}_N^{GL_N}, Q)$ is, by definition, the algebra of GL_N-invariant local observables on the Dirac-reduced phase space of commuting matrix pairs. This algebra is the open trace algebra of the defect; it is what one is forced to write down once the probe stack and the constraint have been named. The full BV master action is the canonical lift of (1) to ghost-number-zero fields; this lift is recorded in Lemma 0.2.5 and Proposition 0.2.7.

0.1.2 *The Hamiltonian Lie algebra of the symplectic disk*

The corresponding closed system is dictated by the same canonical mechanics. A closed infinitesimal symmetry of the formal symplectic disk is a Hamiltonian vector field $X_f = \{f, -\}$. Constants act trivially: $X_c = 0$ for $c \in \mathbb{C}$. The closed Hamiltonian Lie algebra is therefore the formal power series modulo constants,

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1, \qquad \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1,$$

with bracket

$$\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g \pmod{\mathbb{C}}.$$

Polynomial calculations use $\bar{A}$; closed Chevalley–Eilenberg/Koszul calculations use the completed $\widehat{\mathfrak{h}}$. Continuous duality on $\widehat{\mathfrak{h}}$ produces $\mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$, with the Taylor-monomial basis $\delta_{k,l}$ for $k + l > 0$. We use $\bar{A}_{\mathrm{desc}}$ for an abstract copy of $\bar{A}$ whose nonconstant basis vectors are represented on the brane by the first descendant cycles $\Psi_{k,l}$ (Construction 0.2.43).

Remark 0.1.1 *(Scalar Hamiltonian and center-of-mass term).* The quotient by constants is load-bearing. In the reduced Hamiltonian algebra of the formal disk,

$$\{z_1, z_2\} = 1 \equiv 0 \quad \text{in } \bar{A}.$$

On the unreduced brane phase space, however,

$$\{\mathrm{Tr}(\phi_1), \mathrm{Tr}(\phi_2)\} = N.$$

This is the center-of-mass symplectic pair of the stack. Every bracket statement below involving linear Hamiltonians is therefore a statement after projection to the Hamiltonian quotient by scalar trace classes. The same term reappears quantum mechanically as the Capelli shift in Remark 0.2.86.

The closed canonical theory must promote $\widehat{\mathfrak{h}}$ to a local field theory whose phase space is the cotangent of the Hamiltonian-deformation moduli; the cotangent of canonical transformations is the unique covariant choice. This is what BCOV [1] prescribes for the closed B-model on a Calabi–Yau, with the polyvector cotangent realised by the divergence operator. On a holomorphic-symplectic surface the local cotangent theory is the Hamiltonian BF theory for the Tate Lie algebra

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee[1], \tag{2}$$

where the shift [1] records the BV degree of the cotangent field and the Lie bracket on $\mathfrak{g}$ is the semi-direct sum of the Hamiltonian bracket on $\widehat{\mathfrak{h}}$ with the coadjoint action on $\mathfrak{h}^\vee[1]$.

The closed local observables are the continuous Chevalley–Eilenberg cochains

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}} = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big),$$

and their Koszul dual is the completed graded enveloping algebra

$$\big(\mathcal{O}^{\mathrm{cl}}_{\mathrm{grav}}\big)^! \simeq \widehat{U(\mathfrak{g})}.$$

The PBW Rees algebra $U_\hbar(\mathfrak{g})$ has completed special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big). \tag{3}$$

The structural content of (2)–(3) is that the closed sector *is the shifted cotangent theory of canonical transformations of the disk*; Proposition 0.2.13 below derives $\widehat{\mathfrak{h}}$ from divergence-closed polyvectors on the formal disk.

0.1.3 *Closed-open measurement and the derived Poisson centre*

A closed Hamiltonian $f(z_1, z_2)$ is *measured* by the brane by substitution of the matrix-valued normal coordinates:

$$J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

At finite N one chooses a word ordering. In $H^0(\mathcal{R}_N^{GL_N}, Q)$ adjacent swaps are exact, $Q\,\mathrm{Tr}(\psi vu) = \mathrm{Tr}([\phi_1, \phi_2]vu)$, so the ordering ambiguity disappears modulo the constraint. In the stable connected limit $N \to \infty$, $J(f)$ is a well-defined Hamiltonian observable on the brane. The matrix Poisson bracket gives

$$\big\{J(f), J(g)\big\}_\partial = J\big(\{f, g\}_{\mathfrak{h}}\big), \tag{4}$$

after removal of the scalar Hamiltonian. Equation (4) is the ordinary canonical bracket of the probe system: it is not a string-theoretic input; it is the statement that the probe brane is a representation of the Hamiltonian Lie algebra of the disk. On monomials,

$$\big\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\big\} = \overline{(kn - lm)\,\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}. \tag{5}$$

The map $f \mapsto J(f)$ is the *moment-map shadow* of the full closed-open operation. The full operation cannot be a single function on the open algebra: a closed local field that lives in a region of space-time near the brane should produce *a central operation on all open factorisation observables*, not merely an element. The correct target is therefore the derived Poisson centre

$$Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}).$$

The structural form of this identification is the content of Theorem 0.2.27 below: the closed Chevalley–Eilenberg cochains on $\mathfrak{g}$ identify with the reduced derived P_0-Poisson centre of the stable open Hamiltonian algebra, and the boundary observable $J(f) = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is the image, under that identification, of the coordinate dual to the shifted cotangent generator (*not* of the coordinate dual to f itself); the closed Hamiltonian ghost becomes the Hamiltonian vector-field operation on the brane.

0.1.4 Scalar reduction and the U(1) center anomaly

Throughout this manuscript constants in $\mathbb{C}[z_1, z_2]$ and the finite-rank scalar trace $\mathrm{Tr}(1) = N$ are removed before the bracket is recorded. This convention has algebraic content. The class of the removed cocycle is the U(1) center anomaly: the unique nontrivial class $[c] \in H^2_{\mathrm{Lie}}(\mathfrak{h}, \mathbb{C})$ obstructing the lift of the scalar-reduced Hamiltonian Poisson bracket to the unreduced bracket on $\mathbb{C}[z_1, z_2]$ on the closed side, and the unique obstruction to lifting the trace algebra Lie bracket to $\mathfrak{gl}_N$ rather than to its quotient $\mathfrak{sl}_N \oplus \mathbb{C} \cdot I$ on the open side. Theorem 0.1.3 makes this precise; Theorem 0.1.6 identifies the same class with the Moyal/Capelli $\hbar N$ shift on the brane. The flag "before/after killing the scalar" is propagated explicitly through every later theorem statement that touches $\mathfrak{h}$ or $\mathcal{A}_\partial$.

0.1.4.0.1 Notation. Write $\mathfrak{h}_{\mathrm{poly}} = \mathbb{C}[z_1, z_2]_+$ for the polynomial Hamiltonian Lie algebra without the constant subalgebra removed; the bracket on $\mathfrak{h}_{\mathrm{poly}}$ is $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ and takes values in $\mathbb{C}[z_1, z_2]_+ \oplus \mathbb{C} \cdot 1$. Define

$$\omega : \mathfrak{h}_{\mathrm{poly}} \otimes \mathfrak{h}_{\mathrm{poly}} \longrightarrow \mathbb{C}, \qquad \omega(f, g) = [\, \{f, g\} \,]_0,$$

the constant Taylor coefficient of $\{f, g\}$. On monomials,

$$\omega(z_1^k z_2^l, z_1^m z_2^n) = \begin{cases} kn - lm, & (k + m, l + n) = (1, 1), \\ 0, & \text{otherwise}, \end{cases}$$

hence ω is concentrated in the unique bidegree where the bracket multidegree $(k + m - 1, l + n - 1)$ is $(0, 0)$; this is exactly the bidegree the manuscript discards as "scalar trace".

LEMMA 0.1.2. ω is a Lie 2-cocycle on $\mathfrak{h}_{\mathrm{poly}}$, and its cohomology class

$$[c] := [\omega] \in H^2_{\mathrm{Lie}}(\mathfrak{h}_{\mathrm{poly}}, \mathbb{C})$$

is nontrivial. After projection to $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C}$, the same formula represents a nontrivial class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$.

Proof. The Jacobi identity for $\{-, -\}$ on $\mathfrak{h}_{\mathrm{poly}}$ asserts

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0,$$

and projecting each summand to its constant Taylor coefficient gives the cocycle identity

$$\omega(f, \{g, h\}) + \omega(g, \{h, f\}) + \omega(h, \{f, g\}) = 0.$$

Antisymmetry $\omega(g, f) = -\omega(f, g)$ is inherited from antisymmetry of the bracket. Hence $\omega \in Z^2_{\mathrm{Lie}}(\mathfrak{h}_{\mathrm{poly}}, \mathbb{C})$. Suppose $\omega = \delta\eta$ for some linear $\eta : \mathfrak{h}_{\mathrm{poly}} \to \mathbb{C}$. Then $\omega(f, g) = -\eta(\{f, g\})$. Test on $f = z_1$, $g = z_2$: $\omega(z_1, z_2) = 1$ while $\{z_1, z_2\} = 1$, so $\eta(1) = -1$. But $z_1 \cdot 1$ and $1 \cdot z_1$ both bracket to zero

with any $g \in \mathfrak{h}_{\mathrm{poly}}$ except via the same calculation, so the system $\eta(\{f, g\}) = -\omega(f, g)$ overdetermines η: testing on $f = z_1^2, g = z_2$ gives $\eta(2z_1) = 0$, hence η vanishes on $z_1\mathbb{C}[z_1, z_2]_+$, contradicting $\eta(1) = -1$. Therefore $[\omega] \neq 0 \in H^2_{\mathrm{Lie}}(\mathfrak{h}_{\mathrm{poly}}, \mathbb{C})$. The same argument descends to $\bar{A} = \mathfrak{h}_{\mathrm{poly}}/\mathbb{C}$, since ω vanishes on the line $\mathbb{C} \cdot 1$ already, so it descends to a cocycle on the quotient and remains nontrivial. □

Theorem 0.1.3 *(U(1) center anomaly: closed side).* The class $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ of Lemma 0.1.2 is the unique obstruction class to lifting the Lie algebra structure on $\bar{A}$ to a Lie algebra structure on the unreduced polynomial Hamiltonians $\mathfrak{h}_{\mathrm{poly}}$ along the abelian extension

$$0 \longrightarrow \mathbb{C} \cdot 1 \longrightarrow \mathfrak{h}_{\mathrm{poly}} \longrightarrow \bar{A} \longrightarrow 0.$$

Equivalently, $\mathfrak{h}_{\mathrm{poly}}$ is isomorphic to the central extension $\bar{A}_{[\bar{c}]} = \bar{A} \oplus \mathbb{C}K$ with bracket

$$[\,(f, aK), (g, bK)\,] = (\{f, g\}_{\bar{A}},\ \bar{c}(f, g)\,K),$$

and $[\bar{c}]$ generates a one-dimensional sub-line of $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ that is the obstruction.

Proof. Central extensions of $\bar{A}$ by $\mathbb{C}$ are classified by $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ via the standard Chevalley-Eilenberg argument. The exact sequence above is a central extension because the constant Hamiltonian $1 \in \mathfrak{h}_{\mathrm{poly}}$ has zero Hamiltonian vector field, so $\mathbb{C} \cdot 1$ is central in $\mathfrak{h}_{\mathrm{poly}}$. The class of this extension is the obstruction to a Lie-algebra splitting; computing the class on monomial generators,

$$[z_1^k z_2^l, z_1^m z_2^n]_{\mathfrak{h}_{\mathrm{poly}}} = \{z_1^k z_2^l, z_1^m z_2^n\} = (kn - lm) z_1^{k+m-1} z_2^{l+n-1},$$

and projecting to $\mathbb{C} \cdot 1$ keeps only the $(0, 0)$-Taylor coefficient ω of Lemma 0.1.2. Thus the extension class is $[\bar{c}]$, and Lemma 0.1.2 shows it is nontrivial. The same lemma shows the line $\mathbb{C} \cdot [\bar{c}]$ is one-dimensional inside $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$, and that splitting along this line recovers $\mathfrak{h}_{\mathrm{poly}}$. □

Theorem 0.1.4 *(U(1) center anomaly: open side).* At finite N the trace algebra Lie bracket $\{F, G\} = \mathrm{Tr}(\partial_{\phi_1} F\, \partial_{\phi_2} G - \partial_{\phi_2} F\, \partial_{\phi_1} G)$ on $\mathbb{C}[\mathfrak{gl}_N \oplus \mathfrak{gl}_N]^{GL_N}$ satisfies

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$$

in the unreduced ring, and zero in the scalar-reduced quotient $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$. The cohomology class of this central scalar in H^2_{Lie} of the unreduced trace algebra is the image of $[\bar{c}]$ under the boundary evaluation map

$$\partial^{\mathrm{full}}_{\mathrm{bb},N} : \mathbb{C}[z_1, z_2] \longrightarrow \mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}, \qquad f \longmapsto \mathrm{Tr}\, f(\phi_1, \phi_2),$$

i.e. the same anomaly class is realised on the brane as the U(1) center of $\mathfrak{gl}_N$ with central element I.

Proof. The matrix Poisson bracket gives, on monomials of bidegree $(1, 0)$ and $(0, 1)$,

$$\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = \mathrm{Tr}\big((\partial_{\phi_1}\mathrm{Tr}\,\phi_1)(\partial_{\phi_2}\mathrm{Tr}\,\phi_2) - (\partial_{\phi_2}\mathrm{Tr}\,\phi_1)(\partial_{\phi_1}\mathrm{Tr}\,\phi_2)\big) = \mathrm{Tr}(I \cdot I) - 0 = N.$$

Hence the unreduced bracket realises the constant N as a scalar element. In $H^0_{\mathrm{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$ this scalar is killed, matching the projection on the closed side. The boundary evaluation map $\partial^{\mathrm{full}}_{\mathrm{bb},N}$ sends $1 \in \mathbb{C}[z_1, z_2]$ to $\mathrm{Tr}(I) = N$ and satisfies, for monomials with $(k + m, l + n) = (1, 1)$,

$$\partial^{\mathrm{full}}_{\mathrm{bb},N}(\{z_1^k z_2^l, z_1^m z_2^n\}) = (kn - lm) \cdot \mathrm{Tr}(I) = (kn - lm) \cdot N \cdot I,$$

which equates the closed cocycle ω with the open trace bracket on the central direction. Hence the extension class on the brane is the image of $[\bar{c}]$. Since $\mathbb{C} \cdot I \subset \mathfrak{gl}_N$ is the U(1) centre with character Tr, the brane realisation of $[\bar{c}]$ is exactly the U(1) centre of $\mathfrak{gl}_N$. □

Remark 0.1.5. Every theorem in this manuscript that uses $\mathfrak{h}$, $\bar{A}$, or $\mathcal{A}_\partial = H^0_{\mathrm{red}}$ lives *after* killing the scalar. The unreduced statement requires keeping the central line $\mathbb{C} \cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ on the closed side and $\mathbb{C} \cdot \mathrm{Tr}(1)$ on the open side, with the U(1) center anomaly class $[\bar{c}]$ recording the obstruction to a Lie-algebra splitting. Theorem 0.1.22, Theorem 0.2.81, Corollary 0.2.82, and Proposition 0.2.62 are all stated on the reduced side; their unreduced refinements are governed by $[\bar{c}]$ via the central extensions of Theorems 0.1.3 and 0.1.4.

QUANTUM-CLASSICAL EQUIVALENCE OF THE U(1) CENTER COCYCLE

THEOREM 0.1.6 *(Quantum-classical equivalence of the U(1) center class).* Let $\Phi = (\phi_1, \phi_2)$ be the classical brane field on the formal local disk and $\Phi_\hbar = (\phi_1^\hbar, \phi_2^\hbar)$ its Weyl quantization at finite N, with the Moyal product $f * g = m \circ \exp\left(\frac{\hbar}{2} P\right)(f \otimes g)$ for $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Then on the brane the quantization satisfies the Capelli/canonical commutator identity

$$Y_\hbar X_\hbar - X_\hbar Y_\hbar + \hbar N I = 0, \qquad X_\hbar = \phi_1^\hbar,\ Y_\hbar = \phi_2^\hbar,$$

which evaluates to $\{\phi_2, \phi_1\} = -\partial = N$ on the trace generators $X = \mathrm{Tr}\,\phi_1$, $Y = \mathrm{Tr}\,\phi_2$. In the classical limit $\hbar \to 0$, the bracket $\{\mathrm{Tr}\,\phi_1, \mathrm{Tr}\,\phi_2\} = N$ recovers the central scalar of Theorem 0.1.4. The classical Poisson cocycle ω of Lemma 0.1.2 and the quantum Capelli shift $\hbar N$ represent the *same cohomology class* $[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$ under classical-to-quantum reduction, i.e. via the Rees deformation $U_\hbar(\bar{A}_{[\bar{c}]})$ specialising to $\widehat{\mathbf{S}}(\bar{A}_{[\bar{c}]})$ at $\hbar = 0$.

Proof. The Weyl convention gives $[X_\hbar, Y_\hbar]_* = \hbar\{X, Y\} + O(\hbar^3)$. On the trace generators, Proposition 0.2.62 at $(k, l, m, n) = (1, 0, 0, 1)$ gives $\{X, Y\} = N$ before scalar reduction. Hence

$$X_\hbar Y_\hbar - Y_\hbar X_\hbar = \hbar N + O(\hbar^3),$$

and at finite N the higher-order terms vanish on these linear generators because P^r requires at least r derivatives in each tensor factor. Rearranging gives the displayed Capelli/canonical identity. This is the classical-to-quantum reduction in the sense of Razmyslov [19] and Procesi [18]: the trace identities of $\mathfrak{gl}_N$ at level N are exactly the Capelli identities [26, 27], and the abelian Capelli at $r = 1$ is the canonical-commutator shift.

For the cohomology-class identification, observe that the Rees algebra $U_\hbar(\bar{A}_{[\bar{c}]})$ has central fibre $\widehat{\mathbf{S}}(\bar{A}_{[\bar{c}]})$ at $\hbar = 0$ and generic fibre containing $\bar{A}_{[\bar{c}]}$ as Lie algebra at $\hbar = 1$. The PBW filtration's associated graded carries the Lie 2-cocycle as the primary obstruction to PBW-splitting the deformation. Both sides of the classical-to-quantum reduction therefore record the same class in $H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C})$: classically ω is the obstruction to splitting the central extension $\mathfrak{h}_{\mathrm{poly}} \to \bar{A}$, quantum-mechanically the same ω is the leading-order coefficient of the Moyal star commutator on the central direction. □

Remark 0.1.7. The class $[\bar{c}]$ is therefore a physical observable of the brane: the U(1) center of $\mathfrak{gl}_N$ acts on the brane Hilbert space with character Tr, and the canonical/Capelli shift $\hbar N$ is the charge of this central direction. The scalar reduction in the body of the manuscript is the operation of restricting to the kernel of this character; it is not a vacuous bookkeeping step.

0.1.5 The two algebras

The local model contains two algebras, both *derived* from the canonical mechanics above rather than postulated:

- The algebra $\mathcal{A}_N$ of GL_N-invariant local trace observables on the Dirac-reduced probe stack of (1); equivalently, the algebra of local operators of the open BV theory on the brane line $\mathbb{R} \times p$.
- The algebra $\mathcal{B}$ of local operators of the closed Hamiltonian BF sector (2); the operator-product expansion in the $\mathbb{R}$-direction equips $\mathcal{B}$ with an associative product, derived from the locally constant factorisation structure of the topological direction.

Conjecture 0.1.8 *(Stable bulk–boundary Koszul duality).* In the stable large-N limit and within the Hamiltonian sector,

$$\lim_{N\to\infty} \mathcal{A}_N \simeq \mathcal{B}^!,$$

the Koszul dual of $\mathcal{B}$. Theorem 0.1.22 is the classical Hamiltonian shadow of this conjecture; the quantum refinement is the formal Weyl/Moyal target of §0.1.13 below.

This recovers the open/closed string duality of [4], but here it is no longer a string-theoretic ansatz: it is the statement that the closed local theory is the derived Poisson centre of the open local theory.

The provenance of this local model in higher-dimensional physics is twisted M-theory. The following conjecture states, in precise form, the universality claim that the formal local construction here is the Hamiltonian-sector restriction of every twisted M-theory background of the indicated type.

Conjecture 0.1.9 *(Twisted-M-theory universality of the local model).* Let X be a holomorphic-symplectic surface and let $\mathcal{T}_{\mathrm{tw}M}(\mathbb{R} \times X; \mathbb{R} \times p)$ denote the perturbative twisted M-theory data on $\mathbb{R} \times X$ in the presence of a stack of N one-dimensional topological branes supported on $\mathbb{R} \times p$, in the sense of the twisted holomorphic-topological theory [25] whose protected sector restricts to the local Hamiltonian sector here. Hypotheses: (i) the BV-cohomology vanishing of [14, 15] on the formal local model; (ii) the Costello locality theorem ensuring that restriction to a formal neighbourhood of $\mathbb{R} \times p$ is well-defined; (iii) the Hamiltonian reduction of the holomorphic-symplectic data to the formal symplectic disk at p is regular. Then the Hamiltonian-sector restriction

$$\mathcal{T}_{\mathrm{tw}M}(\mathbb{R} \times X; \mathbb{R} \times p)\big|_{\mathrm{Ham},\widehat{p}}$$

is canonically equivalent, as a pair (open BV theory; closed Hamiltonian BF theory together with open trace data), to the formal local model constructed in this manuscript: open side $\big(\Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1],\ S(\alpha)\big)$, closed side $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ with $\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \mathfrak{h}^\vee[1]$, and bulk-to-boundary evaluation $f \mapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$. In particular every such twisted M-theory restriction satisfies Theorem 0.1.22.

Remark 0.1.10. Conjecture 0.1.9 replaces the BCOV analogy as the primary physical motivation. The original BCOV analysis is written for Calabi–Yau threefolds ([1]), where the target dimension is $d = 3$; the local model here is a holomorphic-symplectic surface ($d = 2$) and is therefore not a reduction of BCOV in any direct sense. The Hamiltonian BF formulation (2) is the correct primary physical name. The deciding evidence for the conjecture is (a) the explicit BV-cohomology computation of [14, 15]; (b) a reduction lemma for protected twisted-M-theory operators on $\mathbb{R} \times X$ to the formal disk at p; (c) compatibility with the scalar/center-of-mass anomaly term of Remark 0.1.1.

0.1.6 Three notions of locality

Locality is not one notion in this model. It is three.

0.1.6.0.1 (a) Support locality along the brane. For an open interval $I \subset \mathbb{R}$ on the world-line, define the compactly supported Hamiltonian current algebra $\mathfrak{h}_I = C_c^\infty(I) \otimes \widehat{\mathfrak{h}}$ with bracket

$$[a \otimes f,\, b \otimes g] = ab \otimes \{f, g\}_{\widehat{\mathfrak{h}}}.$$

The open observables on I are $\mathcal{A}_\partial(I) = \mathbf{S}(\mathfrak{h}_I)$, generated by the smeared trace functionals

$$O_f(a) = \int_I a(t)\, \mathrm{Tr}\, f\big(\phi_1(t), \phi_2(t)\big)\, dt.$$

Their bracket is the canonical equal-time bracket:

$$\big\{O_f(a), O_g(b)\big\} = O_{\{f,g\}_{\widehat{\mathfrak{h}}}}(ab).$$

Disjoint supports multiply, since $ab = 0$ outside the intersection; overlapping supports give a delta-function contact term. This is the operational meaning of locality for the open factorisation algebra: disjoint independent measurements multiply, and Poisson brackets are supported where the measurements meet.

0.1.6.0.2 (b) Topological locality. In the topological direction (the $\mathbb{R}^2$ factor), translations are Q-exact:

$$L_v = [Q, \iota_v].$$

Equivalently, positive jets in the topological direction vanish in local-operator cohomology by the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$ (Lemma 0.2.39). Therefore the precise position of an insertion is not physical; only the collision pattern of insertions matters. This is the structural reason the factorisation algebra is locally constant in the topological direction.

0.1.6.0.3 (c) Formal holomorphic locality. In the holomorphic direction (the $\mathbb{C}^2$ factor), the model has been completed at the brane point $p \in \mathbb{C}^2$. It is a theory of holomorphic jets at p, not a theory of finite propagation through $\mathbb{C}^2$. Antiholomorphic translations are Q-exact:

$$L_{\partial/\partial\bar{z}_i} = [Q, \iota_{\partial/\partial\bar{z}_i}].$$

The kinematic differential of the closed sector is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, and on the formal-jet model $\bar{\partial}$-cohomology is concentrated in degree zero. The cotangent boundary sector is *distributionally local, not polynomially local*: a coadjoint mode is a transverse principal part $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1\, dz_2$, paired with a brane Hamiltonian by residue,

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \delta_{a,m}\delta_{b,n}.$$

This is exactly what one expects for a momentum dual field attached to a defect: support on the defect, transverse behaviour encoded by residues. The polynomial one-ψ descendants $\Psi_{k,l}$ realise the PBW special fibre of $\mathfrak{h}^\vee[1]$ (Proposition 0.2.63); Corollary 0.2.64 records the deformation-level mismatch with the coadjoint module, exactly the ordinary distinction between polynomial and distributional realisations.

The factorisation-algebra structure is now *forced*: it is the algebraic expression of (a)+(b)+(c). Disjoint supports multiply along the brane; topological motion is Q-exact and changes nothing in cohomology; transverse holomorphic dependence is distributional. No external categorical input has been imposed; the structure is the local algebra of measurements in a constrained canonical system.

0.1.7 Unitarity

The model is a twisted BV sector. It is not a Lorentzian unitary quantum mechanics. There is no positive-definite Hilbert space, no Wightman positivity, and no reflection positivity. The fields include ghosts and antifields; topological time has been made cohomologically trivial.

The replacements for unitarity are the consistency conditions of the BV formalism: the classical master equation

$$QI + \tfrac{1}{2}\{I, I\} = 0$$

and the quantum master equation

$$QI + \tfrac{1}{2}\{I, I\} + \hbar\, \Delta I = 0,$$

together with associativity of the resulting factorisation/OPE algebra. If the local model arises as a protected sector of a larger unitary theory (for example, of a topological twist of a supersymmetric quantum field theory in which the local observables of the present model are the cohomology with respect to a supercharge), then unitarity is carried by the larger theory, and the local sector retains only the cohomological algebra of protected observables. Within the present manuscript we do not impose unitarity, and we make no Lorentzian claims.

0.1.8 Thesis

> A stack of branes is a system of probes for the formal symplectic normal disk. Dirac reduction of the probe phase space gives the open trace algebra. Hamiltonian BF theory is the shifted cotangent theory of canonical transformations. The closed observables are precisely the derived Poisson central operations on the stable open Hamiltonian brane algebra.

0.1.9 Mixed holomorphic-topological strings

A local string theory in this paper means the part of string field theory visible to local operators. The closed fields form a BV theory; local observables form a factorization algebra; a brane contributes an open cyclic field theory and a bulk-to-boundary map. The words topological and holomorphic describe how these local operators vary in the spacetime directions. This is a local string-field-theoretic use of the word "string", not a definition of a global theory of worldsheet moduli, amplitudes on all backgrounds, or a completed quantum anomaly theory.

Gwilliam–Williams [5] are the precedent for formulating a holomorphic string in the Costello BV/factorization formalism. Their object is the holomorphic bosonic string on a Riemann surface, with Dolbeault fields (γ, β, c, b), a one-loop quantization in the critical complex dimension, and a factorization algebra recovering the BRST cohomology. The object here is not that theory and not a replacement definition. It is a formal local target-space sector: de Rham/Floer directions, a holomorphic symplectic formal disk, Hamiltonian cotangent BF fields, and a brane large-N operator algebra.

The topological word refers to locally constant factorization in the de Rham/Floer directions. The holomorphic word refers to the Dolbeault presentation: the kinetic operator contains $\bar{\partial}$, the fields are holomorphic jets after local formal completion, and anti-holomorphic translations act by cochain-homotopic zero. A holomorphic theory may acquire further topological descent after quantization; that extra descent is not part of the definition used here.

Definition 0.1.11 *(Working local topological factorization sector).* Let M be a smooth manifold. A local topological closed-string sector on M is a classical BV theory whose local observables form a locally

constant factorization algebra on M: for inclusions of sufficiently small disks $U \subset V$, the factorization structure map

$$\mathrm{Obs}^{\mathrm{cl}}(U) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(V)$$

is a quasi-isomorphism. In this paper, attaching branes to such a sector means specifying a cyclic open BV theory and the closed-open operations actually constructed. In a de Rham model the closed fields carry the differential d_M plus the BV linearized differential, and the locally constant factorization product records collision of operators up to isotopy.

Remark 0.1.12. The exact A-model factor used below is a local topological sector in this sense: the self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ is represented by the de Rham complex $\Omega^*(\mathbb{R})$, and its local observables are insensitive to the metric geometry of the interval. This is the topological part of the mixed holomorphic-topological example.

Definition 0.1.13 *(Working local holomorphic factorization sector).* Let Y be a complex manifold. A local holomorphic sector on Y is a classical BV theory whose fields are Dolbeault complexes on Y, whose differential has the form $\bar{\partial}$ plus the BV linearized differential, whose interactions are holomorphic polydifferential operators, and whose local observables form a holomorphic factorization algebra. Here holomorphic factorization means that, on coordinate polydisks, the factorization products and restriction maps are represented by holomorphic polydifferential operations and every anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables. In the target-space Kodaira-Spencer examples the fields are Dolbeault-valued polyvectors; in the holomorphic symplectic examples below they are Hamiltonian polyvectors after removing constants. In this paper branes are treated through cyclic open BV theory and the bulk-to-boundary operations constructed below.

Remark 0.1.14. BCOV/Kodaira-Spencer theory is the guiding threefold example [1]: its fields are Dolbeault-valued polyvectors, and the Calabi-Yau volume form supplies the divergence operator. In the holomorphic symplectic surface model used here, the analogous divergence-free vector fields are represented locally by Hamiltonians modulo constants. The surface model is not a truncation of threefold BCOV theory.

Remark 0.1.15. The classical local observable theory underlying Gwilliam–Williams is an example of a pure holomorphic BV/factorization sector on a Riemann surface, with additional worldsheet ghost, BRST, and critical-dimension quantum structure. Their critical-dimension and BRST-cohomology theorems are not imported here. The model below is a different formal local target-space sector whose holomorphic sector is Hamiltonian BF, or the holomorphic symplectic Kodaira-Spencer analogue, and whose topological sector is de Rham/Floer. In particular, the critical complex dimension in Gwilliam–Williams is a worldsheet ghost-anomaly statement for the holomorphic bosonic string; it is not a constraint on the local $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian sector studied here.

Definition 0.1.16 *(Working classical local mixed holomorphic-topological sector).* Let M be a smooth manifold and let Y be a complex manifold. A classical local mixed holomorphic-topological closed sector on $M \times Y$ consists of:

(i) a sheaf of cochain complexes of fields $\mathcal{E}$ which, on product coordinate opens $D \times B$, has the form

$$\mathcal{E}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B, \mathcal{V})$$

for a graded holomorphic vector bundle or formal holomorphic vector space $\mathcal{V}$;

(ii) a degree -1 local BV pairing on $\mathcal{E}$ and a differential

$$Q = d_M + \bar{\partial}_Y + Q_0,$$

where Q_0 is holomorphic over Y and is linear over de Rham forms on M;

(iii) a degree-zero local functional $I \in \mathcal{O}_{\mathrm{loc}}(\mathcal{E})$, built from differential operators in the M directions and holomorphic polydifferential operators in the Y directions, satisfying the classical master equation

$$QI + \frac{1}{2}\{I, I\} = 0;$$

(iv) the associated classical observables

$$\mathrm{Obs}^{\mathrm{cl}}(U) = \left(\widehat{\mathrm{Sym}}(\mathcal{E}^!_c(U)), Q_{\mathrm{obs}}\right), \qquad Q_{\mathrm{obs}} = Q^! + \{I, -\},$$

with the standard factorization product for disjoint supports. Here $\mathcal{E}^!_c(U)$ denotes compactly supported continuous linear observables, i.e. the density-dual cosheaf of fields; extension by zero gives the covariant structure maps used by the factorization algebra.

These data must satisfy the mixed locality condition. For every inclusion of small disks $D \subset D'$ in M and every holomorphic polydisk $B \subset Y$, the structure map

$$\mathrm{Obs}^{\mathrm{cl}}(D \times B) \longrightarrow \mathrm{Obs}^{\mathrm{cl}}(D' \times B)$$

is a quasi-isomorphism. In holomorphic coordinates on B, each anti-holomorphic translation $\partial/\partial\bar{z}_i$ acts null-homotopically on observables: there is a degree -1 derivation η_i with

$$[Q_{\mathrm{obs}}, \eta_i] = \mathcal{L}_{\partial/\partial\bar{z}_i}.$$

A mixed brane for such a sector is supported on $N \times Z$, where $N \subset M$ is a smooth submanifold and $Z \subset Y$ is a holomorphic submanifold or a formal holomorphic neighbourhood of one, and is specified by an object $\mathcal{F}$ of the local holomorphic brane category on Z. Its open fields have the derived form

$$\Omega^*(N)\widehat{\otimes}\mathrm{RHom}_Y(\mathcal{F}, \mathcal{F}) \otimes \mathrm{End}(V)[1],$$

with a cyclic trace. For a smooth brane this recovers the Dolbeault model $\Omega^*(N)\widehat{\otimes}\Omega^{0,*}(Z, \mathcal{W})$; for a skyscraper brane it produces the Ext algebra. A full closed-open mixed datum would further require a morphism of factorization algebras

$$\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed}}\big|_{N\times Z} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}),$$

from closed observables to the Poisson/BV factorization center of the open brane observables, compatible with the BV brackets and factorization products. In the Hamiltonian model below we construct the degree-zero Hamiltonian current shadow of this map; the full closed CE/cotangent factorization-center lift is Problem 0.2.35.

Remark 0.1.17. A quantum mixed holomorphic-topological string datum would be the corresponding $\hbar$-adic factorization algebra obtained from a renormalized BV action satisfying the quantum master equation. This paper constructs no such quantum datum for the Hamiltonian model. It proves the classical local closed-open Hamiltonian theorem and formulates the quantum upgrade in Problem 0.2.93.

Definition 0.1.16 separates three levels. A pure topological sector is locally constant in all real directions; in examples this includes the A-model local/Floer direction and the Chern-Simons-type open field theory on a topological brane. A pure holomorphic sector is Dolbeault in its complex directions and produces a holomorphic factorization algebra; Gwilliam–Williams' holomorphic bosonic string is a worldsheet example, while BCOV/Kodaira-Spencer gives the guiding target-space closed-string example. The Hamiltonian BF complex below is the formal local target-space sector used in this paper. A mixed holomorphic-topological sector lives on $M \times Y$ with locally constant factorization in M and holomorphic factorization in Y. The A/B terminology used below is one source of such examples: the A-side supplies the de Rham/Floer direction and the B-side supplies the Dolbeault or formal holomorphic direction.

0.1.10 The local mixed model

The paper studies the product local model

$$\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}.$$

The theorem is not a global statement about all twisted M-theory backgrounds. It is the formal local calculation obtained after completing at a brane $\mathbb{R} \times p$ and taking the holomorphic symplectic normal disk at $p \in \mathbb{C}^2$, with

$$\omega = dz_1 \wedge dz_2.$$

The closed sector is the Hamiltonian Kodaira-Spencer/BF theory on this mixed space, not a pure A-model or pure B-model. The open sector is the cyclic open string field theory on the mixed brane: topological along $\mathbb{R}$ and formal holomorphic in the normal $\mathbb{C}^2$ directions.

Remark 0.1.18 *(Global obstruction).* The Hamiltonian identification used here is local. On a holomorphic symplectic surface X, a holomorphic vector field v is symplectic when

$$\alpha_v = -\iota_v \omega$$

is a closed holomorphic one-form, and it is Hamiltonian only after choosing f with $df = \alpha_v$, in the convention $\iota_{X_f}\omega = -df$. Thus globalizing the Hamiltonian sector requires either a sheaf of local Hamiltonians or vanishing of the corresponding holomorphic de Rham obstruction. The formal disk has no such obstruction by the holomorphic Poincare lemma; a general X need not.

PROPOSITION 0.1.19 *(Global local-Hamiltonian restriction).* Let (X, ω) be a connected holomorphic symplectic surface and let $p \in X$. Put

$$\mathcal{H}_X = \mathcal{O}_X/\mathbb{C}_X, \qquad \widehat{\mathfrak{h}}_{X,p} = \widehat{\mathcal{O}}_{X,p}/\mathbb{C}.$$

The differential identifies $\mathcal{H}_X$ with the sheaf of closed holomorphic one-forms, hence with the sheaf of holomorphic symplectic vector fields by $\alpha = -\iota_v\omega$. For every $p \in X$ there is a Lie algebra restriction

$$H^0(X, \mathcal{H}_X) \longrightarrow \widehat{\mathfrak{h}}_{X,p}.$$

The formal neighbourhood $(\widehat{X}_p, \omega)$ defines the same stable Hamiltonian brane sector as the local model of Theorem 0.1.22, after choosing a formal Darboux coordinate system. The resulting boundary map

$$\widehat{\mathfrak{h}}_{X,p} \longrightarrow H^0_{\mathrm{Ham}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{brane}})$$

is independent of this choice up to the natural action of the formal symplectomorphism group. The subspace coming from global Hamiltonian functions is the image of

$$H^0(X, \mathcal{O}_X)/\mathbb{C} \longrightarrow H^0(X, \mathcal{H}_X).$$

The obstruction for a locally Hamiltonian symplectic vector field to come from a global Hamiltonian function is its period class in $H^1(X, \mathbb{C}_X)$.

Proof. The holomorphic Poincare lemma gives an isomorphism

$$d : \mathcal{O}_X/\mathbb{C}_X \xrightarrow{\sim} \Omega^1_{X,\mathrm{cl}}.$$

The holomorphic symplectic form identifies a symplectic vector field v with the closed one-form $\alpha_v = -\iota_v\omega$. Thus a section of $\mathcal{H}_X$ is exactly a local Hamiltonian primitive, modulo local constants, and the Poisson bracket is independent of the local constants. Taking the germ at p and completing gives the displayed restriction to $\widehat{\mathfrak{h}}_{X,p}$.

The formal holomorphic Darboux theorem identifies the completed local symplectic algebra at p with $\mathbb{C}[[z_1, z_2]]$ equipped with $dz_1 \wedge dz_2$. Under this identification Hamiltonians modulo constants become $\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C}$, and the point-brane Ext algebra is the coordinate-free exterior algebra $\mathbf{Ext}^*_X(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda^* T_p X$, with cyclic trace fixed by ω_p. Hence the open BV reduction and the Hamiltonian boundary evaluation are exactly the formal local construction after choosing Darboux coordinates. A different Darboux coordinate system differs by a formal symplectomorphism, which acts on Hamiltonians and on the normal coordinate algebra; the substitution map $z_i \mapsto \phi_i$ is equivariant for this action.

The exact sequence

$$0 \to \mathbb{C}_X \to \mathcal{O}_X \to \mathcal{H}_X \to 0$$

gives

$$0 \to H^0(X, \mathcal{O}_X)/\mathbb{C} \to H^0(X, \mathcal{H}_X) \xrightarrow{\delta} H^1(X, \mathbb{C}_X) \to H^1(X, \mathcal{O}_X).$$

For a locally Hamiltonian vector field, δ is the period class of the closed one-form $-\iota_v\omega$. Vanishing of this class is exactly the existence of a single global holomorphic Hamiltonian primitive, unique modulo constants. □

Remark 0.1.20. This is the maximal global statement used in the paper. If $H^1(X, \mathbb{C}) = 0$, global symplectic vector fields which are locally Hamiltonian have global Hamiltonian primitives. If X is Stein, every finite jet needed for a finite local calculation can be lifted to a global holomorphic Hamiltonian, noncanonically: apply Cartan's theorem B to

$$0 \to \mathfrak{m}_p^{k+1} \to \mathcal{O}_X \to \mathcal{O}_X/\mathfrak{m}_p^{k+1} \to 0.$$

This does not imply surjectivity onto $\widehat{\mathcal{O}}_{X,p}$. If X is compact connected, then $H^0(X, \mathcal{O}_X)/\mathbb{C} = 0$; abelian surfaces still have nonzero locally Hamiltonian translations, obstructed by their period classes. A global cochain-level Poisson factorization-center map or a global quantum deformation-quantization statement requires additional descent, anomaly, and background-choice data.

The closed Hamiltonian coefficient Lie algebra is the reduced Tate vector space

$$\widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1.$$

We write a class by its unique zero-constant Taylor representative. The bracket is the projected Poisson bracket

$$\{f, g\} = \partial_{z_1} f \, \partial_{z_2} g - \partial_{z_2} f \, \partial_{z_1} g \mod \mathbb{C} \cdot 1.$$

Thus $\{z_1, z_2\} = 0$ in $\widehat{\mathfrak{h}}$, although the unreduced Poisson bracket of the chosen representatives is 1. The cotangent BF field contributes the continuous dual, so the residual local closed Lie algebra is

$$\mathfrak{g} = \widehat{\mathfrak{h}} \ltimes \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[1].$$

On product opens in $\mathbb{R}^2 \times \mathbb{C}^2$ the closed fields have the mixed form

$$\Omega^*(D) \widehat{\otimes} \Omega^{0,*}(B) \widehat{\otimes} \big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

with differential $d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ and BF interaction $\frac{1}{2} \int \langle \beta, \{\alpha, \alpha\} \rangle dz_1 dz_2$. Proposition 0.2.1 checks that this is a local mixed holomorphic-topological BV sector in the sense above.

The open brane is the product of the exact local A-brane $\mathbb{R} \subset T^*\mathbb{R}$ and the B-brane $\mathcal{O}_p$ at the formal holomorphic point. The Yoneda algebra of the skyscraper at the origin is computed in Proposition 0.2.2; for a cyclic Calabi-Yau A_∞ category the open field of an object $\mathcal{F}$ is $\mathrm{RHom}(\mathcal{F}, \mathcal{F})[1]$, and the formal Chern-Simons action

$$S(a) = \sum_{n \geq 1} \frac{1}{n+1} \mathrm{Tr}\big(a \, m_n(a, \ldots, a)\big), \qquad a \in \mathrm{RHom}(\mathcal{F}, \mathcal{F})[1],$$

is the algebraic core of Witten's open string field theory [4] and of the AKSZ packaging [6]. In the present local model

$$\mathrm{RHom}_{\mathbb{C}[z_1, z_2]}(\mathcal{O}_p, \mathcal{O}_p) \cong \Lambda(\varepsilon_1, \varepsilon_2),$$

and the ghost-zero open field is

$$\alpha_0 = A + \phi_1 \varepsilon_1 + \phi_2 \varepsilon_2.$$

The ghost-zero action is the first-order gauged matrix model of (1), where ϕ_1, ϕ_2 are adjoint scalar normal coordinates and A imposes the moment-map equation $[\phi_1, \phi_2] = 0$.

0.1.11 Hamiltonian sector and stable large N

Everything below is over $\mathbb{C}$. The phrase *formal local model* means the completed Taylor model at the chosen point of $\mathbb{C}^2$; equivalently, one may work polynomially degree by degree and then complete at the augmentation ideal. The symbol $\widehat{\mathbf{S}}$ denotes this completed symmetric algebra, and all Chevalley-Eilenberg cochains are continuous cochains for the same filtration.

Shifts follow the BV convention used in the field descriptions: an element of $V[1]$ has parity reversed and cohomological degree one lower than the corresponding element of V. The notation $\mathbf{S}(V)$ always means the free graded-commutative algebra on the graded vector space V.

Let

$$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1, \qquad \mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1, \qquad \mathfrak{h}^\vee = \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}.$$

Constants are removed because a constant Hamiltonian has zero Hamiltonian vector field. We use zero-constant Taylor representatives for calculations, but the Lie bracket is always the Poisson bracket

followed by projection modulo constants. On the brane side $\mathrm{Tr}_N(1) = N$ is the empty trace, a central scalar depending on the rank. We do not quotient the associative operator algebra by its unit. The Hamiltonian comparison uses the connected stable trace sector: the unit is retained in the symmetric algebra, but the empty trace is not a primitive generator and constants project to zero in the Hamiltonian Lie/vector quotient. If T is a positive word-degree trace expression, $\overline{T}$ denotes its connected trace class; constants have zero connected class.

At a fixed point of the brane line the reduced BV algebra is generated by two even matrices ϕ_1, ϕ_2 and the odd antifield $\psi = A^*$, with

$$Q\phi_1 = Q\phi_2 = 0, \qquad Q\psi = [\phi_1, \phi_2].$$

Thus the moment-map equation is imposed homologically. Boundary terms on $\mathbb{R}$ are discarded by taking compactly supported variations, or equivalently by working on an interval with fixed boundary values and passing to local operators in the interior.

The term *stable large-N* means the degreewise stable value of the invariant algebras under block restriction as $N \to \infty$. Inside a fixed finite matrix algebra all finite-N trace identities remain present. In the stable limit, Procesi–Razmyslov invariant theory [18, 19] says that invariant functions of two matrices are freely generated by cyclic trace words. The differential $Q\psi = [\phi_1, \phi_2]$ then makes adjacent swaps exact, so ghost-zero cyclic words reduce to ordinary commutative monomials

$$\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \qquad k + l > 0.$$

The primitive one-ψ Koszul homology supplies a second nonconstant family $\Psi_{k,l}$, obtained by inserting one ψ and symmetrising over all words with k copies of ϕ_1 and l copies of ϕ_2. The scalar class $\mathrm{Tr}(\psi)$ is excluded from the Hamiltonian comparison because constants have been removed from $\bar{A}$; higher ψ-count homology lies outside this Hamiltonian sector. The collapse of cyclic words to commutative monomials on $H^0(Q)$ is the algebraic shadow of Dirac reduction: the constraint $Q\psi = [\phi_1, \phi_2]$ is the Koszul resolution of the moment-map equation.

The large-N principle is the string-field-theoretic form of the Loday-Quillen-Tsygan principle [2, 3]; see also [22, Ch. 13] for the operadic packaging. Open strings on a stack produce matrix algebras; closed modes appear in the primitive stable part. LQT controls the stable invariants of the *unconstrained* matrix algebra; the moment-map equation $[\phi_1, \phi_2] = 0$ is the first-class constraint whose Koszul resolution is exactly the BV reduction $Q\psi = [\phi_1, \phi_2]$. The theorem below verifies the principle directly in the present connected Hamiltonian trace sector.

Theorem 0.1.21 *(Loday-Quillen-Tsygan).* Let A be a unital associative algebra over a field of characteristic zero. The stable Chevalley-Eilenberg homology of $\mathfrak{gl}_\infty(A)$ is the free graded-commutative algebra generated by shifted cyclic homology:

$$H_*^{\mathrm{CE}}\big(\mathfrak{gl}_\infty(A)\big) \cong \mathbf{S}\big(HC_{*-1}(A)\big).$$

Here this theorem is a template rather than a direct input: stable matrix constructions are controlled by cyclic words, while the particular reduced model is computed by Procesi–Razmyslov invariant theory and the explicit Koszul differential above. No global HC_{*-1} convention is imported into the main theorem; cyclic homology supplies the guiding stable-trace analogy, not the proof of the Hamiltonian sector.

0.1.12 *The concrete dictionary*

We write $\bar{A}_{\mathrm{desc}}$ for an abstract copy of $\bar{A}$ whose basis vector labelled by $z_1^k z_2^l$, with $k + l > 0$, is represented on the brane by the first descendant $\Psi_{k,l}$. On the closed side the cotangent field gives the continuous Taylor dual $\mathfrak{h}^\vee$. Choosing the monomial Taylor basis identifies $\mathfrak{h}^\vee$ with $\bar{A}_{\mathrm{desc}}$ as a filtered vector space. The closed module structure remains the coadjoint module structure.

For $k, l \geq 0$ and $k + l > 0$, set

$$H_{k,l} = z_1^k z_2^l \in \bar{A}, \qquad E_{k,l} = \delta_{k,l}[1] \in \mathfrak{h}^\vee[1],$$

where $\delta_{k,l}$ is the coefficient functional dual to the Taylor monomial $z_1^k z_2^l$:

$$\delta_{k,l}(z_1^a z_2^b) = \delta_{k,a}\delta_{l,b}.$$

On the brane side set

$$O_{k,l} = \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)},$$

and let $\Psi_{k,l}$ be the symmetrised trace with one ψ, k copies of ϕ_1, and l copies of ϕ_2. The theorem proves that

$$H_{k,l} \longmapsto O_{k,l}, \qquad E_{k,l} \longmapsto \Psi_{k,l}$$

extends to an isomorphism of the connected stable Hamiltonian descendant sector with

$$\mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]) \xrightarrow{\sim} O^{\mathrm{cl}}_{\mathrm{brane},H}$$

in the sector considered here. On the closed side the symbols $H_{k,l}$, $E_{k,l}$ form the degreewise polynomial PBW generators inside the special fibre of $U_\hbar(\mathfrak{g})$.

Each later step has a concrete target: construct the symbols $O_{k,l}$, $\Psi_{k,l}$ from the open fields; construct $H_{k,l}$, $E_{k,l}$ from Hamiltonians and cotangent Hamiltonians; prove that commutators of words reduce to commutative monomials in $H^0(Q)$; and verify that the bracket coefficient is $kn - lm$.

0.1.13 *Status of assertions and the analytic problem*

The local theorem stated below is the precise local assertion proved in these notes; the terms appearing in it are constructed in the subsequent sections. The proof consists of: the affine **Ext** calculation (Proposition 0.2.2); the open BV reduction (Lemma 0.2.5, Proposition 0.2.7); the Dirac reduction of the probe phase space (Lemma 0.2.6); the stable invariant-theory calculation of the Hamiltonian trace algebra (Proposition 0.2.40); the computation of the primitive one-antifield Koszul homology (Proposition 0.2.45); the formal Hamiltonian BF model for the closed side (Proposition 0.2.13); the construction of the bulk-to-boundary Hamiltonian evaluation map (Construction 0.2.43, Theorem 0.2.81); the CE/Koszul-duality computation; and the first bracket calculation (5).

The formal Weyl/Moyal calculation below is an all-order algebraic deformation-quantisation target for the constant Poisson formal disk; its monomial coefficients are checked by an exact script. A closed-string Feynman-integral derivation of those coefficients is a separate analytic problem, requiring the global BV/gauge-fixing data of Problem 0.2.103: a global elliptic complex extending the formal jet model; boundary conditions on $\mathbb{C}^2$; a gauge-fixing Q^{GF} and the heat-kernel propagator $P_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t \, dt$; zero-mode treatment and the IR limit; UV renormalisation scheme and local counterterms; quantum master equation or anomaly class; bulk-to-boundary kernel for the brane; trace normalisation and large-N weighting; and the connected/single-trace extraction convention. Costello's renormalized BV formalism supplies the general graph/RG package; Costello–Li prove the flat odd-dimensional BCOV/HCS case [14, 15]. The proof of Theorem 0.1.22 uses the formal local algebra alone and is independent of these analytic choices.

0.1.14 *The local theorem*

The theorem has three layers. First, the open calculation says that the ghost-zero brane theory is the algebra of invariant functions on commuting matrix pairs, together with the first Koszul descendants of the commuting-pair equation. Second, the closed calculation says that the same generators are the PBW symbols of the Hamiltonian cotangent Lie algebra. Third, the boundary map is not an abstract isomorphism: it is the physical operation of evaluating a Hamiltonian on the matrix-valued normal coordinates of the brane.

The local theorem is the umbrella of six interlocking statements: a cochain-level open-closed derived-center isomorphism Φ at universal generality (any pro-finite-dimensional Tate-conilpotent Lie algebra), its principal-part / coadjoint identification on the cotangent direction, its factorization-algebra upgrade on the brane line, its all-order quantum deformation $\Phi_\hbar$ on the Hamiltonian sector, and its U(1) center anomaly $[\bar{c}]$ at the boundary between scalar-reduced and unreduced sectors. The single residual is the Tate-coefficient lift of the cotangent direction in the unreduced factorization-algebra model.

Theorem 0.1.22 *(Local Hamiltonian closed-open derived center theorem).* Consider the formal local mixed holomorphic-topological BV brane on $\mathbb{R}^2 \times \mathbb{C}^2$ with brane $\mathbb{R} \times p$, in the convention of Section 0.2.8. Let $\mathfrak{h} = (z_1, z_2)\,\mathbb{C}[[z_1, z_2]]$ be the reduced Hamiltonian Lie algebra of the formal symplectic disk, let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be its shifted-cotangent semidirect extension, and for an interval $I \subset \mathbb{R}$ let $\mathfrak{h}_I = C^\infty_c(I)\,\widehat{\otimes}\,\mathfrak{h}$ and $\mathfrak{g}_I = \mathfrak{h}_I \ltimes \mathcal{P}_I[1]$ with $\mathcal{P}_I = \mathcal{D}'_c(I)\,\widehat{\otimes}\,\mathcal{P}$ as in Construction 0.2.19. The following six statements hold and together constitute the local Hamiltonian closed-open derived center theorem.

(i) *Universal cochain-level isomorphism.* For any pro-finite-dimensional Lie algebra $\mathfrak{h}$ in **TateVec** over $\mathbb{C}$, concentrated in degree zero, satisfying the Tate-conilpotency hypothesis of Lemma 0.2.57, there is a canonical isomorphism of completed dg graded-commutative P_0-algebras

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big) \simeq Z^{P_0}_{\mathrm{fact}}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big),$$

determined on generators by $\Phi(c^I) = \theta^I$, $\Phi(u_I) = O_I$, where c^I, u_I are the CE coordinates dual to $H_I \in \mathfrak{h}$ and $\eta_{\delta_I} \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ and θ^I, O_I are the corresponding Schouten/Lichnerowicz coordinates. The cochain identity $\Phi \circ d_{\mathrm{CE}} = d_\pi \circ \Phi$ is verified on generators and propagated by graded Leibniz to the completed algebra. (See 0.2.29, 0.2.59, and 0.2.57.)

(ii) *Principal-part coadjoint identification.* The continuous dual $\mathfrak{h}^\vee_{\mathrm{cont}}$ of the reduced Hamiltonian Lie algebra is canonically isomorphic to the principal-part residue module $\mathcal{P} = \bigoplus_{a+b>0} \mathbb{C}\,\rho_{a,b}$ with $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\,dz_1\,dz_2$, by the unique $\mathfrak{h}$-equivariant continuous pairing of total degree zero

$$\langle \rho_{a,b},\, z_1^m z_2^n \rangle = \mathrm{Res}_{z_1=z_2=0}\, z_1^{m-a-1} z_2^{n-b-1}\,dz_1\,dz_2 = \delta_{a,m}\,\delta_{b,n}$$

(modulo a single nonzero scalar, by Schur rigidity). The coadjoint action on $\mathcal{P}$ is the lowering action

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\,\rho_{a-p+1,\,b-q+1},$$

verified on three explicit monomial test cases and a 1225-case sweep (`scripts/check_one_psi_homology.py`). The polynomial $\Psi_{a,b} \in \bar{A}_{\mathrm{desc}}$ basis acts by symmetric raising and represents the PBW special-fibre at $\hbar = 0$; the $\rho_{a,b}$ basis acts by coadjoint lowering and represents the deformation-correct module. The two are not in conflict: they are different bases of the same $\mathfrak{g}$-module, interpolated by the Rees parameter $\hbar$ of the formal Moyal deformation. (See the principal-part block around 0.2.67.)

(iii) *Factorization-algebra upgrade on the brane line.* The Hamiltonian factor admits an interval-wise quasi-isomorphism of locally constant P_0 factorization algebras

$$\Phi^{\mathrm{Ham}}_{\mathrm{fact}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}\big|_{\mathbb{R}\times p} \xrightarrow{\sim} Z^{P_0}_{\mathrm{fact}}(A^{\mathrm{Ham}}_{\partial}),$$

where on each interval $I \subset \mathbb{R}$, $A^{\mathrm{Ham}}_{\partial}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ with the strict P_0 bracket $\{B_f(a), B_g(b)\}_{A^{\mathrm{Ham}}_{\partial}} = B_{\{f,g\}_{\bar{A}}}(ab)$, supplied on length one by

$$(a \otimes f)^{\vee}[-1] \longmapsto B_f(a) := \int_I a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt.$$

The map is compatible with extension by zero, with disjoint-interval factorization products, with the locally constant structure on $\mathbb{R}$, and recovers the stalkwise central operation $f \mapsto L_{B_f}$ at the point limit $I \to \{t_0\}$. The locality input is the canonical equal-time bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j\, \delta(t-s)$, and the strict (non-homotopical) P_0 structure follows from the delta-function locality. (See 0.2.22, 0.2.20, 0.2.24, and 0.2.82.)

(iv) *Cotangent factor at the post-contraction reduced level.* The principal-part / cotangent factor extends the Hamiltonian factorization algebra to the P_0 factorization algebra $A^{\mathrm{pp}}_{\partial}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I) \widehat{\otimes} \widehat{\mathbf{S}}(\mathcal{P}_I[1])$ on the reduced post-de Rham contraction model, with brackets

$$\{B_f(a), \Theta_\rho(B)\}_{A^{\mathrm{pp}}_{\partial}} = \Theta_{f\cdot\rho}(aB), \qquad \{\Theta_\rho(B), \Theta_\sigma(C)\}_{A^{\mathrm{pp}}_{\partial}} = 0,$$

and the bulk-to-boundary morphism $\widetilde{\partial}_{\mathrm{bb}} : \bar{A} \oplus \mathcal{P}[1] \to A^{\mathrm{pp}}_{\partial}$ extends 0.2.81 to the closed-coadjoint sector $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^{\vee}_{\mathrm{cont}}[1]$. In the reduced model the cochain-level isomorphism of part (a) extends to the cotangent factor; in the unreduced factorization-algebra model the cotangent factor is conditional on 0.2.35.

(v) *All-order quantum deformation $\Phi_\hbar$ on the degree-zero Hamiltonian sector.* The classical Φ deforms to all orders in $\hbar$ on the degree-zero Hamiltonian sector, giving a $\mathbb{C}[[\hbar]]$-linear cochain isomorphism

$$\Phi^{(0)}_{\hbar} : \big[C^{\bullet}_{\mathrm{CE},\mathrm{cont},\hbar}(\mathfrak{g}_\hbar)\big]^{(0)}[[\hbar]] \xrightarrow{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^{\hbar}_{\mathrm{open}}\big)^{(0)},$$

where on degree-zero generators $\Phi^{(0)}_{\hbar}(f) = [\,\overline{J_N(f)}\,]_{N\to\infty} = \partial_{\mathrm{bb},\hbar}(f)$ with $J_N(f)$ the Capelli-renormalized stable connected Weyl trace. The cochain identity $\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f,g\}_\hbar)$ holds at all orders. The boundary product is the formal Weyl/Moyal product of $\bar{A}_\hbar$ in the unique unconditional normalization with cross-contraction kernel $P_\partial = P$ at weight $\hbar/2$; the first and third Costello graph normalizations agree with the Moyal coefficients 1 and 1/24 respectively. The associated graded at $\hbar = 0$ recovers the degree-zero restriction of Φ. (See the quantum sector around 0.2.115.)

(vi) *U(1) center anomaly.* The scalar-reduction direction $\mathbb{C}\cdot 1 \subset \mathfrak{h}_{\mathrm{poly}}$ and $\mathbb{C}\cdot \mathrm{Tr}(1) \subset H^0$ carries a single nontrivial cohomology class

$$[\bar{c}] \in H^2_{\mathrm{Lie}}(\bar{A}, \mathbb{C}),$$

represented by the constant Taylor coefficient $\omega(z_1^k z_2^l, z_1^m z_2^n) = (kn - lm)\,\mathbf{1}[(k+m, l+n) = (1,1)]$ of the Poisson bracket on $\mathfrak{h}_{\mathrm{poly}}$. The class $[\bar{c}]$ is realized identically at classical level (the Poisson cocycle ω, obstructing the Lie-algebra splitting of $\mathfrak{h}_{\mathrm{poly}} \to \bar{A}$) and at quantum level (the

Capelli shift $\hbar N$ in $YX - XY + \hbar N\, I = 0$, leading coefficient of the Moyal commutator on linear traces). The equivalence Φ holds modulo $[\bar{c}]$, i.e. on the scalar-reduced quotient; the U(1) anomaly $[\bar{c}]$ is the unique obstruction to lifting Φ to the unreduced sector on the Hamiltonian-constants axis. The healed local bulk-boundary coupling carries an explicit central piece

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\mathrm{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N \cdot \omega(f,g) \int_{\mathbb{R}} a(t)b(t)\, dt,$$

which vanishes after projection to H^0_{red}. (See 0.2.82.)

Statement (a) is the universal cochain-level identification, of which the formal symplectic disk in dimension two is one corollary. Statement (b) supplies the deformation-correct realization of the cotangent direction $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$. Statement (c) upgrades the local fiber to a factorization-algebra equivalence on the brane line in the connected Hamiltonian sector. Statement (d) extends the equivalence to the cotangent factor at the post-contraction reduced level. Statement (e) deforms the equivalence to all orders in $\hbar$ on the degree-zero Hamiltonian sector. Statement (f) names the unique obstruction class along the scalar-reduction axis.

The moment-map shadow at degree zero, $f \mapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)} = O_f$, recovers the existing classical bulk-to-boundary map of 0.2.81.

Proof. Each statement is established by the cited constituent result, with cross-consistency of signs, Tate completions, factorization sectors, anomaly classes, and conditionality classifications recorded throughout the manuscript.

Statement (a) is 0.2.29 applied to $\mathfrak{h} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$, which is pro-finite-dimensional in Taylor filtration, perfect, and Tate-conilpotent (the bracket strictly lowers Taylor degree, so descending Lie powers vanish on each finite Taylor window).

Statement (b) is the existence and Schur uniqueness of the residue pairing combined with the induced $\mathfrak{h}$-module isomorphism with explicit lowering action; the PBW-vs-deformation comparison interpolates the polynomial $\Psi_{a,b}$ raising basis (PBW special fibre) and the principal-part $\rho_{a,b}$ lowering basis (deformation-correct).

Statement (c) is 0.2.22 on the connected Hamiltonian sub-factorization algebra $A^{\mathrm{Ham}}_\partial(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$, with locality of the strict P_0 bracket on disjoint supports (0.2.20) and stalk recovery via 0.2.39.

Statement (d) combines the factorization companion to the local cotangent identification with the principal-part boundary cotangent operators. In the reduced post-contraction model the cochain isomorphism of (a) extends to the cotangent factor by the embedding $\bar{A} \oplus \mathcal{P}[1] \hookrightarrow A^{\mathrm{pp}}_\partial(I)$, with brackets matching $\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ on the smeared sector. The unreduced factorization-algebra extension is the content of 0.2.35.

Statement (e) is the all-order quantum upgrade, with the Capelli renormalization absorbing the $\hbar N$ contact term into the change of basis $T_{a,b} \leftrightarrow J_N(z_1^a z_2^b)$, the radial-parts identity producing the all-orders bracket equation, the unconditional boundary product normalization fixing the Moyal/Weyl coefficient at every order, and the matching with Costello's first and third graph normalizations. The classical limit $\hbar \to 0$ recovers the degree-zero restriction of (a) through the moment-map shadow.

Statement (f) is 0.1.2 (the Poisson cocycle ω is closed and nontrivial), 0.1.3 (closed-side central extension $0 \to \mathbb{C} \cdot 1 \to \mathfrak{h}_{\mathrm{poly}} \to \bar{A} \to 0$ is classified by $[\bar{c}]$), 0.1.4 (open-side U(1) centre of $\mathfrak{gl}_N$ realizes $[\bar{c}]$ via the boundary evaluation), 0.1.6 (Rees specialization identifies the classical Poisson cocycle ω with the quantum Capelli shift $\hbar N$ as the same class), and the healed 0.2.82 which exhibits the central piece explicitly with the constant N.

The six statements interlock without circularity: (a) supplies the cochain identity at universal generality; (b) supplies the deformation-correct cotangent realization (independent of (a) but compatible with it via the dualization Φ_{coad}); (c) upgrades (a) to the brane-line factorization algebra on the Hamiltonian factor (using only the classical local equal-time bracket); (d) extends (c) to the cotangent factor at the reduced level (using (b)); (e) deforms (a)+(c) at the degree-zero Hamiltonian sector (using the Capelli renormalization, independent of the descendant); (f) names the obstruction class along an orthogonal axis (the scalar-reduction line $\mathbb{C} \cdot 1$, independent of the cotangent direction $\mathfrak{h}^\vee[1]$). □

Remark 0.1.23 *(Universal generality of the umbrella).* Statement (a) of 0.1.22 holds for any pro-finite-dimensional Lie algebra $\mathfrak{h}$ in **TateVec** with the conilpotency hypothesis. The formal symplectic disk in dimension two is one corollary; the formal Hamiltonian Lie algebra of any holomorphic symplectic surface reduced by constants is another. Statements (b)–(f) are dimension-two specific (they use the residue pairing on $\mathbb{C}^2$, the matrix Weyl algebra of pairs (X, Y), and the U(1) center of $\mathfrak{gl}_N$), and recover the higher-dimensional analogues by the same universal cochain identity $\Phi \circ d_{\mathrm{CE}} = d_\pi \circ \Phi$ together with the appropriate higher-dimensional residue pairing and matrix Weyl data.

Remark 0.1.24 *(Dirac thesis, as a consequence of the umbrella).* The umbrella theorem makes precise the Dirac thesis sentence that organizes the manuscript:

> *A stack of branes is a system of probes for the formal symplectic normal disk. Dirac reduction of the probe phase space gives the open trace algebra. Hamiltonian BF theory is the shifted cotangent theory of canonical transformations. The closed observables are precisely the derived Poisson central operations on the stable open Hamiltonian brane algebra.*

This is now a corollary of 0.1.22: the brane stack is the open BV theory of N matrix-valued probes (ϕ_1, ϕ_2) for the formal symplectic disk; Dirac first-class reduction along the moment-map ideal $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)$ produces the stable connected open trace algebra $H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$; the closed Hamiltonian BF theory has fields valued in the shifted-cotangent semidirect product $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ with $\mathfrak{h}$ the Hamiltonian Lie algebra of canonical transformations of the formal disk; and the closed observables $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$ are identified, via Φ, with the derived Poisson factorization-center $Z^{P_0}_{\mathrm{fact}}(A^{\mathrm{Ham}}_\partial)$ on the brane line at the reduced post-contraction classical level (and on the degree-zero Hamiltonian sector at all orders in $\hbar$). The slogan is no longer a slogan; it is the umbrella theorem read in physical language.

The proof uses the formal local model alone. The single residual problem on the unreduced Tate-coefficient cotangent lift is recorded as 0.2.35 below. The quantum calculation later fixes the formal Weyl/Moyal target; the analytic problem is to identify Costello's renormalized BV graph package with this Hamiltonian BF sector and to recover the first and third graph normalizations.

0.2 The Local Model

0.2.1 Setup

Let

$$M = \mathbb{R}^2, \qquad Y = \mathbb{C}^2, \qquad X = M \times Y.$$

The real factor M is the topological direction and the complex factor Y is the holomorphic direction; all Hamiltonian calculations are completed at the formal neighbourhood of $p = 0 \in Y$. We use the exact A-brane given by the standard line $\mathbb{R} \subset M$ with trivial brane data and the B-brane $\mathcal{O}_p \in \mathrm{Perf}(Y)$. Consider the mixed brane

$$\mathbb{R} \times p \rightarrowtail \mathbb{R}^2 \times \mathbb{C}^2$$

for some point $p \in Y$. We take $p = 0$ in coordinates. The holomorphic factor carries the standard holomorphic symplectic form

$$\omega = dz_1 \wedge dz_2.$$

Thus the model is topological along $\mathbb{R}^2$ and holomorphic Hamiltonian along $\mathbb{C}^2$.

Proposition 0.2.1 *(Local mixed model).* The formal local Hamiltonian data used here define a closed mixed holomorphic-topological BV sector and a compatible cyclic open brane BV theory. The constructed closed-open operation is the degree-zero Hamiltonian boundary evaluation. The full Poisson factorization-center morphism is Problem 0.2.35, not part of this proposition.

Proof. On product opens $D \times B$ in the completed formal neighbourhood of $\mathbb{R}^2 \times \{p\} \subset \mathbb{R}^2 \times \mathbb{C}^2$, the closed fields are

$$\mathcal{E}_{\mathrm{cl}}(D \times B) = \Omega^*(D)\widehat{\otimes}\Omega^{0,*}(B)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big).$$

Degreewise polynomial calculations use the dense subspace $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The differential is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$, so $Q_0 = 0$ in Definition 0.1.16. The BV pairing is the local pairing on compactly supported fields, obtained by multiplying the form components, integrating over $D \times B$ against $dz_1\, dz_2$, and using the continuous evaluation pairing $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}} \otimes \widehat{\mathfrak{h}} \to \mathbb{C}$. If $\alpha \in \Omega^p\widehat{\otimes}\widehat{\mathfrak{h}}[1]$ and $\beta \in \Omega^q\widehat{\otimes}\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$, then $|\alpha| = p - 1$ and $|\beta| = q - 2$; the top-form condition $p + q = 4$ makes the pairing degree -1.

The free quadratic term

$$S_{\mathrm{free}} = \int_{D\times B} \big\langle \beta, D\alpha \big\rangle dz_1\, dz_2$$

is the BV Hamiltonian for the differential D. The interaction in Definition 0.1.16 is only

$$I_{\mathrm{BF}} = \frac{1}{2}\int_{D\times B} \big\langle \beta, \{\alpha, \alpha\} \big\rangle dz_1\, dz_2.$$

The total BF action is $S_{\mathrm{free}} + I_{\mathrm{BF}}$. The Poisson bracket on $\bar{A}$ is the holomorphic bracket determined by $dz_1 \wedge dz_2$. Since D is a derivation of this bracket and the bracket satisfies Jacobi,

$$DI_{\mathrm{BF}} + \frac{1}{2}\{I_{\mathrm{BF}}, I_{\mathrm{BF}}\} = 0,$$

so the total BF action satisfies the classical master equation without double-counting the free differential.

Mixed locality is the Cartan homotopy calculation on the interacting observables. For a constant vector field v in the $\mathbb{R}^2$ direction, $[D, \iota_v] = \mathcal{L}_v$; for an anti-holomorphic coordinate vector field $\partial/\partial\bar{z}_i$, $[D, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$. The interaction is built from the de Rham wedge product and holomorphic polydifferential operators, so the contractions are derivations of the local observable algebra and commute with $\{I_{\mathrm{BF}}, -\}$ up to the displayed Lie derivative. Thus $[Q_{\mathrm{obs}}, \iota_v] = \mathcal{L}_v$ and $[Q_{\mathrm{obs}}, \iota_{\partial/\partial\bar{z}_i}] = \mathcal{L}_{\partial/\partial\bar{z}_i}$ on local observables. The de Rham Poincare lemma in the D directions gives the quasi-isomorphism for small disk inclusions, and the Dolbeault Poincare lemma gives the holomorphic dependence in the B direction.

The brane is $N \times Z = \mathbb{R} \times \{p\}$, with Z understood as the formal holomorphic neighbourhood at p. Its open fields are

$$\Omega^*(\mathbb{R}) \otimes \mathbf{Ext}^*_{\mathbb{C}[z_1,z_2]}(\mathcal{O}_p, \mathcal{O}_p) \otimes \mathfrak{gl}_N[1],$$

and Proposition 0.2.2 identifies the Ext algebra with $\Lambda(\varepsilon_1, \varepsilon_2)$. This is the derived $\mathrm{RHom}_Y(\mathcal{O}_p, \mathcal{O}_p)$ allowed in Definition 0.1.16, not an ordinary Dolbeault algebra on the point. The trace on $\mathfrak{gl}_N$ and

the Berezin trace on this exterior algebra give the cyclic open string field theory. Construction 0.2.43 and Theorem 0.2.81 construct the degree-zero Hamiltonian boundary component and prove its bracket compatibility. The full Poisson factorization-center closed-open map is Problem 0.2.35, not a theorem proved here. □

0.2.2 *Open mixed brane states*

The open string states are the tensor product of the topological brane direction and the formal holomorphic normal direction.

- *Topological factor.* In the exact self-Floer model for the graded, spin zero-section $\mathbb{R} \subset T^*\mathbb{R} \cong \mathbb{R}^2$, we use the local exact Fukaya model near a compact interval in the brane. A small compactly supported Hamiltonian perturbation identifies $CF^*(\mathbb{R}, \mathbb{R})$ with the Morse complex on that interval, and passage to local operators in the interior is represented by the de Rham complex $\Omega^*(\mathbb{R})$ with compactly supported variations. No wrapped generators at infinity are included in this local model.
- *Holomorphic factor.* Put affine coordinates at $p = 0$ and write $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $\mathcal{O}_p = R/\mathfrak{m}$. The skyscraper sheaf has open string states

$$\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m}),$$

  computed in Proposition 0.2.2.

PROPOSITION 0.2.2 *(Affine skyscraper self-Ext).* Let $R = \mathbb{C}[z_1, z_2]$, $\mathfrak{m} = (z_1, z_2)$, and $k = R/\mathfrak{m}$. Then the Yoneda algebra of the skyscraper sheaf at the origin is

$$\mathbf{Ext}^*_R(k, k) \cong \Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2),$$

where $|\varepsilon_i| = 1$. In particular this is the free graded-commutative algebra on two odd generators.

Proof. The pair (z_1, z_2) is a regular sequence in the polynomial ring R, so the affine Koszul complex

$$K = \big(0 \to R\, e_{12} \xrightarrow{d_2} R\, e_1 \oplus R\, e_2 \xrightarrow{d_1} R \to 0\big)$$

is a free resolution of k, with

$$d_1(ae_1 + be_2) = az_1 + bz_2, \qquad d_2(e_{12}) = z_2 e_1 - z_1 e_2.$$

Applying $\mathrm{Hom}_R(-, k)$ gives

$$0 \to k \xrightarrow{0} k\,\varepsilon_1 \oplus k\,\varepsilon_2 \xrightarrow{0} k\,\varepsilon_{12} \to 0,$$

because z_1 and z_2 act by zero on k. Hence the graded vector space of Ext groups is k in degree 0, $k\varepsilon_1 \oplus k\varepsilon_2$ in degree 1, and $k\varepsilon_{12}$ in degree 2.

The product is the Yoneda product. The Koszul resolution is the exterior dg-algebra $R \otimes \Lambda(e_1, e_2)$ with $d(e_i) = z_i$; its diagonal sends e_i to $e_i \otimes 1 + 1 \otimes e_i$. Dualising this diagonal gives $\varepsilon_i^2 = 0$ and $\varepsilon_1\varepsilon_2 = -\varepsilon_2\varepsilon_1$, while $\varepsilon_1\varepsilon_2$ spans the top Ext group. Thus the Yoneda algebra is $\Lambda_{\mathbb{C}}(\varepsilon_1, \varepsilon_2)$. □

0.2.3 *Open mixed brane field theory*

The open theory is the minimal local field theory which remembers the brane's two normal holomorphic directions. The two Ext generators are odd because they are derived normal directions of a point brane; after the BV shift their ghost-zero coefficients become the two ordinary matrix coordinates ϕ_1, ϕ_2. The one-form coefficient is the gauge field A along the brane. Thus the formal Chern-Simons action is the mechanism which turns the Ext algebra into first-order matrix mechanics.

The total space of fields (in the Batalin-Vilkovisky formalism) is

$$\alpha \in \Omega^*(\mathbb{R}) \otimes \Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N[1]$$

with action

$$S(\alpha) = \int_{\mathbb{R}\times\mathbb{C}^{0|2}} \frac{1}{2}\operatorname{Tr}(\alpha\, d\alpha) + \frac{1}{3}\operatorname{Tr}\alpha^3.$$

Here $d = d_{\mathrm{dR}}$; the Ext algebra $\Lambda(\varepsilon_1, \varepsilon_2)$ has zero differential. The integral over $\mathbb{C}^{0|2}$ is the Berezin integral $\int d\varepsilon_1\, d\varepsilon_2$, which picks out the top component in the odd directions.

The cyclic trace is

$$\operatorname{Tr}_{\mathrm{open}}(\eta \otimes u \otimes M) = \int_{\mathbb{R}} \eta\, [\varepsilon_1\varepsilon_2]u\ \operatorname{Tr}_N(M),$$

where $[\varepsilon_1\varepsilon_2]u$ denotes the coefficient of the top Ext class. The top Ext class is the residue orientation dual to $dz_1 \wedge dz_2$. The de Rham integral has degree -1 and the Ext residue has degree -2; after the field shift by $[1]$, the induced BV pairing has degree -1.

Construction 0.2.3 *(The open field from the Ext basis).* Write $\varepsilon_{12} = \varepsilon_1\varepsilon_2$. Before imposing ghost number, an open field is a finite sum

$$\alpha = a_0 + a_1\varepsilon_1 + a_2\varepsilon_2 + a_{12}\varepsilon_{12},$$

where each a_I is a de Rham form on $\mathbb{R}$ with values in $\mathfrak{gl}_N$, shifted by $[1]$. The ghost-number-zero components are exactly

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N, \qquad \phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and they appear as

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2.$$

The Berezin integral kills every monomial except the coefficient of ε_{12}, so every term in the ghost-zero action is obtained by multiplying the displayed three summands and keeping exactly one real one-form and exactly the product $\varepsilon_1\varepsilon_2$.

Remark 0.2.4. The shift $[1]$ acts on the total cohomological degree: a form of de Rham degree p and Grassmann degree q has total degree $p + q$, which after the shift sits in ghost number $p + q - 1$. In particular, the 1-form component of α (de Rham degree 1, Grassmann degree 0) sits in ghost number 0 and is the bosonic gauge field. The scalar components proportional to ε_i (de Rham degree 0, Grassmann degree 1) also sit in ghost number 0 and represent the motion of the brane in the two normal directions.

0.2.3.1 Ghost-zero open field content

The ghost-number-zero part of the BV field is

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2,$$

where $A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N$ and $\phi_i \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$.

LEMMA 0.2.5. With the Berezin convention $\int_{\mathbb{C}^{0|2}} \varepsilon_1\varepsilon_2 = 1$, the restriction of the BV action to α_0 is, up to a boundary term on $\mathbb{R}$,

$$S(\alpha_0) = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Proof. Only monomials containing one real one-form and the factor $\varepsilon_1\varepsilon_2$ survive the integration over $\mathbb{R} \times \mathbb{C}^{0|2}$. In the quadratic term this leaves

$$\frac{1}{2}\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 - \phi_2\, d\phi_1\big),$$

where the sign is the Koszul sign from interchanging the two odd generators. Since $d\,\mathrm{Tr}(\phi_1\phi_2) = \mathrm{Tr}(d\phi_1\,\phi_2 + \phi_1\, d\phi_2)$, the second summand differs from $\mathrm{Tr}(\phi_1\, d\phi_2)$ by an exact one-form. In the cubic term the six orderings of A, $\phi_1\varepsilon_1$, $\phi_2\varepsilon_2$ reduce by cyclicity of the trace and the odd relation $\varepsilon_2\varepsilon_1 = -\varepsilon_1\varepsilon_2$ to

$$3\,\mathrm{Tr}(A\phi_1\phi_2 - A\phi_2\phi_1).$$

The coefficient $\frac{1}{3}$ in the Chern-Simons action therefore gives exactly $\mathrm{Tr}(A[\phi_1, \phi_2])$. □

LEMMA 0.2.6 *(Dirac reduction of the probe phase space).* Let

$$\mathcal{P}_N = \mathfrak{gl}_N \oplus \mathfrak{gl}_N$$

with coordinates (ϕ_1, ϕ_2) and symplectic form

$$\Omega_N = \mathrm{Tr}(d\phi_1 \wedge d\phi_2).$$

With the convention $\iota_{X_f}\Omega_N = -df$, simultaneous conjugation by $\mathfrak{gl}_N$ is Hamiltonian with moment map components

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

They satisfy the first-class identity

$$\{\mu_\epsilon, \mu_\eta\} = \mu_{[\epsilon,\eta]}.$$

The action of Lemma 0.2.5 is the corresponding Dirac first-order action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A,$$

and varying A imposes the constrained phase space

$$\mu^{-1}(0)/GL_N = \{[\phi_1, \phi_2] = 0\}/GL_N.$$

Proof. The symplectic form gives

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l\delta^k_j.$$

For $v_\epsilon(\phi_i) = [\epsilon, \phi_i]$, cyclicity of the trace gives

$$d\mu_\epsilon(\delta\phi_1, \delta\phi_2) = -\mathrm{Tr}\big(\epsilon([\delta\phi_1, \phi_2] + [\phi_1, \delta\phi_2])\big) = -\Omega_N(v_\epsilon, (\delta\phi_1, \delta\phi_2)).$$

Thus $X_{\mu_\epsilon} = v_\epsilon$. The moment-map bracket is therefore the infinitesimal equivariance identity $\{\mu_\epsilon, \mu_\eta\} = \mu_{[\epsilon,\eta]}$. Substituting $\epsilon = A(t)$ gives

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2) - \mu_A = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big),$$

and the A-equation is $[\phi_1, \phi_2] = 0$. □

Proposition 0.2.7 *(Classical BV truncation).* The subspace of ghost-number-zero fields

$$\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$$

with all nonzero ghost-number components set to zero is a consistent classical truncation of the open BV theory. The induced classical action is the first-order gauged matrix model of Lemma 0.2.5.

Proof. Decompose the BV field as $\alpha = \sum_r \alpha^{(r)}$ by ghost number. The differential $d_{\mathbb{R}}$ and the product on $\Lambda(\varepsilon_1, \varepsilon_2) \otimes \mathfrak{gl}_N$ preserve total degree, and the BV action has ghost number zero. Hence every monomial in the Euler-Lagrange expression for a component of ghost number $r \neq 0$ contains at least one field component of nonzero ghost number when evaluated at $\alpha = \alpha_0$. Setting all such components to zero therefore makes the discarded equations vanish. The remaining equation is exactly the variation of $S(\alpha_0)$, computed in Lemma 0.2.5.

Concretely, the full Euler-Lagrange expression is the curvature

$$F(\alpha) = d\alpha + \alpha^2.$$

For $\alpha_0 = A + \phi_1\varepsilon_1 + \phi_2\varepsilon_2$ on a line,

$$F(\alpha_0) = (d\phi_1 + [A, \phi_1])\varepsilon_1 + (d\phi_2 + [A, \phi_2])\varepsilon_2 + [\phi_1, \phi_2]\varepsilon_1\varepsilon_2.$$

The BV pairing keeps exactly real-form degree one and $\varepsilon_1\varepsilon_2$. Hence the displayed three components pair exactly with the variations of A, ϕ_2, ϕ_1 respectively, while variations of nonzero ghost number see complementary curvature components that are zero for α_0 by real-form degree or ε-degree. This is a physical ghost-zero truncation; the residual ordinary GL_N gauge symmetry remains. □

The ghost-number-zero truncation of the BV master action is closed under the classical equations and gives a quantum-mechanical system with a connection form

$$A \in \Omega^1(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

two adjoint scalars

$$\phi_1, \phi_2 \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N,$$

and action

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

This action is invariant under

$$\delta_\varepsilon \phi_i = [\varepsilon, \phi_i], \qquad \delta_\varepsilon A = d\varepsilon + [A, \varepsilon],$$

for $\varepsilon \in \Omega^0(\mathbb{R}) \otimes \mathfrak{gl}_N$. Indeed the variation of $\mathrm{Tr}(\phi_1 d\phi_2)$ containing $d\varepsilon$ is $-\mathrm{Tr}(d\varepsilon[\phi_1, \phi_2])$, and this cancels the $d\varepsilon$ variation of $\mathrm{Tr}(A[\phi_1, \phi_2])$; the remaining terms cancel by invariance of the trace pairing.

Remark 0.2.8. The same local open model is obtained from the Calabi-Yau 2 point-brane residue calculation by tensoring the skyscraper object with $\mathbb{C}^N$ and passing to endomorphisms. This is a local CY_2 pairing statement, not a global CY-to-chiral input. Global transfer would require the corresponding negative cyclic Calabi-Yau class, framing data, and descent; none of that is used in the proof below.

Remark 0.2.9. In AKSZ language this is the one-dimensional gauged sigma model with Hamiltonian target $\mathfrak{gl}_N \oplus \mathfrak{gl}_N$ and symplectic potential $\mathrm{Tr}(\phi_1\, d\phi_2)$. We use the resulting Poisson bivector

$$\Pi = \sum_{i,j} \frac{\partial}{\partial(\phi_1)^i{}_j} \wedge \frac{\partial}{\partial(\phi_2)^j{}_i},$$

so $\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$. Equivalently, with the convention $\iota_{X_f}\omega = -df$ used throughout, the symplectic form is $\mathrm{Tr}(d\phi_1 \wedge d\phi_2)$. For $\epsilon \in \mathfrak{gl}_N$, the Hamiltonian

$$\mu_\epsilon = -\mathrm{Tr}\big(\epsilon[\phi_1, \phi_2]\big)$$

generates $\delta_\epsilon \phi_i = [\epsilon, \phi_i]$. Its zero locus is the commuting-pair equation $[\phi_1, \phi_2] = 0$, and the classical reduced phase space is the stack of commuting pairs of matrices modulo conjugation.

Remark 0.2.10. Equivalently, take Chern-Simons theory on $\mathbb{R} \times T^2$ and restrict to the harmonic forms $1, \eta_1, \eta_2, \eta_1\eta_2$ on T^2. Writing $A_{3d} = A_t\, dt + \phi_1\eta_1 + \phi_2\eta_2$ and integrating over T^2 gives the same action; the odd variables ε_i represent the degree-one cohomology classes η_i.

0.2.4 Closed mixed Hamiltonian sector

The closed field is the infinitesimal geometry of the formal normal surface. Because the surface is holomorphic symplectic, an infinitesimal canonical transformation is a Hamiltonian function modulo constants. The BF partner is the cotangent variable which makes the theory local and BV. This is the closed counterpart of the open first-order form: both sides are governed by the same symplectic normal plane.

The closed sector used here is a mixed field theory, not a tensor product of two independent closed-string state spaces. Its fields have a topological de Rham factor along $\mathbb{R}^2$ and a holomorphic Dolbeault factor along $\mathbb{C}^2$:

$$\Omega^*(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,*}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big).$$

The coefficient $\widehat{\mathfrak{h}}$ is the completed Lie algebra of nonconstant Hamiltonian functions on the formal holomorphic symplectic disk. The continuous dual copy $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$ is the cotangent/BF field. Thus the closed theory is the Hamiltonian Kodaira-Spencer/BF sector of the mixed holomorphic-topological string.

Remark 0.2.11. The original BCOV analysis [1] is written for Calabi-Yau threefolds; the higher-genus B-model on Calabi–Yau manifolds is taken up at the quantum level in [24]. On a holomorphic symplectic surface the natural local cotangent theory is the BF theory for the dg Lie algebra of Hamiltonian vector fields. The holomorphic symplectic form supplies the Poisson bracket, while the holomorphic volume form supplies the divergence operator on polyvectors.

We consider the space of polyvector fields

$$\mathbf{PV}^{*,*}(\mathbb{C}^2) \cong \Omega^{0,*}(\mathbb{C}^2)[\partial_{z_1}, \partial_{z_2}]$$

where ∂_{z_i} are odd generators of poly-vector degree 1 (corresponding to the two holomorphic vector fields on $\mathbb{C}^2$). The operator

$$\partial : \mathbf{PV}^{i,*} \to \mathbf{PV}^{i-1,*}$$

is the divergence operator determined by the holomorphic volume form $\Omega = dz_1 \wedge dz_2$:

$$\partial_\Omega = \iota_\Omega^{-1} \circ \partial_{\mathrm{hol}} \circ \iota_\Omega, \qquad \iota_\Omega : \mathbf{PV}^{i,*} \xrightarrow{\sim} \Omega^{2-i,*}.$$

Thus for $X = a_1\partial_{z_1} + a_2\partial_{z_2}$,

$$\partial_\Omega X = (\partial_{z_1} a_1 + \partial_{z_2} a_2),$$

while ∂_Ω vanishes on functions.

We work in the formal polynomial local model at $0 \in \mathbb{C}^2$, so Dolbeault cohomology is represented by holomorphic Taylor polynomials. Here ∂_Ω is the divergence constraint selecting the Hamiltonian BF sector. The kinematic differential of the closed theory below is $D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$. Thus the Poincare lemma identifies divergence-free vector fields with Hamiltonian potentials. The quotient by ∂_Ω-exact Hamiltonians is absent. Analysing degree by degree:

- $\mathbf{PV}^{0,*}(\mathbb{C}^2) = \Omega^{0,*}(\mathbb{C}^2) \subset \mathbf{Ker}\partial$ trivially.
- $\mathbf{PV}^{1,*}(\mathbb{C}^2) \cong \Omega^{1,*}(\mathbb{C}^2)$ via $\iota_{(-)}\omega$. Under this identification, with the Hamiltonian convention $\iota_{X_f}\omega = -df$ used below, $\mathbf{Ker}\partial_\Omega$ is the space of closed holomorphic one-forms. Lemma 0.2.12 identifies these with df, uniquely modulo constants.
- The constant Hamiltonians act trivially, since their Hamiltonian vector fields vanish. We therefore use the reduced polynomial algebra

  $$\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$$

  for nontrivial Hamiltonian fields.

Lemma 0.2.12 *(Polynomial Poincare primitive).* Let

$$\theta = a(z_1, z_2)\, dz_1 + b(z_1, z_2)\, dz_2$$

be a polynomial one-form on $\mathbb{C}^2$. If $\partial_{z_2} a = \partial_{z_1} b$, then there is a polynomial $f \in \mathbb{C}[z_1, z_2]$, unique modulo constants, such that $\theta = df$.

Proof. Define

$$f(z_1, z_2) = \int_0^1 \big(z_1 a(tz_1, tz_2) + z_2 b(tz_1, tz_2)\big)\, dt.$$

Since a and b are polynomials, the integral is termwise division of each homogeneous term by its degree plus one, hence f is polynomial. Differentiate under the integral sign. Closedness says $\partial_{z_2} a = \partial_{z_1} b$, so

$$\begin{aligned}
\partial_{z_1} f &= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_1} b(tz)\big)\, dt \\
&= \int_0^1 \big(a(tz) + tz_1 \partial_{z_1} a(tz) + tz_2 \partial_{z_2} a(tz)\big)\, dt \\
&= \int_0^1 \frac{d}{dt}\big(t\, a(tz)\big)\, dt = a(z).
\end{aligned}$$

The same calculation, with a and b interchanged, gives $\partial_{z_2} f = b(z)$. Therefore $df = \theta$. If $df = dg$, then $d(f - g) = 0$, hence $f - g$ is constant. □

Proposition 0.2.13 *(Hamiltonian polyvector reduction).* In the formal polynomial model on $(\mathbb{C}^2, \omega = dz_1 \wedge dz_2)$, the divergence-closed polyvector-one fields reduce to Hamiltonian vector fields:

$$\mathbf{Ker}(\partial_\Omega : \mathbf{PV}^1 \to \mathbf{PV}^0) \cong \{df : f \in \mathbb{C}[z_1, z_2]\} \cong \bar{A}.$$

Under the convention

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df,$$

the commutator of vector fields is

$$[X_f, X_g] = X_{\{f,g\}_{\bar{A}}}.$$

Proof. Let $X = a_1\partial_{z_1} + a_2\partial_{z_2}$. The divergence operator is $\partial_\Omega X = \partial_{z_1} a_1 + \partial_{z_2} a_2$. Contracting with ω gives

$$\iota_X \omega = a_1\, dz_2 - a_2\, dz_1.$$

The equation $\partial_\Omega X = 0$ is exactly the closedness of this one-form:

$$d(\iota_X\omega) = (\partial_{z_1} a_1 + \partial_{z_2} a_2)\, dz_1 \wedge dz_2 = 0.$$

By Lemma 0.2.12, there is a polynomial f, unique modulo constants, such that $\iota_X\omega = -df$. Therefore

$$a_1 = -\partial_{z_2} f, \qquad a_2 = \partial_{z_1} f,$$

so $X = X_f$. Constants are exactly the kernel of $f \mapsto X_f$, giving $\bar{A}$.

Finally, for any polynomial h,

$$\begin{aligned}[X_f, X_g](h) &= \{f, \{g, h\}\} - \{g, \{f, h\}\} \\ &= \{\{f, g\}, h\}\end{aligned}$$

by the Jacobi identity for the constant Poisson bracket. Thus $[X_f, X_g] = X_{\{f,g\}}$, and projecting constants gives the bracket on $\bar{A}$. □

For the Hamiltonian sector of the closed mixed field theory we replace $\mathbf{Ker}\partial$ (on poly-1-vectors) by the Dolbeault-valued Hamiltonian vector fields:

$$\Omega^{0,*}(\mathbb{C}^2)\text{-valued Hamiltonian vector fields,}$$

i.e. sections of $\Omega^{0,*}(\mathbb{C}^2)$ identified with Hamiltonian vector fields via the symplectic form $\omega = dz_1 \wedge dz_2$. Concretely, a section $f \in \Omega^{0,*}(\mathbb{C}^2)$ corresponds to the Hamiltonian vector field $X_f = \{f, -\}_\omega$, where

$$\{f, g\}_\omega = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

With this convention

$$X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}, \qquad \iota_{X_f}\omega = -df.$$

Remark 0.2.14. The replacement is the elementary statement of Proposition 0.2.13. Constants are removed because $X_f = 0$ exactly when f is constant.

0.2.5 *Hamiltonian BF closed sector*

0.2.5.1 The Action Functional

Let

$$D = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

Write

$$\alpha \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}[1]$$

for the Hamiltonian connection field and

$$\beta \in \Omega^*(\mathbb{R}^2) \hat{\otimes} \Omega^{0,*}(\mathbb{C}^2) \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]$$

for its cotangent field. The bracket on form-valued Hamiltonians is obtained by wedging the form parts and taking the Poisson bracket of the Hamiltonian parts with the tensor-product Koszul sign:

$$[\eta \otimes f, \theta \otimes g] = (-1)^{|f||\theta|}(\eta \wedge \theta) \otimes \{f, g\},$$

for homogeneous form η, θ and homogeneous Hamiltonian coefficients f, g. In the ghost-zero Hamiltonian calculations used below the coefficients are even, so this sign is $+1$. The action is

$$S = \int_{\mathbb{R}^2 \times \mathbb{C}^2} \left\langle \beta,\ D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle dz_1\, dz_2.$$

The pairing is the continuous evaluation pairing between $\widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$ and $\widehat{\mathfrak{h}}$. The dense polynomial submodel replaces these by $\bar{A}^\vee$ and $\bar{A}$ in each finite degree. For ghost-number-zero fields, α has total real/Dolbeault form degree 1 and β has total degree 2. Hence $\beta \wedge D\alpha$ and $\beta \wedge \{\alpha, \alpha\}$ have total real/Dolbeault degree 4; multiplying by $dz_1 \wedge dz_2$ gives a top real degree 6 form on $\mathbb{R}^2 \times \mathbb{C}^2$.

Remark 0.2.15. The open mixed brane system lives on the brane line. The closed Hamiltonian BF theory lives on the full mixed space-time $\mathbb{R}^2 \times \mathbb{C}^2$.

0.2.5.2 Classical field content

This theory is closely related to a 4-dimensional BF theory: β plays the role of the B-field (Lagrange multiplier) and the equations of motion of β enforce $F_\alpha = 0$, where

$$F_\alpha = D\alpha + \tfrac{1}{2}\{\alpha, \alpha\}$$

is the "curvature" of α (viewed as a connection on a bundle of holomorphic-symplectic surfaces over $\mathbb{R}^2$).

Consider the space-time $\mathbb{R}^2_{x_i} \times \mathbb{C}^2_{z_i}$. The ghost-number-zero components of α and β are:

$$\alpha = \alpha_{x_i}\, dx_i + \alpha_{\bar{z}_j}\, d\bar{z}_j, \qquad \beta = \beta_{x_1 x_2}\, dx_1\, dx_2 + \beta_{x_i \bar{z}_j}\, dx_i\, d\bar{z}_j + \beta_{\bar{z}_1 \bar{z}_2}\, d\bar{z}_1\, d\bar{z}_2.$$

Here the coefficients of α are Hamiltonian functions modulo constants. The gauge symmetries are the BF-type transformations

$$\delta_\varepsilon \alpha = D\varepsilon + \{\alpha, \varepsilon\}, \qquad \delta_\varepsilon \beta = \mathrm{ad}^\vee_\varepsilon \beta,$$

and

$$\delta_\lambda \alpha = 0, \qquad \delta_\lambda \beta = D_\alpha \lambda,$$

where $D_\alpha = D + \{\alpha, -\}$ and λ is a one-form valued in $\bar{A}^\vee$. The second symmetry is reducible on shell: if $\lambda = D_\alpha \mu$, then $D_\alpha \lambda = D_\alpha^2 \mu = \{F_\alpha, \mu\}$, which vanishes when $F_\alpha = 0$.

Analysing α as a Hamiltonian function on $\mathbb{C}^2$: the component $\alpha_{\bar{z}_j} d\bar{z}_j$ is the Hamiltonian potential whose vector field $X_{\alpha_{\bar{z}_j}} d\bar{z}_j$ is the Beltrami differential encoding a deformation of the complex structure of $\mathbb{C}^2$; the component $\alpha_{x_i} dx_i$ similarly defines a connection on $\mathbb{R}^2$ valued in Hamiltonian vector fields on $\mathbb{C}^2$, encoding a bundle of $\mathbb{C}^2$s over $\mathbb{R}^2$. Together, the equation of motion $F_\alpha = 0$ says this data constitutes an integrable deformation of complex structure with flat holonomy. The field β appears linearly in the action and enforces exactly this equation.

Remark 0.2.16. This is entirely analogous to BF theory, where B is a Lagrange multiplier enforcing $F_A = 0$, and the phase space is $T^* \mathcal{M}_{\text{flat}}$.

0.2.5.3 PHASE SPACE

As an optional analytic model for propagators, formally compactify one real direction and put the perturbative local model on $\mathbb{R} \times S^1 \times Y_{\text{hol}}$, where Y_{hol} is holomorphic symplectic and the calculation is completed at the trivial solution. This is a compactified avatar of the formal local model, not a change of notation for the product $X = M \times Y$ used above. The equations of motion $F_\alpha = 0$ cut out the Maurer-Cartan moduli problem for Hamiltonian vector fields on $S^1 \times Y_{\text{hol}}$. Fix the analytic package of Problem 0.2.103 in this compact or formal setting: elliptic complex, brane-defect or boundary condition, gauge fixing, propagator, renormalization scheme, and anomaly cancellation. The perturbative phase space is the derived cotangent stack

$$T^*_{\text{der}} \mathrm{MC}\big(\Omega^*(S^1) \widehat{\otimes} \Omega^{0,*}(Y_{\text{hol}}, \mathrm{Ham}(Y_{\text{hol}})), D, \{-,-\}\big).$$

After passing to the underived tangent space at the trivial solution this reduces to the ordinary cotangent bundle of the linearized deformation complex.

0.2.6 Closed-open brane coupling

This is the point at which the physical picture becomes a formula. A closed Hamiltonian is not observed by the brane as an abstract vector field; it is observed by evaluating it on the brane's matrix-valued normal coordinates and taking the trace over Chan-Paton indices.

The local classical bulk-boundary operation is the Hamiltonian shadow of the desired closed-open morphism: evaluate a closed Hamiltonian on the brane normal fields. If the brane lies along x_1 and the closed field is $dx_1\, z_1^k z_2^l$, the local boundary functional is

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big) + \int_{\mathbb{R}} dx_1\, \mathrm{Tr}\, \phi_1^k \phi_2^l.$$

This is a statement inside the formal local algebraic model: the closed Hamiltonian is pulled back along the matrix-valued normal coordinate map $z_i \mapsto \phi_i$ and then traced. Construction 0.2.43 is the algebraic form of this bulk-to-boundary prescription. Theorem 0.2.81 below proves that the resulting Hamiltonian evaluation preserves the classical bracket. This is the degree-zero Hamiltonian component expected from a full closed-open map; the full Poisson factorization-center morphism is not constructed here.

Construction 0.2.17 *(Reduced line-defect current kernel).* Let

$$X = \mathbb{R}^2_{\text{top}} \times \mathbb{C}^2_{\text{hol}}, \qquad L = \mathbb{R}_{x_1} \times \{x_2 = 0\} \times \{p\} \subset X,$$

and let $i : L \hookrightarrow X$ be the inclusion. This is a line defect, not a spacetime boundary. In the reduced formal local model, a test density $a(t)\,dt$ on L defines the defect-supported current $i_*(a(t)\,dt)$, locally written as $a(x_1)\delta(x_2)\delta_p\,dx_1dx_2$ together with the formal holomorphic delta-current at p.

The degree-zero Hamiltonian part of the reduced PBW/Rees defect-current vertex is

$$K_L^{\mathrm{cl}}(a \otimes f) = \int_L a(t)\,\overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\,dt, \qquad f \in \bar{A}.$$

For the cotangent principal-part source, the reduced vertex sends

$$B \otimes \rho[1] \longmapsto \Theta_\rho(B), \qquad B \in \mathcal{D}'_c(L), \quad \rho \in \mathcal{P},$$

using Theorem 0.2.77. These formulas define a reduced PBW/Rees defect-current kernel. They are not a cochain-level map from closed CE observables, and they are not yet the quantum Costello brane-defect kernel: the Ward identity, anomaly cancellation, finite-scale propagator, and quantum moment-map compatibility remain separate obligations.

Remark 0.2.18. Geometrically, specifying a closed Hamiltonian field on $\mathbb{C}^2$ amounts to equipping $\mathbb{C}^2$ with a connection, so that the brane can parallel-transport its internal degrees of freedom as it moves along $\mathbb{R}$.

Construction 0.2.19 *(Interval Hamiltonian and factorization algebras).* For an interval $I \subset \mathbb{R}$, define

$$\begin{aligned} \mathfrak{h}_I &= C_c^\infty(I)\widehat{\otimes}\bar{A}, \\ \mathcal{P}_I &= \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}, \\ \mathfrak{g}_I &= \mathfrak{h}_I \ltimes \mathcal{P}_I[1], \end{aligned}$$

with brackets

$$\begin{aligned} [a \otimes f, b \otimes g]_{\mathfrak{h}_I} &= ab \otimes \{f, g\}_{\bar{A}}, \\ [a \otimes f, B \otimes \rho[1]]_{\mathfrak{g}_I} &= aB \otimes (f \cdot \rho)[1], \\ [B \otimes \rho[1], C \otimes \sigma[1]]_{\mathfrak{g}_I} &= 0, \end{aligned}$$

where aB is multiplication of a compactly supported distribution by a smooth compactly supported function and $f \cdot \rho = \pi_{\mathrm{pp}}\{f, \rho\}$ is the principal-part coadjoint action of 0.2.67. Inclusions $I' \subset I$ act by extension by zero on test functions and distributions; disjoint unions act by sum. This makes $I \mapsto \mathfrak{h}_I, \mathfrak{g}_I$ cosheaves of (graded) Lie algebras on $\mathbb{R}$.

Define the boundary P_0 factorization algebras on $\mathbb{R}$ in the formal stable Hamiltonian sector by

$$\begin{aligned} A_\partial^{\mathrm{Ham}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I), \\ A_\partial^{\mathrm{pp}}(I) &= \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathcal{P}_I[1]), \end{aligned}$$

with P_0 bracket extended as a biderivation from

$$\begin{aligned} \{a \otimes f, b \otimes g\}_{A_\partial} &= ab \otimes \{f, g\}_{\bar{A}}, \\ \{a \otimes f, B \otimes \rho[1]\}_{A_\partial} &= aB \otimes (f \cdot \rho)[1], \\ \{B \otimes \rho[1], C \otimes \sigma[1]\}_{A_\partial} &= 0. \end{aligned}$$

The factorization product on disjoint intervals is the multiplication in $\widehat{\mathbf{S}}$ after extension by zero. These are the reduced post-contraction targets for the closed-open factorization-center map. The Hamiltonian factor $A_\partial^{\mathrm{Ham}}$ is the connected component of the formal Hamiltonian sector of the open brane factorization algebra; the principal-part factor $A_\partial^{\mathrm{pp}}/A_\partial^{\mathrm{Ham}}$ is the reduced current object of 0.2.77.

PROPOSITION 0.2.20 *(Locality of the brane P_0 bracket).* Let $a \in C_c^\infty(I_1)$, $b \in C_c^\infty(I_2)$ with $I_1 \cap I_2 = \emptyset$. For all $f, g \in \bar{A}$,

$$\{B_f(a), B_g(b)\}_{A_\partial^{\mathrm{Ham}}} = 0.$$

In particular, $A_\partial^{\mathrm{Ham}}$ has the strict factorization product property: the bracket between sections supported on disjoint intervals vanishes.

Proof. By 0.2.19, $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. The pointwise product $ab \equiv 0$ since the supports are disjoint, so $B_{\{f,g\}}(ab) = 0$. □

Remark 0.2.21 *(Locality from the canonical equal-time bracket).* The locality input is the canonical equal-time bracket of the first-order open BV theory:

$$\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j\, \delta(t-s), \qquad \{(\phi_a)^i{}_j(t), (\phi_a)^k{}_l(s)\} = 0 \ \ (a = 1, 2).$$

Tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(s)$, applying the Leibniz rule, and integrating against $a(t)b(s)\delta(t-s)$ produces exactly the smeared identity $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$. This is the unsmeared content packaged in 0.2.82; the delta-function locality is what produces a strict P_0 factorization algebra rather than a homotopical one.

THEOREM 0.2.22 *(Open-closed derived-center factorization equivalence, Hamiltonian sector).* Let $\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}$ be the locally constant reduced open-brane factorization algebra of 0.2.24. Let $\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}|_{\mathbb{R}\times p}$ denote the formal classical Hamiltonian-mode subfactorization algebra of the cotangent BF theory on the formal neighbourhood of $\mathbb{R} \times p$ obtained by restricting to the connected Hamiltonian sector (the $\bar{A}$-modes), before adjoining the cotangent generators. Define the interval-wise CE cochain factorization algebra by

$$\mathcal{F}_{\mathrm{cl}}(I) = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h}_I) = \widehat{\mathbf{S}}\big(\mathfrak{h}^\vee_{I,\mathrm{cont}}[-1]\big),$$

with structure maps induced from extension by zero on C_c^∞ cochain-by-cochain. Then:

(i) there is a continuous interval-wise morphism of P_0 factorization algebras

$$\Phi^{\mathrm{Ham}}_{\mathrm{fact}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H,\mathrm{Ham}}\big|_{\mathbb{R}\times p} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}\big(A_\partial^{\mathrm{Ham}}\big),$$

whose underlying CE cochain map on each interval I is given on length-one CE coordinates by

$$(a \otimes f)^\vee[-1] \ \longmapsto\ B_f(a) := \int_I a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt;$$

(ii) this morphism is a quasi-isomorphism in each interval and is compatible with extension by zero and disjoint factorization products;

(iii) the induced equivalence is locally constant in the topological direction $\mathbb{R}$ in the sense of [16, 17] (Lurie, *Higher Algebra*, §5.5): every inclusion $I' \hookrightarrow I$ of nonempty open intervals induces a quasi-isomorphism of the structure maps;

(iv) the factorization product $A_\partial^{\mathrm{Ham}}(I_1) \otimes A_\partial^{\mathrm{Ham}}(I_2) \to A_\partial^{\mathrm{Ham}}(V)$ for disjoint $I_1 \sqcup I_2 \subset V$ corresponds under $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}$ to the CE coproduct on the closed side induced by extension by zero $C_c^\infty(I_j) \hookrightarrow C_c^\infty(V)$;

(v) at the stalk $I \to \{t_0\}$, $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}$ restricts to the stalkwise central operation

$$f \longmapsto L_{B_f} \in \mathrm{Hoch}^0\big(A_N^{\mathrm{pt}}\big)$$

of 0.2.50.

Proof. Step 1. Identify the interval-wise central factorization algebra. In the formal stable Hamiltonian sector, the boundary factorization algebra $A_\partial^{\mathrm{Ham}}(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$ is graded-commutative, with strict P_0 bracket inherited from $\mathfrak{h}_I$ as a biderivation. By the Hochschild–Kostant–Rosenberg theorem in the smooth/algebraic setting, applied here to the free graded-commutative algebra on the Tate vector space $\mathfrak{h}_I$, the Hochschild cochain complex $\mathrm{Hoch}^\bullet\big(\widehat{\mathbf{S}}(\mathfrak{h}_I)\big)$ is quasi-isomorphic to the polyvector fields $\mathrm{PV}^\bullet\big(\widehat{\mathbf{S}}(\mathfrak{h}_I)\big) = \widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}_{I,\mathrm{cont}}^\vee[-1])$ with the Schouten bracket. The locally constant factorization-center stalk on the line records exactly the Hochschild cochains in the corresponding E_1 algebra; for graded-commutative algebras this collapses to PV. Restricting to the polynomial subalgebra $\mathbf{S}(\mathfrak{h}_I) \subset \widehat{\mathbf{S}}(\mathfrak{h}_I)$ and to the constant polyvectors gives the central multiplication operations, which are recorded as the length-one CE coordinates dual to elements of $\mathfrak{h}_I$. The bracket of two such cocycles is the Schouten bracket on PV restricted to one-vectors, which is the Lie bracket of $\mathfrak{h}_I$.

Step 2. Construct the CE source. By 0.2.57, $C_{\mathrm{CE,cont}}^\bullet(\mathfrak{h}_I) = \widehat{\mathbf{S}}(\mathfrak{h}_{I,\mathrm{cont}}^\vee[-1])$, and its CE differential is the dual of the Lie bracket of $\mathfrak{h}_I$. The continuous dual $\mathfrak{h}_{I,\mathrm{cont}}^\vee$ is computed by the residue-Taylor pairing on the $\bar{A}$ factor and by the density-distribution pairing on the $C_c^\infty(I)$ factor: every continuous functional on $C_c^\infty(I)$ is a compactly supported distribution, and the polynomial subspace $\bar{A} \subset \widehat{\bar{A}}$ pairs with the discrete dual generated by the $\delta_{a,b}$. This gives

$$\mathfrak{h}_{I,\mathrm{cont}}^\vee = \mathcal{D}_c'(I)\widehat{\otimes}\bar{A}_{\mathrm{cont}}^\vee.$$

Step 3. Define $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}(I)$ on generators. By 0.2.55, the closed Hamiltonian cotangent BF observables on the formal neighbourhood of $\mathbb{R} \times p$ in the Hamiltonian sector are $\widehat{\mathbf{S}}(\mathfrak{h}_{I,\mathrm{cont}}^\vee[-1])$, with CE differential dual to the Hamiltonian Lie bracket on $\mathfrak{h}_I$. The interval-wise map sends a length-one CE coordinate $(a \otimes f)^\vee[-1] \in \mathfrak{h}_{I,\mathrm{cont}}^\vee[-1]$ to the element of the central operations factorization algebra given by the multiplication operator

$$L_{B_f(a)} \in \mathrm{Hoch}^0\big(A_\partial^{\mathrm{Ham}}(I)\big) \subset \mathrm{Z}_{\mathrm{fact}}^{P_0}\big(A_\partial^{\mathrm{Ham}}\big)(I),$$

with $B_f(a)$ as displayed. By 0.2.50, this operator is closed (Hochschild 0-cocycle on the graded-commutative open stalk). Extend $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}(I)$ as a continuous algebra morphism on $\widehat{\mathbf{S}}$.

Step 4. Verify CE differential compatibility. On length two, the CE differential of a generator $\xi \in \mathfrak{h}_{I,\mathrm{cont}}^\vee[-1]$ is the dual of the Lie bracket on $\mathfrak{h}_I$. By 0.2.82 (its sharpened form below),

$$\{L_{B_f(a)}, L_{B_g(b)}\}_{A_\partial^{\mathrm{Ham}}} = L_{B_{\{f,g\}}(ab)}.$$

Pulling back through $\Phi_{\mathrm{fact}}^{\mathrm{Ham}}$ matches this with the dual of $[a \otimes f, b \otimes g]_{\mathfrak{h}_I} = ab \otimes \{f, g\}$. Inductive extension by length and the biderivation property propagates this to all CE lengths.

Step 5. Quasi-isomorphism on each interval. Both sides are free graded-commutative algebras on the same continuous Tate vector space $\mathfrak{h}_{I,\mathrm{cont}}^\vee[-1]$, with the same CE differential induced from the Lie

bracket of $\mathfrak{h}_I$ (closed side: by definition; brane side: through the HKR identification of Step 1 and the Schouten bracket). The map respects the symmetric-algebra generators by construction. Thus it is an isomorphism of underlying graded vector spaces and intertwines the differentials: a quasi-isomorphism, in fact an isomorphism, of P_0-cdg algebras on each interval.

Step 6. Factorization product compatibility. For disjoint $I_1, I_2 \subset V$, the CE source extension by zero gives $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. The cobracket on the closed side is the dual of this: it is the splitting $\widehat{\mathbf{S}}(\mathfrak{h}_V^\vee[-1]) \to \widehat{\mathbf{S}}(\mathfrak{h}_{I_1}^\vee[-1])\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}_{I_2}^\vee[-1])$ obtained from the dual surjection $\mathfrak{h}_V^\vee \twoheadrightarrow \mathfrak{h}_{I_1}^\vee \oplus \mathfrak{h}_{I_2}^\vee$. On the brane side, $A_\partial^{\mathrm{Ham}}(I_1) \otimes A_\partial^{\mathrm{Ham}}(I_2) \to A_\partial^{\mathrm{Ham}}(V)$ is the multiplication map under the inclusion of $\mathfrak{h}_{I_1} \oplus \mathfrak{h}_{I_2} \hookrightarrow \mathfrak{h}_V$. By 0.2.20, the bracket between the two factors vanishes on disjoint supports, so the product is strictly multiplicative. The induced map between central-operation factorization algebras is the corresponding tensor product of multiplication operators. This is exactly the dual of the CE-source extension by zero. Therefore the diagram

$$\begin{array}{ccc} \mathcal{F}_{\mathrm{cl}}(I_1) \otimes \mathcal{F}_{\mathrm{cl}}(I_2) & \xrightarrow{\Phi(I_1)\otimes\Phi(I_2)} & \mathrm{Z}_{\mathrm{fact}}^{P_0}(A_\partial^{\mathrm{Ham}})(I_1) \otimes \mathrm{Z}_{\mathrm{fact}}^{P_0}(A_\partial^{\mathrm{Ham}})(I_2) \\ \uparrow & & \downarrow \\ \mathcal{F}_{\mathrm{cl}}(V) & \xrightarrow{\Phi(V)} & \mathrm{Z}_{\mathrm{fact}}^{P_0}(A_\partial^{\mathrm{Ham}})(V) \end{array}$$

commutes after vertical reversal of the left arrow (CE coproduct = dual of inclusion).

Step 7. Locally constant structure. By 0.2.39 the de Rham contraction in t kills all positive jets in $C_c^\infty(I)$, leaving values at points. The translation vector field ∂_t is Q-exact through $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence the structure maps for inclusions of intervals are quasi-isomorphisms on both sides; the E_1-equivalent associative algebra structure is the underlying graded-commutative product on the stalk, equipped with the strict P_0 bracket of 0.2.19.

Step 8. Stalk recovery. As I shrinks to a point $t_0 \in \mathbb{R}$, $C_c^\infty(I)$ contracts onto evaluation at t_0 in cohomology (0.2.39). The image of a length-one CE coordinate becomes the multiplication operator L_{B_f} at the stalk, recovering 0.2.50. □

Remark 0.2.23 *(Locally constant equivalence and E_1-algebra translation).* In the locally constant setting on $\mathbb{R}$, factorization algebras are equivalent to homotopy associative algebras up to translation invariance [16, 17] (Lurie, *Higher Algebra*, §5.5.4). The Hochschild cochain complex of the E_1 algebra computes the factorization-center stalk. For graded-commutative E_1 algebras (which the open Hamiltonian stalk is, in the reduced model), the Hochschild cochain complex is quasi-isomorphic to polyvector fields by HKR, and its strict P_0 subalgebra of multiplication operators with constant differential is the polynomial Hamiltonian sector $\widehat{\mathbf{S}}(\mathfrak{h}_I)\widehat{\otimes}\widehat{\mathbf{S}}(\mathfrak{h}_I^\vee[-1])$. The CE source identification of 0.2.22 is exactly the closed dual of this PV picture in the connected Hamiltonian sector.

For a factorization algebra $\mathcal{A}$ on the line, write $\mathrm{Z}_{\mathrm{fact}}(\mathcal{A})$ for the factorization algebra of E_1 central operations. In the locally constant one-dimensional case its stalk is the Hochschild cochain complex of the corresponding E_1 algebra. The notation $\mathrm{Z}_{\mathrm{fact}}^{P_0}(\mathcal{A})$ means the same central operations with the additional compatibility homotopies for the classical Poisson/BV bracket. The proposition below records the degree-zero current shadow as a corollary of 0.2.22.

PROPOSITION 0.2.24 *(Healed Hamiltonian-current central operation).* 0.2.22 restricted to the length-one CE component on each interval recovers the degree-zero Hamiltonian current shadow: the assignment

$$\Phi_{I,N}(a \otimes f)(u) = B_f(a) \cdot u = \Big(\int_I a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt\Big) u$$

is a Hochschild 0-cocycle in $\mathrm{Z}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N})(I)$, the assignments are compatible with interval inclusions and disjoint factorization products, and the induced operations satisfy

$$\{\Phi(a \otimes f), \Phi(b \otimes g)\}_{\mathrm{open}} = \Phi(ab \otimes \{f, g\}_{\bar{A}}).$$

The earlier degree-zero current shadow is therefore the length-one shadow of the present quasi-isomorphism, not a tautology of graded commutativity: its content is the strict P_0 identity, which is one of the structure equations defining $\Phi^{\mathrm{Ham}}_{\mathrm{fact}}$ of 0.2.22.

Proof. In the reduced open model $Q\phi_1 = Q\phi_2 = 0$. Hence $\overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is Q-closed for every Hamiltonian f, and the compactly supported integral is Q-closed. Since the reduced open observable algebra is graded-commutative, multiplication by this closed element is a Hochschild 0-cocycle; this is the interval-wise version of 0.2.50. Extension by zero of the test function is compatible with restriction to a larger interval. For disjoint intervals 0.2.20 gives vanishing of the P_0 bracket between the two factors, so the factorization product of two such operations is the strict multiplication by $I_\partial(a, f)I_\partial(b, g)$; this is the multiplicative extension from $\mathfrak{h}_I$ to $A^{\mathrm{Ham}}_\partial(I) = \widehat{\mathbf{S}}(\mathfrak{h}_I)$, not the linear current bracket.

The Poisson statement is local on the line. For compactly supported test functions a, b, 0.2.82 (the strict P_0 identity on the brane factorization algebra) gives

$$\{I_\partial(a, f), I_\partial(b, g)\}_{\mathrm{open}} = I_\partial(ab, \{f, g\}_{\bar{A}}).$$

Passing to the connected stable Hamiltonian quotient removes only the scalar Hamiltonian line, which is central for the bracket. Thus the current bracket is preserved. Taking a and b supported in a small coordinate interval and then passing to the locally constant stalk recovers 0.2.81. □

Proposition 0.2.25 *(CE-source obstruction, reframed).* Let $\mathfrak{h}$ be a perfect Lie algebra in **TateVec** (i.e. $[\mathfrak{h}, \mathfrak{h}] = \mathfrak{h}$). Then no length-one CE coordinate c^I dual to a Hamiltonian generator $H_I \in \mathfrak{h}$ is closed under d_{CE}:

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C^K_{IJ}\, c^I c^J \neq 0.$$

Equivalently, $H^1(\mathfrak{h}; \mathbb{C}) = 0$.

This obstruction is the structural reason that the boundary observable $O_I = J(H_I) \in A_\partial$ cannot be the image under Φ of c^I. Instead, under 0.2.29,

$$O_I = \Phi(u_I),$$

the image of the CE coordinate dual to the shifted cotangent generator $\eta_{\delta_I} \in \mathfrak{h}^\vee[1]$. The boundary observable lives in the cotangent direction of the CE cochain algebra, not in the Hamiltonian direction. Its CE differential is not zero but $d_{\mathrm{CE}} u_K = C^L_{KJ} u_L c^J$, mapping under Φ to the Hamiltonian vector field $d_\pi O_K = C^L_{KJ} O_L \theta^J$ of O_K in the Schouten algebra.

This is exactly what a P_0-algebra map should produce: a function whose Schouten differential is its Hamiltonian vector field.

Proof. $\mathfrak{h}$ perfect means $C^K_{IJ} c^I c^J \neq 0$ for some triple (I, J, K) realizing a non-trivial bracket; H^1 vanishing follows. Identification $O_I = \Phi(u_I)$ is the content of 0.2.30 combined with $\Phi(u_I) = O_I$ from 0.2.29. Differentials match on generators by 0.2.59. □

Remark 0.2.26. The reduced Hamiltonian Lie algebra $\mathfrak{h} = \widehat{A}/\mathbb{C}\cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ of the formal symplectic disk in dimension two is perfect: for every monomial one has, before quotienting constants, $z_1^a z_2^b = \left\{z_1, \frac{1}{b+1} z_1^a z_2^{b+1}\right\}$, so $[\mathfrak{h}, \mathfrak{h}] = \mathfrak{h}$. In particular $[H_{1,0}, H_{m,n}] = n\, H_{m,n-1}$ and $[H_{0,1}, H_{m,n}] = -m\, H_{m-1,n}$. The hypothesis of 0.2.25 is met. The CE-length-one cochain $\xi_{a,b} = c^{a,b}$ dual to $H_{a,b} = z_1^a z_2^b$ is therefore not closed, and no closed CE cochain has linear Hamiltonian part $c^{a,b}$, recovering the earlier source-obstruction observation.

THEOREM 0.2.27 *(Reduced derived Poisson center).* Let

$$\mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee$$

be the product/direct-sum Tate Hamiltonian Lie algebra of the formal symplectic disk and its continuous dual. Put

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

Let $A_\partial = \widehat{\mathbf{S}}(\mathfrak{h})$ be the completed stable degree-zero Hamiltonian trace algebra, with generators O_i corresponding to a homogeneous topological basis H_i of $\mathfrak{h}$. If

$$[H_i, H_j] = C_{ij}^k H_k,$$

with the constant target projected out, equip A_∂ with the linear Poisson bracket

$$\{O_i, O_j\}_\partial = C_{ij}^k O_k.$$

Let

$$\mathrm{PV}_{\mathrm{cont}}(A_\partial) = A_\partial \widehat{\otimes} \widehat{\Lambda}(\mathfrak{h}^\vee_{\mathrm{cont}})$$

be the completed algebra of continuous polyvectors, with odd generators θ_i dual to the coordinates O_i. The Poisson bivector is

$$\pi_\partial = \frac{1}{2} C_{ij}^k O_k\, \theta_i \theta_j,$$

and the Poisson differential is $d_\pi = [\pi_\partial, -]$ in the Schouten convention.

If c_i is the CE coordinate dual to $H_i \in \mathfrak{h}$ and u_i is the CE coordinate dual to the shifted cotangent generator $\delta_i[1] \in \mathfrak{h}^\vee_{\mathrm{cont}}[1]$, then the assignment

$$c_i \longmapsto \theta_i, \qquad u_i \longmapsto O_i$$

extends to an isomorphism of completed graded-commutative dg algebras

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} \big(\mathrm{PV}_{\mathrm{cont}}(A_\partial), d_\pi\big).$$

Proof. By the product/direct-sum convention already used in Proposition 0.2.60, the continuous CE algebra of $\mathfrak{g}$ is freely generated, after completion, by the odd coordinates c_i and the even coordinates u_i. The CE differential is dual to the brackets of $\mathfrak{g}$:

$$d_{\mathrm{CE}} c_k = -\frac{1}{2} C_{ij}^k c_i c_j, \qquad d_{\mathrm{CE}} u_k = C_{kj}^\ell u_\ell c_j.$$

The second formula is the coadjoint action of $\mathfrak{h}$ on $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$ written on the dual coordinates.

The linear Poisson bivector gives the Lichnerowicz differential [7]

$$d_\pi \theta_k = -\frac{1}{2} C_{ij}^k \theta_i \theta_j, \qquad d_\pi O_k = C_{kj}^\ell O_\ell \theta_j.$$

Hence $c_i \mapsto \theta_i$ and $u_i \mapsto O_i$ intertwine the two differentials. The underlying completed graded-commutative algebras are freely generated by corresponding variables, so this is an isomorphism.

Continuity is the same coefficient-finiteness used in Lemma 0.2.57: for each fixed output coefficient in a bracket of Hamiltonians only finitely many homogeneous input-degree pairs contribute, and the coadjoint input has finite support in $\mathfrak{h}^\vee_{\mathrm{cont}}$. Thus the displayed sums define continuous derivations of the completed CE and polyvector algebras. □

Remark 0.2.28. The theorem is the reduced algebraic center calculation. It does not say that the scalar boundary function $O_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ is the image of the nonclosed CE coordinate c_f. It says that O_f is the image of the CE coordinate u_f dual to the shifted cotangent generator, while c_f maps to the constant Hamiltonian vector-field operation θ_f. This is why Proposition 0.2.25 remains correct and why the full BV/factorization lift below has nontrivial content.

Theorem 0.2.29 *(Open-closed derived center, cochain level).* Let $\mathfrak{h}$ be any pro-finite-dimensional Lie algebra in **TateVec** over $\mathbb{C}$, concentrated in degree zero, satisfying the Tate-conilpotency hypothesis of 0.2.57. Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the shifted-cotangent extension. Let $A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$ be the Tate-completed Kirillov–Poisson algebra of 0.2.59.

Then there is a canonical isomorphism of completed dg graded-commutative P_0-algebras

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} \mathrm{PV}(A_\partial),$$

determined on generators by

$$\Phi(c^I) = \theta^I, \qquad \Phi(u_I) = O_I,$$

where c^I is the CE coordinate dual to $H_I \in \mathfrak{h}$ and u_I is the CE coordinate dual to the shifted cotangent generator $\eta_{\partial_I} \in \mathfrak{h}^\vee[1]$.

Moreover, $\mathrm{PV}(A_\partial)$ identifies with the derived P_0-factorization center of the constant factorization algebra A_∂, so

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \xrightarrow{\sim} Z^{P_0}_{\mathrm{fact}}(A_\partial)$$

is the cochain-level open-closed derived center isomorphism in the formal/local sector.

Proof. Generators and free extension. By 0.2.57,

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}_{\mathrm{Tate}}\big((\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1])^\vee_{\mathrm{cont}}[-1]\big) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}\big(\{u_I\}\big)\widehat{\otimes}\widehat{\Lambda}\big(\{c^I\}\big),$$

with $|u_I| = 0$ and $|c^I| = 1$. On the open side, $\mathrm{PV}(A_\partial) = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\{O_I\})\widehat{\otimes}\widehat{\Lambda}(\{\theta^I\})$, with $|O_I| = 0$ and $|\theta^I| = 1$. The same generator data on both sides freely generate completed graded-commutative algebras in **TateVec** of the same shape; the assignment $c^I \mapsto \theta^I$, $u_I \mapsto O_I$ extends uniquely to a continuous bijection of completed dg-commutative algebras.

Cochain identity on generators. The CE differential is

$$d_{\mathrm{CE}} c^K = -\tfrac{1}{2} C_{IJ}^K c^I c^J, \qquad d_{\mathrm{CE}} u_K = C_{KJ}^L u_L c^J.$$

By 0.2.59,

$$d_\pi \theta^K = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J, \qquad d_\pi O_K = C^L_{KJ} O_L \theta^J.$$

Therefore

$$\Phi(d_{\mathrm{CE}} c^K) = -\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J = d_\pi \theta^K = d_\pi \Phi(c^K),$$
$$\Phi(d_{\mathrm{CE}} u_K) = C^L_{KJ} O_L \theta^J = d_\pi O_K = d_\pi \Phi(u_K).$$

Cochain identity on the full completed algebra. Both d_{CE} and d_π are degree-1 derivations of the completed graded-commutative algebra structures on each side. They satisfy the graded Leibniz rule with the same Koszul sign convention. Φ is an algebra homomorphism, so

$$\Phi(d_{\mathrm{CE}}(xy)) = \Phi\big((d_{\mathrm{CE}} x) y + (-1)^{|x|} x\, d_{\mathrm{CE}} y\big) = \Phi(d_{\mathrm{CE}} x)\Phi(y) + (-1)^{|x|}\Phi(x)\Phi(d_{\mathrm{CE}} y),$$

while

$$d_\pi \Phi(xy) = d_\pi\big(\Phi(x)\Phi(y)\big) = (d_\pi \Phi(x))\Phi(y) + (-1)^{|x|}\Phi(x) d_\pi \Phi(y).$$

Generator-level agreement plus Leibniz/algebra-homomorphism gives agreement on every word in $\{c^I, u_J\}$ at fixed total length; Tate-completeness of both sides plus continuity of $d_{\mathrm{CE}}, d_\pi, \Phi$ in the Tate filtration extends the equality to the completed algebras.

Inverse. The assignment $\theta^I \mapsto c^I, O_I \mapsto u_I$ extends to a continuous algebra homomorphism in the opposite direction. By the same Leibniz argument, it intertwines d_π and d_{CE}. The two compositions fix generators, hence fix the completed algebras. So Φ is an isomorphism of complete dg graded-commutative algebras.

P_0-bracket compatibility. The CE side carries the canonical (-1)-shifted Poisson structure on $C^\bullet_{\mathrm{CE,cont}}(T^*[1]\mathfrak{h})$ with

$$\{c^I, u_J\}_{\mathrm{CE}} = \delta^I_J, \quad \{c^I, c^J\}_{\mathrm{CE}} = 0, \quad \{u_I, u_J\}_{\mathrm{CE}} = 0.$$

These are the standard Poisson brackets on the cochain algebra of a (-1)-shifted cotangent of a Lie algebra (cf. Costello–Gwilliam factorization-algebra conventions). The Schouten bracket on $\mathrm{PV}(A_\partial)$ satisfies

$$[\theta^I, O_J]_S = \delta^I_J, \quad [\theta^I, \theta^J]_S = 0, \quad [O_I, O_J]_S = 0,$$

the last because O_I, O_J are functions on the formal affine space $\mathfrak{h}^\vee$ and the Schouten bracket of two functions is zero. Generator-level matching gives $\Phi\big(\{c^I, u_J\}_{\mathrm{CE}}\big) = \Phi(\delta^I_J) = \delta^I_J = [\theta^I, O_J]_S = [\Phi(c^I), \Phi(u_J)]_S$, and similarly for the other two pairs. Both brackets satisfy the graded Leibniz rule; bilinear extension on completed words gives compatibility on the full completed algebra.

Identification with the derived P_0-factorization center. For a Poisson algebra A regarded as a constant $\mathbb{R}^d$-factorization algebra (or as a stalk of a locally constant factorization algebra), the derived P_0-factorization center is computed by the Poisson cochain complex $\mathrm{PV}(A)$ with the Lichnerowicz differential $[\pi, -]_S$. This is the local stalk version of the Poisson-CE identification of Cattaneo–Felder / Costello–Gwilliam (cf. [16], central operations for factorization algebras with Poisson bracket; the identification is local: at a point, the factorization-center is the Poisson-cochain complex). Therefore $\mathrm{PV}(A_\partial) \simeq Z^{P_0}_{\mathrm{fact}}(A_\partial)$ as completed dg-commutative P_0-algebras, and Φ is the cochain-level open-closed derived center isomorphism in this local stalk sense. □

Corollary 0.2.30 *(Moment-map shadow).* In the situation of 0.2.29, the degree-zero specialization of Φ on the cotangent direction is the moment-map / Kirillov assignment

$$J : \mathfrak{h} \longrightarrow A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h}), \qquad J(f) = O_f.$$

The closed-side Lie bracket $\{f, g\}_{\mathfrak{h}}$ matches the Kirillov–Poisson bracket on A_∂:

$$\{J(f), J(g)\}_\partial = J\big(\{f, g\}_{\mathfrak{h}}\big).$$

Proof. J is the restriction of Φ to the cotangent direction $u_I \mapsto O_I$ followed by linear extension along $\mathfrak{h} \hookrightarrow A_\partial$ (the canonical embedding of $\mathfrak{h}$ as the linear functions on $\mathfrak{h}^\vee$). Bracket compatibility is the (c^J, u_K)-versus-(θ^J, O_K) identity from 0.2.29 read off in degree zero. Equivalently, this is the standard Kirillov–Kostant–Souriau theorem for linear Poisson structures. □

Corollary 0.2.31 *(Derived center, formal symplectic disk).* Let $\mathfrak{h} = \widehat{A}/\mathbb{C} \cdot 1 = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ be the reduced Hamiltonian Lie algebra of the formal symplectic disk on $\mathbb{C}^2$, with bracket $\{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g$ modulo constants. Choose the Taylor basis $H_{a,b} = z_1^a z_2^b$ for $a, b \geq 0$, $a + b > 0$. Structure constants:

$$[H_{a,b}, H_{c,d}] = (ad - bc)\, H_{a+c-1,b+d-1},$$

with the convention that $H_{m,n} = 0$ if $\min(m, n) < 0$ or $m + n = 0$.

Set $A_\partial = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$, with linear coordinates $O_{a,b}$ on $\mathfrak{h}^\vee$ and Kirillov–Poisson bracket $\{O_{a,b}, O_{c,d}\}_\partial = (ad - bc)O_{a+c-1,b+d-1}$. 0.2.29 gives the cochain-level derived-center isomorphism

$$\Phi : C^\bullet_{\mathrm{CE,cont}}(\mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]) \xrightarrow{\sim} \mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big), \qquad c^{a,b} \mapsto \theta^{a,b}, \quad u_{a,b} \mapsto O_{a,b}.$$

Its moment-map shadow is the boundary evaluation map $J(f) = O_f = \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}$ of 0.2.81 after identifying A_∂ with the stable connected scalar-reduced Hamiltonian trace algebra of 0.2.51.

Proof. The structure constants are an instance of 0.2.13 read on monomials. Apply 0.2.29 to this $\mathfrak{h}$ (perfect; pro-finite-dimensional in Taylor filtration; Tate-conilpotent; see 0.2.26). The moment-map shadow follows from 0.2.30 together with 0.2.81. □

Example 0.2.32 *(Low-degree monomial verification).* We verify Φ on small monomials of the formal symplectic disk.

- *Degree-1 generators.* Take $H_{1,0} = z_1$, $H_{0,1} = z_2$. The bracket $[H_{1,0}, H_{0,1}] = 1 \cdot 1 - 0 \cdot 0 = 1$ lifts to the constant $H_{0,0}$, which is set to zero in $\mathfrak{h}$. Hence $C^{0,0}_{(1,0),(0,1)} = 0$ in $\mathfrak{h}$. Consequently $d_{\mathrm{CE}} u_{0,0}$ is not present (no such generator); the $u_{0,0}$ degree of freedom does not appear, consistent with constants being removed.
- *Degree-2 calculation.* Take $H_{2,0} = z_1^2$, $H_{0,2} = z_2^2$. The bracket $\{z_1^2, z_2^2\} = 2z_1 \cdot 2z_2 - 0 \cdot 0 = 4z_1 z_2 = 4H_{1,1}$, so $C^{1,1}_{(2,0),(0,2)} = 4$ and the antisymmetric companion is $C^{1,1}_{(0,2),(2,0)} = -4$. The cochain identity reads

  $$d_{\mathrm{CE}} u_{1,1} = 4\, u_{2,0}\, c^{0,2} - 4\, u_{0,2}\, c^{2,0} + \cdots,$$

  where $\cdots$ denotes terms from other index triples (K, J, L) with $K = (1, 1)$. Each such term $C^L_{(1,1),J}\, u_L\, c^J$ arises from $\{H_{1,1}, H_J\} = \sum_L C^L_{(1,1),J}\, H_L$. For example, $\{H_{1,1}, H_{1,0}\} =$

$\partial_{z_1}(z_1 z_2)\cdot 0 - \partial_{z_2}(z_1 z_2)\cdot 1 = -z_1 = -H_{1,0}$, so $C^{1,0}_{(1,1),(1,0)} = -1$, contributing $-u_{1,0}\, c^{1,0}$ to $d_{\mathrm{CE}} u_{1,1}$. On the open/PV side,

$$d_\pi O_{1,1} = 4\, O_{2,0}\, \theta^{0,2} - 4\, O_{0,2}\, \theta^{2,0} - O_{1,0}\, \theta^{1,0} + O_{0,1}\, \theta^{0,1} + \cdots,$$

the same expression with $u \to O, c \to \theta$. Hence $\Phi(d_{\mathrm{CE}} u_{1,1}) = d_\pi \Phi(u_{1,1})$. The script `scripts/check_moyal_coefficients.py` verifies the same structure-constant input on every monomial pair with exponents up to 6 via the first-order Moyal coefficient $kn - lm$ at $\hbar^1$, which equals C^K_{IJ} for these indices.

- *Higher-order word.* For $K = (1,1), J = (1,0), J' = (0,1)$, the cochain element $u_{1,1} c^{1,0} c^{0,1}$ has degree $0 + 1 + 1 = 2$. Its CE differential, by graded Leibniz, is

$$d_{\mathrm{CE}}\big(u_{1,1} c^{1,0} c^{0,1}\big) = (d_{\mathrm{CE}} u_{1,1}) c^{1,0} c^{0,1} - u_{1,1}(d_{\mathrm{CE}} c^{1,0}) c^{0,1} + u_{1,1} c^{1,0} (d_{\mathrm{CE}} c^{0,1}),$$

with each summand applying the generator formulas above. The same word on the PV side is $O_{1,1}\theta^{1,0}\theta^{0,1}$, and d_π applied with the same Leibniz convention produces the corresponding triple sum with $u \to O$ and $c \to \theta$. Term-by-term equality follows from the generator identities; the full word equality is the algebra-homomorphism Leibniz argument from 0.2.29.

Construction 0.2.33 *(Compact current PBW/Rees source).* Let $\mathcal{P}$ denote the principal-part module

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

identified with $\mathfrak{h}^\vee_{\mathrm{cont}}$ by the residue pairing of Proposition 0.2.67. Put

$$\mathfrak{g}_{\mathrm{pp}} = \mathfrak{h} \ltimes \mathcal{P}[1],$$

with zero differential and bracket

$$[f, g] = \{f, g\}, \qquad [f, \rho[1]] = (f\cdot\rho)[1], \qquad [\rho[1], \sigma[1]] = 0,$$

where $f\cdot\rho = \pi_{\mathrm{pp}}\{f,\rho\}$. On monomials this action is

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with negative-index targets zero.

For an interval $I \subset \mathbb{R}$, define the compactly supported current Lie algebra

$$\mathfrak{J}_{\mathfrak{g}}(I) = C_c^\infty(I)\widehat{\otimes}\mathfrak{g}_{\mathrm{pp}}, \qquad [\alpha\otimes x, \beta\otimes y] = \alpha\beta\otimes[x,y].$$

Its PBW/Rees algebra is

$$U_\hbar\big(\mathfrak{J}_{\mathfrak{g}}(I)\big) = \widehat{T}\big(\mathfrak{J}_{\mathfrak{g}}(I)\big)[[\hbar]] \big/ \langle xy - (-1)^{|x||y|} yx - \hbar[x,y]\rangle.$$

The PBW filtration is the tensor-word filtration. Its special fibre is

$$U_\hbar\big(\mathfrak{J}_{\mathfrak{g}}(I)\big)/(\hbar) \cong \widehat{\mathbf{S}}\big(\mathfrak{J}_{\mathfrak{g}}(I)\big),$$

equipped with the semiclassical P_0 bracket

$$\{\bar{X}, \bar{Y}\}_{P_0} = \big(\hbar^{-1}(XY - (-1)^{|X||Y|} YX)\big) \bmod \hbar.$$

This is the PBW/Koszul-dual source for Hamiltonian and principal-part modes. It is not a closed CE cochain algebra map; the CE comparison must be made separately by bar-cobar twisting or by a completed pairing.

Problem 0.2.34 *(PBW-source P_0 center action and CE comparison).* Construct a filtered factorization-center action of the compactly supported current PBW/Rees source $U_\hbar(\mathfrak{J}_\mathfrak{g})$ of Construction 0.2.33. The target cannot be the polynomial open observable algebra alone if the cotangent generators are included; it must use the enlarged principal-part boundary object of Theorem 0.2.77. In the classical special fibre the action must send

$$f \longmapsto B_f = \overline{\operatorname{Tr} f(\phi_1, \phi_2)}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b},$$

and satisfy the stalkwise bracket identities

$$\{B_f, B_g\}_{\text{open}} = B_{\{f,g\}}, \qquad \{B_f, \Theta_\rho\}_{\text{open}} = \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\}_{\text{open}} = 0.$$

This special-fibre statement is the natural P_0 theorem. A strict action of the full filtered Rees algebra on the same commutative reduced target is not a consequence of these identities: in the source

$$XY - (-1)^{|X||Y|} YX = \hbar[X, Y],$$

while multiplication operators on a commutative classical target commute. Thus the Rees-level theorem requires either a deformed boundary target or an explicit homotopical quantization of the central operations. The target data must include differential compatibility, factorization centrality homotopies, coherent P_0 bracket homotopies, and compact-support naturality for interval inclusions and disjoint unions before the de Rham contraction.

Separately compare this PBW/Koszul-dual source with the closed CE observable algebra by an explicit twisting or pairing construction. Without this comparison, the result is a PBW/Koszul-dual boundary action, not a cochain-level map from closed CE observables.

Problem 0.2.35 *(Unreduced Tate-coefficient cotangent lift).* This is the single residual of the umbrella theorem 0.1.22. Construct a Tate or weighted-coefficient extension of the BV heat-kernel calculus on $\mathbb{R}^2 \times \mathbb{C}^2$ in which:

(i) the formal Casimir $C_\mathfrak{h} = \sum_{d\geq 1} \sum_i e_{d,i} \otimes e^i_d$ converges as an element of $\widehat{\mathfrak{h}} \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\text{cont}}$;

(ii) the Costello propagator $P_{\epsilon,L}$ extends to that coefficient category;

(iii) the unreduced principal-part current sub-factorization algebra $\widetilde{\mathrm{Obs}}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N}(I) \widehat{\otimes} \widehat{\mathbf{S}}(\mathcal{M}^{\mathrm{pp,unred}}_\partial(I))$ admits a chain-level primitive projection π_{prim} with verified Q-compatibility, factorization compatibility, and P_0-bracket homotopies;

(iv) statement (d) of 0.1.22 lifts to the unreduced factorization-algebra model and statement (e) extends to the cotangent (one-antifield) sector at all orders in $\hbar$.

Equivalently, lift the cochain-level 0.2.29 and the Hamiltonian-sector factorization-algebra 0.2.22 to a cochain-level morphism of factorization algebras

$$\Phi_{\mathrm{cl}} : \mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}\big|_{\mathbb{R}\times p} \longrightarrow \mathrm{Z}^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open},N})$$

on the unreduced (cotangent-included) sector, with all P_0-center homotopies, factorization centrality homotopies, the stable connected quotient, and a boundary realization of the shifted cotangent generators. Any global version must additionally pass through the local-Hamiltonian descent and period obstruction of 0.1.19; a stalkwise formal disk map is not a Weiss/Ran descent theorem for factorization centers.

Two routes are named, neither yet executed.

- *Route R1: weighted Tate completion.* Replace the direct sum $\bigoplus_{d\geq 1} H_d^\vee$ by a weighted Tate completion in which the Casimir converges. The $\mathfrak{m}$-adic completion fails by the finite-window obstruction; a candidate is a graded-Mittag-Leffler completion controlled by a polynomial weight in the degree filtration, checked compatible with the Costello RG flow at every order.
- *Route R2: nilpotent truncation.* Restrict to the smaller Lie sector $\mathfrak{m}^2/\mathfrak{m}^{N+1}$ or analogous nilpotent truncation, declaring that translations and the constant-Hamiltonian $\hbar$-normalization have been removed (the Costello–Li $\mathfrak{gl}(N \mid N)$ precedent). This loses the fully stable connected statement on $\mathfrak{h}$ but recovers a meaningful nilpotent sector unconditionally.

The deciding evidence to close: either (R1) yields a Tate-coefficient category with the displayed Casimir convergence and BV propagator extension, or (R2) gives an explicit obstruction class for the unreduced cotangent lift (which would itself be a major theorem and would not invalidate the reduced umbrella 0.1.22, parts (a)–(f)).

The residual is identified across four independent obstruction routes as the same obstruction. The cotangent direction $\mathfrak{h}^\vee_{\mathrm{cont}}[1]$ supports the deformation-correct realization $\mathcal{P}$ at the cochain level; fails to admit a polynomial sub-bimodule realization in any locally-nilpotent category; admits a factorization-algebra realization at the reduced post-contraction level; and obstructs the unreduced lift at the analytic level. The umbrella theorem is unconditional in parts (a), (b), (c), (e) on the degree-zero Hamiltonian sector, and (f); part (d) is unconditional at the reduced post-contraction level and conditional on this problem for the unreduced extension; part (e) is unconditional on the degree-zero Hamiltonian sector at all orders in $\hbar$ and conditional on this problem for the descendant (one-antifield) sector.

Remark 0.2.36 *(The residual is conditional, the umbrella is not).* 0.1.22 is unconditional in parts (a), (b), (c), (e) (degree zero), (f). Part (d) is unconditional at the reduced post-contraction level and conditional on 0.2.35 for the unreduced factorization-algebra extension. Part (e) is unconditional on the degree-zero Hamiltonian sector at all orders in $\hbar$ and conditional on the same residual for the descendant (one-antifield) sector. Cross-volume comparisons (Vol III CY-to-chiral, Igusa Δ_5 / BKM, twisted-M-theory universality) are frontier obligations recorded in the convention contract appendix; they are outside the dependency closure of the present local theorem.

Remark 0.2.37 *(Weiss/Ran descent firewall).* A global closed-open map is not obtained by choosing Darboux coordinates at each point and gluing the resulting stalk maps. The obstruction lives in the Weiss/Ran total complex

$$\mathrm{Tot}\, C^\bullet_{\mathrm{WR}}\Big(\mathrm{Ran}(L); \underline{\mathrm{RHom}}\Big(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{closed},H}, Z^{P_0}_{\mathrm{fact}}(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}})\Big)\Big).$$

Vanishing of the local Hamiltonian period class is only the first condition. One must also prove source comparison, cotangent boundary realization, all P_0-center homotopies, and factorization descent. Until then the Hamiltonian boundary map, the Θ-currents, and the PBW/Rees action are formal local or reduced-current statements.

0.2.7 Operators in the open brane gauge theory and the large N limit

The local-operator calculation is performed in the reduced BV model with fields ϕ_1, ϕ_2 and the antifield $\psi = A^*$ for the Lagrange multiplier A. We take GL_N-invariants in characteristic zero, so the Chevalley part of the gauge BRST differential acts trivially on trace functions by cyclicity.

The gauge field has one algebraic job: it imposes the moment-map equation $[\phi_1, \phi_2] = 0$. In BV language this equation is imposed by the Koszul differential $Q\psi = [\phi_1, \phi_2]$. This is why the stable trace

algebra starts noncommutative, as cyclic words in two matrices, and then collapses in cohomology to commutative monomials on the formal symplectic plane.

Construction 0.2.38 *(The reduced BV algebra at the insertion point).* At a fixed point of the brane line, introduce three matrix variables

$$\phi_1, \phi_2 \in \mathfrak{gl}_N, \qquad \psi \in \mathfrak{gl}_N[1].$$

The reduced local polynomial algebra is

$$\mathcal{R}_N = \mathbb{C}[(\phi_1)^i{}_j, (\phi_2)^i{}_j] \otimes \Lambda\big((\psi)^i{}_j\big).$$

Its BRST differential is the derivation determined on generators by

$$Q(\phi_1)^i{}_j = 0, \qquad Q(\phi_2)^i{}_j = 0, \qquad Q(\psi)^i{}_j = ([\phi_1, \phi_2])^i{}_j.$$

This formula is the algebraic shadow of the A-equation of motion: varying

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives $[\phi_1, \phi_2] = 0$, and the antifield $\psi = A^*$ records that equation as a Koszul differential.

Lemma 0.2.39 *(De Rham jets).* Positive jets in the topological direction vanish in local operator cohomology. Equivalently, every cohomology class has a representative determined by the values of the fields and antifields at the insertion point.

Proof. For a coordinate t on the line, the de Rham part of the BRST operator is $d_{\mathbb{R}}$. On the jet algebra one has the contraction homotopy $[d_{\mathbb{R}}, \iota_{\partial_t}] = \partial_t$. Hence a local expression containing a positive t-derivative is cohomologous to zero after applying the usual Euler homotopy on jet degree. This kills topological-direction jets and leaves the holomorphic z_i-derivatives used later on the closed side. □

Proposition 0.2.40 *(Stable trace invariants).* Let $F = \mathbb{C}\langle x_1, x_2\rangle$ and let $\mathrm{Cyc}(F) = F/[F, F]$ be the vector space of cyclic words. For each N, GL_N acts on $\mathfrak{gl}_N^2$ by simultaneous conjugation. The degreewise stable invariant algebra

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} := \bigoplus_{d\geq 0} \varprojlim_N \big(\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}\big)_d$$

with respect to block-restriction maps in each word-degree is the free commutative algebra generated by the nonempty cyclic words:

$$\mathbb{C}[\mathfrak{gl}_\infty^2]_{\mathrm{st}}^{GL_\infty} \cong \mathbf{S}\big(\mathrm{Cyc}(F)_{\mathrm{nonempty}}\big),$$

where a cyclic word w maps to the trace function $(\phi_1, \phi_2) \mapsto \mathrm{Tr}\, w(\phi_1, \phi_2)$. More precisely, in total positive word-degree d this map is an isomorphism for every $N \geq d$, after omitting the scalar empty trace as a primitive generator.

Proof. Procesi–Razmyslov invariant theory [18, 19] gives two inputs. First, $\mathbb{C}[\mathfrak{gl}_N^2]^{GL_N}$ is generated by trace functions of words in the two generic matrices. Second, all polynomial relations among these trace functions are consequences of the fundamental trace identities of matrix size N; in particular every

homogeneous trace relation has total word-degree at least $N+1$. Fix a total word-degree d. For every $N \geq d$ the degree-d relations are exactly cyclicity of the trace. Passing degree by degree to the stable inverse limit removes all finite-N trace identities. The surviving identification is $\mathrm{Tr}(uv) = \mathrm{Tr}(vu)$. This is exactly the symmetric algebra on cyclic words.

□

LEMMA 0.2.41 *(Stable marked trace invariants).* In the stable invariant theory used below, the primitive part linear or quadratic in an odd marked matrix ψ is obtained from the corresponding even three-matrix trace invariant by polarization and skew-symmetrisation in the marked letters. The only stable single-trace relation is graded cyclicity: moving one ψ through the trace has sign $+1$, while moving one ψ past another ψ gives a minus sign.

Proof. Apply Proposition 0.2.40 to three even generic matrices and take the homogeneous summand linear, respectively quadratic, in the third matrix. The Procesi–Razmyslov trace identities have no stable relations in the fixed total word-degree once N is large enough. Declaring the marked matrix odd is the usual super-polarization: the linear part is unchanged, and the quadratic part is projected to the alternating component under exchange of the two marked occurrences. The trace relation therefore becomes exactly graded cyclicity. These are the signs used in the finite complex for one- and two-ψ words. □

Remark 0.2.42. The empty trace is the finite-rank scalar $\mathrm{Tr}_N(1) = N$, which changes with N and fails to define a stable primitive generator. The stable symmetric algebra still has its ordinary unit; what is omitted is the empty trace as a generator of the connected primitive part. The class $\mathrm{Tr}(\psi)$ is a genuine stable odd primitive class. It is excluded from the Hamiltonian comparison because constants have been removed from the closed Hamiltonian algebra.

Varying

$$S = \int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big)$$

with respect to A gives the moment-map equation $[\phi_1, \phi_2] = 0$. In the reduced BV complex,

$$Q\psi = [\phi_1, \phi_2], \qquad Q\phi_1 = Q\phi_2 = 0$$

on invariant trace functions.

Construction 0.2.43 *(Boundary evaluation of a closed Hamiltonian).* Let $f = \sum_{k,l} c_{k,l} z_1^k z_2^l \in \mathbb{C}[z_1, z_2]$. The brane replaces the two normal coordinates z_1, z_2 by the two adjoint scalar fields ϕ_1, ϕ_2. Choosing the ordered lift with all ϕ_1's to the left of all ϕ_2's gives the invariant function

$$\widetilde{\partial}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l}\, \mathrm{Tr}(\phi_1^k \phi_2^l) \in \mathcal{R}_N^{GL_N}.$$

Let

$$H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q) := H^0(\mathcal{R}_N^{GL_N}, Q)/\mathbb{C} \cdot \mathrm{Tr}(1)$$

as a vector space, and as a Lie algebra for the Poisson bracket. This is not an associative quotient of the operator algebra by its unit. Constants give the scalar $c_{0,0}N$ and therefore vanish in this Hamiltonian quotient. Thus the construction descends to a linear map

$$\partial_{\mathrm{bb},N} : \bar{A} \to H^0_{\mathrm{Ham}}(\mathcal{R}_N^{GL_N}, Q), \qquad [f] \longmapsto \overline{\widetilde{\partial}_{\mathrm{bb},N}(f)}.$$

Let $\mathsf{W}_{k,l}$ be the set of words in two letters containing k copies of the first letter and l copies of the second. The first descendant is obtained by inserting one Koszul variable and summing over all ordered lifts: For $k + l > 0$ the descendant part of the Hamiltonian comparison is

$$\partial^{(1)}_{\mathrm{bb},N}(\delta_{k,l}[1]) = \Psi_{k,l}, \qquad \Psi_{k,l} := \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big(\psi\, w(\phi_1, \phi_2)\big).$$

The same formula at $(k, l) = (0, 0)$ gives the decoupled scalar antifield $\mathrm{Tr}(\psi)$, which is excluded from the Hamiltonian pairing because constants have been removed from $\bar{A}$.

Lemma 0.2.44 *(First descendants are cycles).* For every $k, l \geq 0$, the symmetrised trace $\Psi_{k,l}$ satisfies $Q\Psi_{k,l} = 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Since x and y are even and $Q\psi = [x, y]$, one has

$$Q\Psi_{k,l} = \sum_{w \in \mathsf{W}_{k,l}} \mathrm{Tr}\big((xy - yx)w(x, y)\big).$$

Group the resulting traces by cyclic word of multidegree $(k + 1, l + 1)$. The coefficient of a cyclic word in the positive sum is the number of cyclic adjacent occurrences of the block xy; the coefficient in the negative sum is the number of cyclic adjacent occurrences of the block yx. On a circle the number of transitions from an x-run to a y-run equals the number of transitions from a y-run to an x-run. The two grouped sums are therefore equal, so $Q\Psi_{k,l} = 0$. □

Proposition 0.2.45 *(Primitive one-antifield Koszul homology).* Fix $k, l \geq 0$. The middle homology of the primitive single-trace complex with one displayed ψ and k, l displayed x, y letters is one-dimensional, generated by the class of $\Psi_{k,l}$. The nonconstant part relevant to the Hamiltonian comparison is the direct sum over $k + l > 0$.

Proof. Write $x = \phi_1$ and $y = \phi_2$. Let $\mathsf{W}_{k,l}$ be the set of ordinary words with k letters x and l letters y, and let $\mathsf{N}_{a,b}$ be the set of cyclic words with a letters x and b letters y. By Lemma 0.2.41, primitive cyclic words are taken modulo the graded cyclic rule

$$\mathrm{Tr}(ab) = (-1)^{|a||b|}\mathrm{Tr}(ba), \qquad |x| = |y| = 0, \quad |\psi| = 1.$$

Graded cyclicity gives a unique representative $\mathrm{Tr}(\psi w)$ for every one-ψ cyclic word. Thus the relevant one-ψ chain group is

$$C_1(k, l) = \mathbb{C}\{\mathsf{W}_{k,l}\}.$$

The zero-ψ target is

$$C_0(k, l) = \mathbb{C}\{\mathsf{N}_{k+1,l+1}\},$$

and the differential is the incidence map

$$\partial_1(w) = [xyw]_{\mathrm{cyc}} - [yxw]_{\mathrm{cyc}}.$$

The two-ψ source is generated by traces $\mathrm{Tr}(\psi u \psi v)$ with u and v ordinary words whose total bidegree is $(k - 1, l - 1)$; if $k = 0$ or $l = 0$ this source is zero. Since ψ is odd, exchanging the two displayed ψ's changes sign. Applying $Q\psi = xy - yx$ gives

$$\begin{aligned}\partial_2(u, v) &= Q\,\mathrm{Tr}(\psi u \psi v)\\ &= \mathrm{Tr}\big(\psi vxyu - \psi vyxu - \psi uxyv + \psi uyxv\big),\end{aligned}$$

so, after removing the common front ψ,

$$\partial_2(u,v) = vxyu - vyxu - uxyv + uyxv.$$

A direct substitution shows $\partial_1\partial_2 = 0$.

If $k = 0$ or $l = 0$, then $C_1(k,l)$ is spanned by x^k or y^l, $\partial_1 = 0$, and $C_2(k,l) = 0$. Hence the middle homology is already the line spanned by $\Psi_{k,l}$. Assume from now on that $k, l > 0$.

We compute the first homology of this finite complex. Form a graph $\Gamma_{k,l}$ whose vertices are the necklaces in $\mathsf{N}_{k+1,l+1}$ and whose oriented edges are the words $w \in \mathsf{W}_{k,l}$, with

$$w : [yxw]_{\mathrm{cyc}} \longrightarrow [xyw]_{\mathrm{cyc}}.$$

Then ∂_1 is the cellular boundary of this graph. For each pair (u,v) as above attach the square whose oriented boundary is $vxyu - vyxu - uxyv + uyxv$. Denote the resulting two-complex by $X_{k,l}$.

Equivalently, $X_{k,l}$ is the two-skeleton of the chip-moving state complex $Y_{k,l}$. The full complex $Y_{k,l}$ has an r-cell for r mutually chosen chip moves on a cycle with $l+1$ oriented edges and $k+1$ indistinguishable chips, modulo cyclic relabelling of the edge labels. A vertex is a cyclic distribution of the chips among the vertices of this cycle, hence a necklace in $\mathsf{N}_{k+1,l+1}$. A one-cell moves one chosen chip across one oriented edge; cutting the cycle at this moving edge records exactly a word $w \in \mathsf{W}_{k,l}$, and its two endpoints are $[yxw]_{\mathrm{cyc}}$ and $[xyw]_{\mathrm{cyc}}$. A two-cell records two chip moves performed independently, and its oriented boundary is precisely $vxyu - vyxu - uxyv + uyxv$. Therefore the cellular chain complex of $Y_{k,l}$ through degree two is exactly

$$C_2(k,l) \longrightarrow C_1(k,l) \longrightarrow C_0(k,l),$$

and the inclusion $X_{k,l} = Y_{k,l}^{(2)} \hookrightarrow Y_{k,l}$ induces an isomorphism on H_1, since cells of dimension at least three do not affect first homology.

The homology of $Y_{k,l}$ is the homology of the cycle. For completeness we recall the elementary Abel-map argument. Before quotienting by cyclic relabelling, send a chip configuration to its Abel coordinate, the sum of the chip positions in $\mathbb{R}/(l+1)\mathbb{Z}$, and extend linearly over each cell. In the universal cover of the target circle, a fibre is described by weakly ordered lifted chip positions with fixed total sum; it is a convex polytope, subdivided by the same chip-crossing cells. Hence the Abel map is a cellular homotopy equivalence to S^1. The finite cyclic relabelling of the y-gaps is equivariant for this Abel map: it rotates the target circle, and every stabilizer acts affinely on a convex fibre, whose quotient is again contractible. Thus the quotient target is still a circle and the quotient complex $Y_{k,l}$ has the homology of S^1. Consequently

$$H_1(X_{k,l};\mathbb{C}) \cong H_1(S^1;\mathbb{C}) \cong \mathbb{C}.$$

Lemma 0.2.44 identifies this representative as a cycle in the original trace complex. Its non-boundary property follows from the functional

$$\epsilon\big(\mathrm{Tr}(\psi w)\big) = 1, \qquad w \in \mathsf{W}_{k,l},$$

kills every displayed two-ψ boundary because each boundary has two positive and two negative summands, while

$$\epsilon(\Psi_{k,l}) = \binom{k+l}{k} \neq 0.$$

Thus the one-ψ homology in bidegree (k,l) is the one-dimensional line generated by $[\Psi_{k,l}]$. □

Remark 0.2.46. The script `scripts/check_one_psi_homology.py` constructs the finite complex $C_2(k,l) \to C_1(k,l) \to C_0(k,l)$ over $\mathbb{Q}$ and verifies $\mathbf{dim}(\ker)\partial_1/\mathrm{im}\,\partial_2 = 1$ for $0 \le k, l \le 5$. It also checks the open Hamiltonian action of Proposition 0.2.63 for exponents at most 3, and checks the closed coadjoint Taylor-dual formula and the principal-part realization of Proposition 0.2.67 for exponents at most 5. These checks verify the displayed finite models without adding assumptions.

Corollary 0.2.47 *(Boundary pairing for cotangent generators).* Let $\delta_{k,l} \in \mathfrak{h}^\vee$ be the functional dual to the monomial $z_1^k z_2^l \in \bar{A}$. The assignment

$$\delta_{k,l}[1] \longmapsto [\Psi_{k,l}]$$

extends to an isomorphism from the shifted Taylor-dual space $\mathfrak{h}^\vee[1]$ to the nonconstant primitive one-ψ brane homology.

Proof. The monomials $z_1^k z_2^l$ with $k + l > 0$ form the polynomial basis of $\bar{A}$ and the topological basis of $\widehat{\mathfrak{h}}$; the $\delta_{k,l}$ form the continuous coefficient-dual basis. By Proposition 0.2.45, the primitive one-ψ homology has one generator in each bidegree, namely $[\Psi_{k,l}]$. Discarding the scalar bidegree $(0,0)$ and sending the dual basis vector in every nonconstant bidegree to that generator gives a filtered vector-space isomorphism. This is the boundary pairing used in the PBW special fibre; the closed module structure remains coadjoint. □

Lemma 0.2.48 *(Adjacent swaps).* In the invariant reduced BV cohomology,

$$\mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v)$$

is Q-exact for all words u, v.

Proof. By cyclicity of the trace,

$$\begin{aligned} Q\,\mathrm{Tr}(\psi v u) &= \mathrm{Tr}([\phi_1, \phi_2] v u) \\ &= \mathrm{Tr}(\phi_1\phi_2 v u - \phi_2\phi_1 v u) \\ &= \mathrm{Tr}(u\phi_1\phi_2 v) - \mathrm{Tr}(u\phi_2\phi_1 v). \end{aligned}$$

□

Corollary 0.2.49 *(Matrix evaluation is commutative in cohomology).* If two words in ϕ_1, ϕ_2 have the same numbers of ϕ_1 and ϕ_2, then their traces define the same class in the reduced BV cohomology. Consequently $\partial_{\mathrm{bb},N}(f)$ is determined by the commutative polynomial $f \in \mathbb{C}[z_1, z_2]$ and is independent of the chosen ordered lift.

Proof. Adjacent transpositions generate all reorderings of a finite word. Each adjacent transposition of the adjacent letters ϕ_1, ϕ_2 changes the trace by a Q-exact term by Lemma 0.2.48; cyclicity moves the chosen adjacent pair into the displayed position of that lemma. Summing the resulting exact differences proves the claim. □

Proposition 0.2.50 *(Stalk central multiplication).* Let

$$A_N^{\mathrm{pt}} = (\mathcal{R}_N^{GL_N}, Q)$$

be the invariant reduced open BV algebra at a brane insertion point. For $f \in \mathbb{C}[z_1, z_2]$, set

$$B_f = \widetilde{\delta}_{\mathrm{bb},N}(f) = \sum_{k,l} c_{k,l} \mathrm{Tr}(\phi_1^k \phi_2^l).$$

Then $QB_f = 0$. Regard B_f as a Hochschild 0-cochain; its associated central operation is multiplication

$$L_{B_f} : A_N^{\mathrm{pt}} \to A_N^{\mathrm{pt}}, \qquad a \longmapsto B_f a.$$

This Hochschild 0-cochain is closed. The same statement holds for multiplication by a closed one-antifield class $\Psi_{k,l}$. After projection to the Hamiltonian quotient and stabilization in N, the degree-zero classes satisfy

$$\{B_f, B_g\}_{\mathrm{open}} = B_{\{f,g\}_{\bar{A}}}.$$

Proof. Since $Q\phi_1 = Q\phi_2 = 0$, every B_f is Q-closed. The algebra A_N^{pt} is graded-commutative, so the Hochschild differential of a 0-cochain b is

$$d_{\mathrm{Hoch}} b(a) = Qb\, a + ba - (-1)^{|b||a|} ab.$$

This vanishes for every closed graded-central element b. The classes $\Psi_{k,l}$ are closed by Lemma 0.2.44 and are central for the same graded-commutative reason. Ordered-lift independence for B_f is cohomological and is exactly Corollary 0.2.49. The Poisson identity is Theorem 0.2.81. □

Let $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C} \cdot 1$. The removed constant corresponds on the gauge side to the scalar $\mathrm{Tr}(1) = N$ and has zero Hamiltonian vector field.

PROPOSITION 0.2.51 *(Connected brane Hamiltonian sector).* In the stable large-N open BV model, the connected Hamiltonian descendant sector freely generated by the nonempty primitive ghost-zero traces and the nonconstant primitive one-ψ Koszul homology is

$$O^{\mathrm{cl}}_{\mathrm{brane},H} \cong \mathbf{S}\big(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]\big).$$

The class of $z_1^k z_2^l$ with $k + l > 0$ corresponds to $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$. The corresponding descendant basis vector corresponds to the symmetrised trace with one ψ and k, l copies of ϕ_1, ϕ_2.

Proof. Proposition 0.2.40 gives the free commutative algebra on cyclic words. Lemma 0.2.48 identifies any two cyclic words with the same numbers of ϕ_1 and ϕ_2, because adjacent transpositions generate all reorderings. Thus the connected ghost-zero cyclic words reduce to the monomials $\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}$ with $k + l > 0$. Proposition 0.2.45 computes the primitive one-ψ homology in each multidegree; the nonconstant generators are $\Psi_{k,l}$ with $k + l > 0$. Products of the connected ghost-zero primitive classes and these one-ψ primitive classes give the free graded-commutative multi-trace algebra in the Hamiltonian sector. By definition of the Hamiltonian sector, single traces containing two or more occurrences of ψ are excluded from this algebra. □

Remark 0.2.52. The word "sector" is literal. The calculation above presents the stable algebra generated by the primitive no-ψ Hamiltonians and the nonconstant primitive one-ψ homology generated by $\Psi_{k,l}$ for $k + l > 0$. The full invariant Koszul homology in higher ψ-count lies outside this sector, as does the decoupled scalar class $\mathrm{Tr}(\psi)$.

Problem 0.2.53 *(Full primitive Koszul homology).* Let

$$A_\psi = \mathbb{C}\langle x, y, \psi\rangle, \qquad |x| = |y| = 0, \quad |\psi| = 1,$$

with differential $d\psi = xy - yx$ and $dx = dy = 0$. Put

$$\mathrm{Cyc}(A_\psi) = A_\psi/[A_\psi, A_\psi]_{\mathrm{gr}}, \qquad [uv] = (-1)^{|u||v|}[vu].$$

Give x, y, ψ weights

$$\mathrm{wt}(x) = (1, 0), \qquad \mathrm{wt}(y) = (0, 1), \qquad \mathrm{wt}(\psi) = (1, 1).$$

For fixed (R, S) define

$$K^p_{R,S} = \mathbb{C}\{[w]_{\mathrm{cyc}} : \#\psi(w) = p,\ \#x(w) = R - p,\ \#y(w) = S - p\},$$

with $0 \le p \le \min(R, S)$. The differential is

$$\partial[w] = \sum_{i:a_i=\psi} (-1)^{\#\{\psi \text{ before } i\}}[a_1 \cdots a_{i-1} xy\, a_{i+1} \cdots a_n \\ - a_1 \cdots a_{i-1} yx\, a_{i+1} \cdots a_n]_{\mathrm{cyc}}.$$

Prove the primitive homology formula

$$H_p(K_{R,S}) \cong \begin{cases} \mathbb{C} \cdot [x^R y^S], & p = 0, \\ \mathbb{C} \cdot [\Psi_{R-1,S-1}], & p = 1,\ R, S \ge 1, \\ 0, & p \ge 2. \end{cases}$$

Equivalently, prove that the primitive stable Koszul homology is $\mathbb{C}[x, y] \oplus \mathbb{C}[x, y]\theta$ with $|\theta| = 1$ and $\mathrm{wt}(\theta) = (1, 1)$.

The required new ingredient is the all-p orientation statement identifying this graded cyclic complex with the cellular chains of the chip-moving necklace state complex, together with the Abel-map proof that the relevant state complex has the homology of S^1. Proposition 0.2.45 proves exactly the $p \le 2$ skeleton needed for the Hamiltonian sector. The complete theorem also requires the all-ψ stable trace presentation and an exact computational check of the finite complex.

0.2.8 Operators in the closed Hamiltonian sector

Let $\mathfrak{h} = \widehat{\mathfrak{h}} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ with the projected Poisson bracket, and let $\mathfrak{h}^\vee$ be its continuous Taylor dual. The cotangent field is a coadjoint module, so the local Hamiltonian Lie algebra is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1].$$

For $f, g \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^\vee$, the brackets are

$$[f, g] = \{f, g\}, \qquad [f, \eta_\lambda] = \eta_{\mathrm{ad}^\vee_f \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0,$$

where $(\mathrm{ad}^\vee_f \lambda)(g) = -\lambda(\{f, g\})$. On polynomial monomials in the dense subalgebra $\bar{A}$ this reads

$$[z_1^k z_2^l, z_1^m z_2^n] = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

where the bar means projection to $\bar{A}$; in particular the constant bracket of z_1 and z_2 is zero in the reduced Hamiltonian Lie algebra.

LEMMA 0.2.54 *(Shifted coadjoint signs).* Let η_λ denote the suspension of $\lambda \in \mathfrak{h}^\vee$ into $\mathfrak{h}^\vee[1]$. Since the Hamiltonians in $\mathfrak{h}$ are even,

$$[f, \eta_\lambda] = \eta_{\mathrm{ad}_f^\vee \lambda}, \qquad [\eta_\lambda, \eta_\mu] = 0.$$

Consequently the enveloping algebra relations involving two shifted coadjoint generators are odd-commutative:

$$\eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda = 0.$$

Proof. The semidirect product bracket before shifting is the ordinary coadjoint action of the even Lie algebra $\mathfrak{h}$ on its continuous dual. Suspending the module raises the degree of every λ by one. The mixed-bracket sign is $+1$ because $|f| = 0$. The shifted module is abelian as a Lie ideal, so its bracket with itself is zero; in the associative enveloping algebra this zero Lie bracket gives $\eta_\lambda \eta_\mu - (-1)^{1\cdot 1} \eta_\mu \eta_\lambda = 0$. □

The kinematic BRST operator is

$$Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}.$$

The same jet contraction as in Lemma 0.2.39, together with the Dolbeault Poincare lemma in the formal local model, leaves precisely the holomorphic Taylor coefficients. Thus the linear local coordinates are the functionals

$$\partial_{z_1}^k \partial_{z_2}^l \alpha|_0, \qquad \partial_{z_1}^k \partial_{z_2}^l \beta|_0.$$

With the coefficient-dual convention fixed above, these are represented by $(k!l!)^{-1} \partial_{z_1}^k \partial_{z_2}^l|_0$. Since zero-form components of α and β have negative field ghost numbers, the corresponding linear functions have positive ghost numbers.

PROPOSITION 0.2.55 *(Closed local operators are continuous CE cochains).* The classical local operators of the formal Hamiltonian cotangent BF sector are the continuous Chevalley-Eilenberg cochains

$$O_{\mathrm{grav}}^{\mathrm{cl}} \cong C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]),$$

with CE differential dual to the Lie brackets above.

Proof. Work at the formal insertion point. The jet algebra of the fields is the completed free graded-commutative algebra on the Taylor coefficients of α and β. The kinematic differential $Q_0 = d_{\mathbb{R}^2} + \bar{\partial}_{\mathbb{C}^2}$ contracts all positive real and anti-holomorphic jets: in the real directions the homotopy is $[d_{\mathbb{R}^2}, \iota_{\partial_{x_i}}] = \partial_{x_i}$, and in the anti-holomorphic directions it is the formal Dolbeault Poincare homotopy. The Q_0-cohomology of the jet algebra is therefore generated by the holomorphic Taylor coefficients alone.

The remaining α-coefficients are Taylor Hamiltonians modulo constants, hence elements of $\mathfrak{h} = \widehat{\bar{\mathfrak{h}}}$; degreewise polynomial coefficients form the dense subalgebra $\bar{A}$. The cotangent field β supplies the odd shifted continuous dual $\mathfrak{h}^\vee[1]$. Thus the linear space of residual local modes is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$.

The nonlinear BRST differential is determined by the BV Hamiltonian

$$S = \int \left\langle \beta, D\alpha + \frac{1}{2}\{\alpha, \alpha\} \right\rangle.$$

The linear part is exactly Q_0, already killed in cohomology. The quadratic term sends two Hamiltonian α-modes f, g to their reduced Poisson bracket $\{f, g\}$, and the action on the cotangent variable is the coadjoint action displayed above. The quadratic term has zero component with two cotangent inputs, so $[\eta_\lambda, \eta_\mu] = 0$. Hence the residual BRST differential on functions is precisely the derivation dual to the Lie bracket on $\mathfrak{g}$, which is the Chevalley-Eilenberg differential. Continuity records the completion over polynomial degree. □

0.2.9 Koszul Duality

Definition 0.2.56. For an augmented complete dg algebra B, its Koszul dual is

$$B^! = \mathrm{RHom}_B(\mathbb{C}, \mathbb{C})$$

[21]. We regard commutative Chevalley-Eilenberg cochains as E_1 algebras by restriction. All bar and cobar constructions below are taken in the category of complete filtered Tate vector spaces, with continuous maps. The completions are by symmetric or tensor length together with the coefficient topology coming from Taylor degree. For a Tate space V, the notation $V^\vee_{\mathrm{cont}}$ means the continuous dual with its dual Tate topology, $\widehat{\otimes}$ means the completed Tate tensor product, and

$$\widehat{\mathbf{S}}^n(V) = (V^{\widehat{\otimes} n})_{\mathbb{S}_n}.$$

Chevalley-Eilenberg chains below are direct sums in CE length with completed coefficients in each length; CE cochains, completed cobar algebras, and completed enveloping algebras are products in the corresponding length filtration.

Lemma 0.2.57 *(Tate CE chains, completed PBW, and Koszul duality).* Work in the symmetric monoidal category **TateVec** of complete filtered Tate $\mathbb{C}$-vector spaces:

$$V = \varprojlim_{w \in \mathbb{N}} V_{\leq w}, \qquad V_{\leq w} \text{ finite-dimensional}, \qquad F^w V = \mathbf{Ker}\big(V \to V_{\leq w-1}\big),$$

with $\widehat{\otimes}$ the completed tensor product $V \widehat{\otimes} W = \varprojlim_w (V_{\leq w} \otimes W_{\leq w})$ and $V^\vee_{\mathrm{cont}} = \varinjlim_w V^*_{\leq w}$ the continuous dual. Admissible quasi-isomorphisms are the filtered quasi-isomorphisms, i.e. chain maps inducing quasi-isomorphisms on every quotient $V/F^w V$ and on the associated graded.

Let $\mathfrak{g}$ be a Tate graded Lie algebra, conilpotent in the sense that the descending Lie powers $\mathfrak{g}^{(\geq m)} = [\mathfrak{g}, \mathfrak{g}^{(\geq m-1)}]$ satisfy $\bigcap_m \big(\mathfrak{g}^{(\geq m)} + F^w \mathfrak{g}\big) = F^w \mathfrak{g}$ for every w. Then:

(i) For $V = \prod_d V_d$ with all V_d finite-dimensional and $W = \bigoplus_d W_d$ with all W_d finite-dimensional,

$$(V \oplus W)^\vee_{\mathrm{cont}} = \bigoplus_d V_d^\vee \oplus \prod_d W_d^\vee,$$

and the dualization $V \mapsto V^\vee_{\mathrm{cont}}$ is exact and continuous in the filtration F^w.

(ii) The continuous Chevalley–Eilenberg cochains

$$C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \mathrm{Hom}_{\mathrm{cont}}\big(\widehat{\mathbf{S}}^c(\mathfrak{g}[1]), \mathbb{C}\big)$$

identify with $\widehat{\mathbf{S}}_{\mathrm{Tate}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big)$, and the coaugmentation filtration on the CE coalgebra $C^{\mathrm{CE}}_*(\mathfrak{g}) = \widehat{\mathbf{S}}^c(\mathfrak{g}[1])$ is conilpotent in **TateVec**: every element of any finite Tate window lies in a finite power of the coaugmentation coideal.

(iii) The continuous bar resolution of the augmentation $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) \to \mathbb{C}$ has dual cobar algebra $\Omega C^{\mathrm{CE}}_*(\mathfrak{g})$, and the canonical Lie bar–cobar map

$$\Omega C^{\mathrm{CE}}_*(\mathfrak{g}) \longrightarrow \widehat{U}(\mathfrak{g})$$

is an admissible (filtered) quasi-isomorphism in **TateVec**.

(iv) The PBW filtration on $\widehat{U}(\mathfrak{g})$ by tensor length is complete and separated. Its associated graded is the completed symmetric algebra

$$\mathrm{gr}_F^\bullet \widehat{U}(\mathfrak{g}) \cong \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g}).$$

Proof. Item (1). A continuous functional on $V = \prod V_d$ depends on finitely many coordinates: the topology on $\mathbb{C}$ is discrete, so the kernel of any continuous map contains some $F^w V$. Therefore $V^\vee_{\mathrm{cont}} = \bigoplus_d V_d^*$. Conversely, a continuous functional on the discrete sum $W = \bigoplus W_d$ is given by an arbitrary product of degreewise functionals, hence $W^\vee_{\mathrm{cont}} = \prod_d W_d^*$. Direct sum of these gives the displayed formula. Exactness of dualization in **TateVec** is preservation of finite limits; continuity in the Tate filtration is direct.

Item (2). The completed CE coalgebra $\widehat{\mathbf{S}}^c(\mathfrak{g}[1]) = \bigoplus_n \widehat{\mathbf{S}}^n(\mathfrak{g}[1])$ has coaugmentation coideal $\widehat{\mathbf{S}}^{\geq 1}(\mathfrak{g}[1])$. An element x in the Tate window $V_{\leq w}$ projects to $\mathfrak{g}_{\leq w}$, whose finite-dimensional symmetric powers are conilpotent because the Lie bracket on $\mathfrak{g}_{\leq w}$ has finite descending central series in any finite-dimensional approximation. Pass to the inverse limit using the Tate-conilpotency hypothesis on $\mathfrak{g}$: the descending powers $(\widehat{\mathbf{S}}^c)^{(\geq m)}$ have empty Tate intersection modulo every finite filtration step. The displayed equality identifies cochains as continuous functionals on the CE coalgebra, using item (1) windowwise.

Item (3). Filter the CE coalgebra by symmetric coalgebra length and by Tate window. Project to a fixed window $\mathfrak{g}_{\leq w}$ and bound symmetric/tensor length to a fixed n. The resulting finite-dimensional diagram is the ordinary Lie bar–cobar comparison [22]: the bar resolution computes $\mathrm{RHom}_{C^\bullet_{\mathrm{CE}}(\mathfrak{g}_{\leq w}, \leq n)}(\mathbb{C}, \mathbb{C})$, and the cobar algebra maps to the universal enveloping algebra of the same window. These finite comparisons are compatible with enlargement of the Tate window and increase of the length cap: the enlargement-of-window map is the dual of the surjection $\mathfrak{g} \to \mathfrak{g}_{\leq w}$, and Lie-algebra surjections induce maps of universal enveloping algebras. Passing to the Tate limit and completing in tensor length gives the asserted filtered quasi-isomorphism in **TateVec**. The PBW filtration is, windowwise, the standard PBW filtration on $U(\mathfrak{g}_{\leq w})$, hence preserved under inverse limits.

Item (4). Tensor length is a separated, descending filtration on $\widehat{U}(\mathfrak{g})$ because the Lie relations $xy - (-1)^{|x||y|} yx - [x, y] = 0$ have leading tensor-length term $xy - (-1)^{|x||y|} yx$ of length 2 and lower-order term $-[x, y]$ of length 1; so the associated graded retains only the symmetric (length-fixed) part. Completeness is clear from the Tate construction. The associated graded is therefore the completed symmetric algebra $\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{g})$. □

Remark 0.2.58. Conilpotency in item (2) is automatic for the Hamiltonian case used in 0.2.29: the reduced Hamiltonian Lie algebra $\mathfrak{h} = (z_1, z_2)\mathbb{C}[[z_1, z_2]]$ of the formal symplectic disk satisfies $\mathfrak{h}^{(\geq m)} \cap \mathfrak{h}_{\leq w} = 0$ for $m > w - 1$, since the bracket strictly lowers Taylor degree. The general hypothesis is named here so that the lemma applies to any pro-finite-dimensional Lie algebra of bounded-below filtration. The earlier explicit product/direct-sum Tate convention of Beilinson–Feigin–Mazur [20] is recovered as a special case: take $\mathfrak{g} = \left(\prod P_d\right) \oplus \left(\bigoplus Q_d\right)$ with the bracket bilinear in the displayed pieces.

LEMMA 0.2.59 *(Linear Poisson Schouten differential).* Let $\mathfrak{h}$ be a pro-finite-dimensional Lie algebra in **TateVec**. The completed symmetric algebra $\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})$ is naturally a Poisson algebra with Kirillov–Kostant–Souriau bracket $\{O_I, O_J\}_\partial = O_{[H_I, H_J]} = C^K_{IJ} O_K$ on a chosen homogeneous basis $\{H_I\}$ of $\mathfrak{h}$, where $O_I \in \mathfrak{h}^*$ is the linear coordinate. The associated Poisson bivector is the linear bivector

$$\pi_\partial = \tfrac{1}{2} C^K_{IJ} O_K \theta^I \wedge \theta^J, \qquad \theta^I = \partial/\partial O_I \in \mathrm{Der}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big),$$

and the Lichnerowicz/Schouten differential $d_\pi = [\pi_\partial, -]_S$ on

$$\mathrm{PV}\big(\widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\big) = \widehat{\mathbf{S}}_{\mathrm{Tate}}(\mathfrak{h})\widehat{\otimes}\widehat{\Lambda}(\theta^I)$$

acts on generators by

$$d_\pi \theta^K = -\tfrac{1}{2}\, C^K_{IJ}\, \theta^I \theta^J, \qquad d_\pi O_K = C^L_{KJ}\, O_L\, \theta^J.$$

Proof. Direct calculation. The Schouten Leibniz rule with $[\theta^I, O_J]_S = \delta^I_J$, $[\theta^I, \theta^J]_S = 0$, $[O_I, O_J]_S = 0$ gives

$$\begin{aligned}
[\pi_\partial, O_K]_S &= \tfrac{1}{2} C^L_{IJ}\, O_L\, [\theta^I \theta^J, O_K]_S \\
&= \tfrac{1}{2} C^L_{IJ}\, O_L\, \big(\delta^I_K \theta^J - \delta^J_K \theta^I\big) \\
&= \tfrac{1}{2}\big(C^L_{KJ} - C^L_{JK}\big) O_L \theta^J = C^L_{KJ} O_L \theta^J,
\end{aligned}$$

using the antisymmetry $C^L_{IJ} = -C^L_{JI}$ in the lower indices. For $[\pi_\partial, \theta^K]_S$, the Schouten bracket with θ^K uses $[\theta^I \theta^J, \theta^K]_S = 0$ on the bivector factor and $[O_L, \theta^K]_S = -\delta^K_L$; the result is $-\tfrac{1}{2} C^K_{IJ} \theta^I \theta^J$. □

Proposition 0.2.60 *(Koszul dual of continuous CE cochains).* Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ be the completed graded Lie algebra above, and let

$$B = C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\mathfrak{g}^\vee_{\mathrm{cont}}[-1]\big).$$

Then

$$B^! \simeq \widehat{U(\mathfrak{g})},$$

the completed graded universal enveloping algebra.

Proof. Put

$$H_d = \mathrm{Span}\{z_1^a z_2^b : a + b = d\}, \qquad \mathfrak{h} = \prod_{d \geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d \geq 1} H_d^\vee.$$

Thus

$$\mathfrak{g} = \left(\prod_{d \geq 1} H_d\right) \ltimes \left(\bigoplus_{d \geq 1} H_d^\vee\right)[1]$$

is a Tate Lie algebra. The bracket is degreewise finite:

$$\{H_p, H_q\} \subset H_{p+q-2},$$

with the degree-zero summand projected to zero. Thus the degree-r coefficient of $\{f, g\}$ depends only on the finitely many pairs $p + q = r + 2$. For the coadjoint action, if $f \in H_p$ and $\lambda \in H_q^\vee$, then

$$\mathrm{ad}^\vee_f\, \lambda \in H^\vee_{q-p+2}$$

when $q - p + 2 \geq 1$, and it is zero otherwise. Finite-support Taylor duals remain finite-support: for fixed $\lambda \in H_q^\vee$ only the Hamiltonian degrees $p \leq q + 1$ can act nontrivially, and for a finite direct-sum support in $\mathfrak{h}^\vee$ the union of those contributing p is finite. Thus the product/direct-sum coefficient-finiteness

hypotheses of Lemma 0.2.57 hold. The abelian bracket on $\mathfrak{h}^\vee[1]$ is trivial. The Taylor coefficient topology is independent of the PBW/order filtration used for the Rees algebra below.

Let

$$C_*^{\mathrm{CE}}(\mathfrak{g}) = \bigoplus_{n\geq 0} \widehat{\mathbf{S}}^n(\mathfrak{g}[1])$$

be the coefficient-completed, length-direct-sum Chevalley-Eilenberg coalgebra, with coderivation determined by the bracket of $\mathfrak{g}$. By definition

$$B = \mathrm{Hom}_{\mathrm{cont}}\big(C_*^{\mathrm{CE}}(\mathfrak{g}), \mathbb{C}\big).$$

The Chevalley-Eilenberg coalgebra is conilpotent for symmetric length. The augmentation $B \to \mathbb{C}$ has the continuous bar resolution in the category of complete filtered Tate spaces, and Lemma 0.2.57 identifies

$$\mathrm{RHom}_B(\mathbb{C}, \mathbb{C}) \simeq \widehat{\Omega}\, C_*^{\mathrm{CE}}(\mathfrak{g}),$$

the completed cobar algebra of the CE coalgebra. The completed Lie bar-cobar comparison in the lemma identifies this cobar algebra with the completed associative algebra generated by $\mathfrak{g}$ modulo the closed graded Lie relations; this is $\widehat{U(\mathfrak{g})}$.

Thus, for the generators $f, g \in \mathfrak{h}$ and $\eta_\lambda, \eta_\mu \in \mathfrak{h}^\vee[1]$, the completed enveloping algebra has relations

$$\begin{aligned} fg - gf &= \{f, g\}, \\ f\eta_\lambda - \eta_\lambda f &= \eta_{\mathrm{ad}_f^\vee \lambda}, \\ \eta_\lambda \eta_\mu + \eta_\mu \eta_\lambda &= 0. \end{aligned}$$

□

For comparison with the classical brane algebra introduce the completed PBW/order Rees enveloping algebra

$$U_\hbar(\mathfrak{g}) = \widehat{T}(\mathfrak{g})[[\hbar]] \Big/ \overline{\big\langle xy - (-1)^{|x||y|} yx - \hbar[x, y] \big\rangle},$$

where the closure is for tensor length and the Tate coefficient topology. This Rees parameter records word length in the enveloping algebra and is independent of Taylor degree. At $\hbar = 0$ one obtains the completed PBW special fibre

$$U_\hbar(\mathfrak{g})/(\hbar) \cong \widehat{\mathbf{S}}(\mathfrak{g}) = \widehat{\mathbf{S}}\big(\widehat{\mathfrak{h}} \oplus \mathfrak{h}^\vee[1]\big).$$

After choosing the monomial-dual Taylor basis $\mathfrak{h}^\vee \cong \bar{A}_{\mathrm{desc}}$, the degreewise polynomial subalgebra $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$ is the stable classical brane Hamiltonian descendant algebra of Proposition 0.2.51. Thus the degreewise polynomial part of the PBW associated graded of the closed Koszul dual is the stable classical brane Hamiltonian descendant algebra. The interacting closed Hamiltonian bracket deforms the closed special fibre to $\widehat{U(\mathfrak{g})}$; on the brane side the deformation checked here is the degree-zero Hamiltonian trace bracket below.

0.2.9.1 First-order corrections, order hbar one

Under the Koszul duality correspondence we have

$$z_1^k z_2^l \leftrightarrow \overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}.$$

The reduced Poisson bracket on the gravity side is:

$$\{z_1^k z_2^l,\ z_1^m z_2^n\}_{\bar{A}} = \overline{(kn - lm)\, z_1^{k+m-1} z_2^{l+n-1}}.$$

If an exponent in the displayed monomial is negative then the coefficient $kn - lm$ is already zero; if the displayed monomial is constant, its class in $\bar{A}$ is zero. The matching bracket on the open side is the canonical bracket induced by the first-order kinetic term $\int \mathrm{Tr}(\phi_1\, d\phi_2)$, with the sign convention $\iota_{X_f}\omega = -df$ fixed in the AKSZ remark above:

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j, \qquad \{(\phi_1)^i{}_j, (\phi_1)^k{}_l\} = 0 = \{(\phi_2)^i{}_j, (\phi_2)^k{}_l\}.$$

Lemma 0.2.61 *(Hamiltonian trace bracket).* The canonical matrix Poisson bracket descends to the invariant reduced BV cohomology and then to the Hamiltonian Lie quotient by constant trace classes.

Proof. For $\epsilon \in \mathfrak{gl}_N$, the component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[\phi_1, \phi_2])$ of the moment map generates the infinitesimal conjugation action, with the sign fixed by the convention for the bracket. Hence $\{F, \mu_\epsilon\} = 0$ for every invariant function F. If I_μ is the ideal generated by all μ_ϵ, then $\{F, \mu_\epsilon K\} = \mu_\epsilon\{F, K\}$ lies again in I_μ. Therefore the bracket of invariant functions descends to $(\mathcal{R}_N/I_\mu)^{GL_N}$, equivalently to the degree-zero reduced BV cohomology. The central line $\mathbb{C} \cdot \mathrm{Tr}(1)$ is central for this Lie bracket, so the Hamiltonian vector-space quotient by constants is compatible with the bracket. No associative quotient by the unit is taken. □

Proposition 0.2.62 *(First-order trace bracket).* In the stable connected Hamiltonian trace cohomology,

$$\{\overline{\mathrm{Tr}(\phi_1^k \phi_2^l)}, \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}\} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1} \phi_2^{l+n-1})}.$$

Proof. Write $F_{k,l} = \mathrm{Tr}(\phi_1^k \phi_2^l)$. The matrix Poisson bracket induced by

$$\{(\phi_1)^i{}_j, (\phi_2)^k{}_l\} = \delta^i_l \delta^k_j$$

is

$$\{F, G\} = \mathrm{Tr}\left(\frac{\partial F}{\partial \phi_1}\frac{\partial G}{\partial \phi_2} - \frac{\partial F}{\partial \phi_2}\frac{\partial G}{\partial \phi_1}\right),$$

where the derivatives are cyclic matrix derivatives. Varying one occurrence at a time gives

$$\frac{\partial F_{k,l}}{\partial \phi_1} = \sum_{a=0}^{k-1} \phi_1^{k-1-a} \phi_2^l \phi_1^a,$$
$$\frac{\partial F_{k,l}}{\partial \phi_2} = \sum_{b=0}^{l-1} \phi_2^{l-1-b} \phi_1^k \phi_2^b.$$

Therefore the first summand in the bracket is a sum of kn traces, each obtained by removing one ϕ_1 from the first word and one ϕ_2 from the second word. The second summand is a sum of lm traces obtained

by removing one ϕ_2 from the first word and one ϕ_1 from the second. Every trace appearing in either nonzero sum has multidegree $(k+m-1, l+n-1)$. Lemma 0.2.48 identifies all cyclic words with this multidegree in reduced BRST cohomology. The coefficient is therefore $kn - lm$, and the common class is $\overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}$. When the common word has multidegree $(0,0)$, this class is the scalar trace and is zero in the Hamiltonian quotient by constants. □

PROPOSITION 0.2.63 *(Open action on primitive descendants).* Let

$$\widetilde{\Psi}_{a,b} = \binom{a+b}{a}^{-1} \Psi_{a,b}$$

be the averaged primitive one-ψ class. In the stable primitive one-ψ homology,

$$\{\overline{\mathrm{Tr}(\phi_1^p \phi_2^q)}, \widetilde{\Psi}_{a,b}\} = (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

with the convention that a term with a negative exponent is zero. Thus the unreduced polynomial descendant classes carry the adjoint polynomial Hamiltonian action. The Hamiltonian descendant sector is the quotient by the scalar line $\mathbb{C}\widetilde{\Psi}_{0,0}$, so the displayed action is projected further by killing the scalar target.

Proof. Put $x = \phi_1$ and $y = \phi_2$. The cyclic derivatives of $F_{p,q} = \mathrm{Tr}(x^p y^q)$ are

$$\frac{\partial F_{p,q}}{\partial x} = \sum_{i=0}^{p-1} x^{p-1-i} y^q x^i, \qquad \frac{\partial F_{p,q}}{\partial y} = \sum_{j=0}^{q-1} y^{q-1-j} x^p y^j.$$

For a one-ψ cyclic word $\mathrm{Tr}(\psi w)$ of bidegree (a,b), differentiating with respect to y gives one summand for each of the b occurrences of y in w, and differentiating with respect to x gives one summand for each of the a occurrences of x. Multiplying by the displayed cyclic derivatives of $F_{p,q}$ therefore produces pb positive one-ψ words and qa negative one-ψ words for each $w \in \mathsf{W}_{a,b}$, all of total bidegree $(a+p-1, b+q-1)$.

The bracket is a cycle because $F_{p,q}$ is invariant, hence $\{F_{p,q}, \mu_\epsilon\} = 0$ for every moment-map component $\mu_\epsilon = -\mathrm{Tr}(\epsilon[x,y])$. Equivalently, the Hamiltonian vector field of $F_{p,q}$ preserves the Koszul differential $Q\psi = [x,y]$ and acts on Koszul homology. By Proposition 0.2.45, the target primitive one-ψ homology in that bidegree is one-dimensional. The functional $\epsilon(\mathrm{Tr}(\psi w)) = 1$ detects its coefficient. Since $\epsilon(\widetilde{\Psi}_{a,b}) = 1$, the averaged input gives

$$\epsilon\big(\{F_{p,q}, \widetilde{\Psi}_{a,b}\}\big) = pb - qa.$$

Also $\epsilon(\widetilde{\Psi}_{a+p-1,b+q-1}) = 1$. Hence the displayed formula follows. If the target bidegree is $(0,0)$, the same formula is still the unreduced primitive one-ψ action; the Hamiltonian descendant quotient sends that scalar class to zero. □

COROLLARY 0.2.64 *(Descendant action and coadjoint action: structural separation).* Under the boundary pairing $\delta_{a,b}[1] \mapsto [\Psi_{a,b}]$ of Corollary 0.2.47, the open Hamiltonian descendant action on the polynomial $\Psi_{a,b}$-basis is the symmetric raising action

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1},$$

while the closed coadjoint action on the principal-part basis $\rho_{a,b} \leftrightarrow \delta_{a,b} \in \mathfrak{h}^\vee_{\mathrm{cont}}$ is the lowering action

$$\mathrm{ad}^\vee_{z_1^p z_2^q}\, \delta_{a,b} = -(pb - qa + p - q)\, \delta_{a-p+1,b-q+1},$$

with negative-index targets zero. The two actions agree only on the trivial Hamiltonian $p = q = 0$. The deformation quantisation of the cotangent factor sees the second action; the special-fibre PBW symmetric algebra sees the first. The structural reason is the PBW versus deformation interpolation of Theorem 0.2.72 below: the $\widetilde{\Psi}_{a,b}$ and the $\rho_{a,b}$ are two different bases of the same $\mathfrak{g}$-module, the special-fibre PBW representative and the deformation-compatible coadjoint representative respectively.

Proof. Proposition 0.2.63 gives the symmetric raising formula on $\widetilde{\Psi}_{a,b}$. The coadjoint action is dual to the Poisson bracket: for $f = z_1^p z_2^q$ and the Taylor-dual basis vector $\delta_{a,b}$,

$$(\mathrm{ad}^\vee_f\, \delta_{a,b})(z_1^m z_2^n) = -\delta_{a,b}\big((pn - qm) z_1^{p+m-1} z_2^{q+n-1}\big),$$

giving the displayed lowering formula. For example, z_1 acts on open descendants by $\widetilde{\Psi}_{a,b} \mapsto b\, \widetilde{\Psi}_{a,b-1}$, whereas it acts coadjointly by $\delta_{a,b} \mapsto -(b+1)\, \delta_{a,b+1}$. The scalar edge $\{\, \mathrm{Tr}(\phi_1), \widetilde{\Psi}_{0,1} \,\} = \Psi_{0,0}$ is removed with the scalar descendant. The two formulas have opposite index shift directions and differ by $-(pb - qa + p - q)\rho_{a-p+1,b-q+1} - (pb - qa)\rho_{a+p-1,b+q-1}$ pulled back through the basis identification $\rho \leftrightarrow \delta \leftrightarrow \Psi$, which is non-zero unless both targets vanish identically; this happens exactly on the trivial $p = q = 0$ action. The corollary uses Corollary 0.2.47 for the PBW special-fibre product identification, and Theorem 0.2.70 for the coadjoint half of the comparison. The descendant deformation match is supplied by the Laurent / distributional principal-part realization of Theorem 0.2.67 together with the interpolation of Theorem 0.2.72. □

The reader who has come this far has met three different things called "local" on the brane line $\mathbb{R}$, and we keep them apart once and for all before identifying the coadjoint module on the principal-part side.

Lemma 0.2.65 *(Three notions of locality).* On the brane line $\mathbb{R}$ with formal holomorphic transverse coordinates z_1, z_2 at the brane point $p \in \mathbb{C}^2$, the following three notions of locality are independent.

(i) *Support locality* — distributional support on the brane: a section $T \otimes m$ of $\mathcal{D}'_c(I) \widehat{\otimes} M$ has support locality whenever $\mathrm{supp}\, T \subset I \subset \mathbb{R}$.

(ii) *Topological locality* — locally constant in the $\mathbb{R}$-direction up to Q-exact translations: every section is identified with its translate by a parameter on $\mathbb{R}$ modulo the open BV differential Q_{open}.

(iii) *Formal holomorphic locality* — polynomial, or Taylor-series, dependence on the transverse coordinates z_1, z_2.

The polynomial trace observables $O_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt$ have all three. The shifted cotangent currents $\Theta_\rho(B)$ of Construction 0.2.76 have (1) and (2) but not (3): they are momentum modes attached to the brane and live in the principal-part complement $\mathrm{Ann}(\mathbb{C} \cdot 1)$ of the polynomial Taylor algebra.

Proof. Independence is a category-theoretic statement: a polynomial in z_1, z_2 with point support is constant in $\mathbb{R}$, so (1) does not imply (3); a delta-current at one point of $\mathbb{R}$ has compact support but only trivial Q-exact translation, so (1) does not imply (2). The cotangent currents $\Theta_\rho(B)$ are smooth or distributional sections of the line $\mathbb{R}$ (hence (1)) and translation-invariant up to Q_{open}-exact cocycles (hence (2)); their transverse content $\rho \in \mathcal{P}$ is a Laurent two-form supported at the principal-part divisor, that is, in the complement of the polynomial algebra in the Laurent algebra (hence not (3)). □

Remark 0.2.66. In canonical quantization of a constrained system, the configuration variables are locally polynomial in spatial coordinates and the conjugate momenta are dual distributions. This is exactly the separation Lemma 0.2.65 records: configuration observables on a brane have formal holomorphic locality (3), and momentum observables have only (1) and (2). The principal-part sector is the algebraic image of this separation in the BCOV Hamiltonian model.

PROPOSITION 0.2.67 *(Principal-part realization of the coadjoint module).* Let

$$\mathcal{P} = \bigoplus_{\substack{a,b\geq 0\\ a+b>0}} \mathbb{C}\,\rho_{a,b}, \qquad \rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2,$$

with residue pairing

$$\langle \rho_{a,b}, z_1^m z_2^n \rangle = \mathrm{Res}_{z_1=z_2=0}\, z_1^{m-a-1} z_2^{n-b-1}\, dz_1 dz_2 = \delta_{a,m}\delta_{b,n}.$$

Let π_{pp} denote projection of Laurent two-forms to the span of the principal parts $\rho_{a,b}$. The action

$$f \cdot \rho = \pi_{\mathrm{pp}}\big(\{f, \rho\}\big)$$

identifies $\mathcal{P}$ with the continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}}$. On monomials,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\rho_{a-p+1,b-q+1},$$

with the convention that a term with a negative index is zero.

Proof. The residue pairing identifies $\rho_{a,b}$ with the Taylor-dual functional $\delta_{a,b}$. Hamiltonian vector fields preserve the holomorphic volume form $dz_1 dz_2$, so integration by parts for the residue pairing gives

$$\langle \{f, \rho\}, g\rangle = -\langle \rho, \{f, g\}\rangle.$$

Hence the Laurent principal-part action realizes the coadjoint action. Directly, for $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2$,

$$\begin{aligned}\{f, z_1^{-a-1} z_2^{-b-1}\} &= p z_1^{p-1} z_2^q\,(-(b+1)) z_1^{-a-1} z_2^{-b-2}\\ &\quad - q z_1^p z_2^{q-1}\,(-(a+1)) z_1^{-a-2} z_2^{-b-1}\\ &= -(pb - qa + p - q) z_1^{p-a-2} z_2^{q-b-2}.\end{aligned}$$

Since $z_1^{p-a-2} z_2^{q-b-2}\, dz_1 dz_2 = \rho_{a-p+1,b-q+1}$ when both indices are nonnegative, principal-part projection gives the displayed formula. □

The existence of the principal-part identification just established is strengthened by two rigidity statements: the residue pairing is canonical up to a single nonzero scalar, and the resulting coadjoint isomorphism $\mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ is the unique continuous $\mathfrak{h}$-equivariant isomorphism modulo that scalar.

THEOREM 0.2.68 *(Canonical residue pairing).* Up to a nonzero scalar, the residue pairing

$$\langle\,\cdot\,,\,\cdot\,\rangle : \mathcal{P} \times \mathbb{C}[[z_1, z_2]] \longrightarrow \mathbb{C}, \qquad \langle \rho_{a,b}, z_1^m z_2^n\rangle = \delta_{a,m}\delta_{b,n},$$

is the unique non-degenerate continuous $\mathfrak{h}$-equivariant pairing graded of total degree zero with the property that $\rho_{0,0}$ pairs to zero with the constant $1 \in \mathbb{C}[[z_1, z_2]]$.

Proof. Existence. The residue pairing is non-degenerate by the displayed Kronecker relation; it is continuous because $\mathcal{P}$ is graded discrete and $\mathbb{C}[[z_1, z_2]]$ carries the Taylor filtration; equivariance is integration by parts for the Poisson bracket, since Hamiltonian vector fields preserve the holomorphic two-form $dz_1 dz_2$:

$$\langle \{f, \rho\}, g\rangle = \operatorname{Res}(\{f, \rho\} g) = -\operatorname{Res}(\rho\, \{f, g\}) = -\langle \rho, \{f, g\}\rangle.$$

Uniqueness. Let $\beta : \mathcal{P} \times \mathbb{C}[[z_1, z_2]] \to \mathbb{C}$ be a second pairing satisfying the same conditions, and write $\beta_{a,b;m,n} = \beta(\rho_{a,b}, z_1^m z_2^n)$. Since β is graded of total degree zero, $\beta_{a,b;m,n} = 0$ unless $a + b = m + n$.

Diagonal weight constraint. The Hamiltonian $z_1 z_2$ acts diagonally on monomials by $\{z_1 z_2, z_1^m z_2^n\} = (n - m) z_1^m z_2^n$. Direct Poisson computation on principal parts uses the Hamiltonian vector field $X_{z_1 z_2} = z_2\, \partial_{z_2} - z_1\, \partial_{z_1}$ (with $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}$), giving $X_{z_1 z_2}(z_1^{-a-1} z_2^{-b-1}) = (a - b)\, z_1^{-a-1} z_2^{-b-1}$, so $\{z_1 z_2, \rho_{a,b}\} = (a - b)\rho_{a,b} = -(b - a)\rho_{a,b}$. Equivariance $\beta(\{z_1 z_2, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1 z_2, z_1^m z_2^n\})$ gives $-(b - a)\beta_{a,b;m,n} = -(n - m)\beta_{a,b;m,n}$, hence $\beta_{a,b;m,n} = 0$ unless $a - b = m - n$. Combined with the weight constraint $a + b = m + n$, this gives $\beta_{a,b;m,n} = 0$ unless $(a, b) = (m, n)$. So β is diagonal in the residue basis: $\beta_{a,b;m,n} = c_{a,b}\, \delta_{a,m} \delta_{b,n}$ for some scalars $c_{a,b} \in \mathbb{C}$.

Coefficients are constant. The Hamiltonian generator $f = z_1$ has $X_{z_1} = \partial_{z_2}$, so $\{z_1, z_1^m z_2^n\} = n\, z_1^m z_2^{n-1}$ and a direct Poisson computation gives $\{z_1, \rho_{a,b}\} = -(b + 1)\rho_{a,b+1}$. Equivariance reads $\beta(\{z_1, \rho_{a,b}\}, z_1^m z_2^n) = -\beta(\rho_{a,b}, \{z_1, z_1^m z_2^n\})$, i.e., $-(b + 1)\beta_{a,b+1;m,n} = -n\, \beta_{a,b;m,n-1}$. Both sides vanish identically unless the diagonality condition is met on each pair. The right-hand side is non-zero only if $(a, b) = (m, n - 1)$, i.e., $m = a$, $n = b + 1$. The left-hand side is non-zero only if $(a, b + 1) = (m, n)$, i.e., $m = a$, $n = b + 1$. Both indices coincide, and we obtain $(b + 1)\, c_{a,b+1} = (b + 1)\, c_{a,b}$, hence $c_{a,b+1} = c_{a,b}$ for all $b \geq 0$ with $a + b + 1 \geq 1$. Symmetrically from $f = z_2$ (using $X_{z_2} = -\partial_{z_1}$, so $\{z_2, z_1^m z_2^n\} = -m\, z_1^{m-1} z_2^n$ and $\{z_2, \rho_{a,b}\} = (a + 1)\rho_{a+1,b}$) we get $c_{a+1,b} = c_{a,b}$ for all $a \geq 0$. Connecting these chains, all $c_{a,b}$ for $a + b \geq 1$ equal a single constant $c \in \mathbb{C}$. Non-degeneracy forces $c \neq 0$.

Therefore $\beta = c\, \langle \cdot\,, \cdot \rangle$ is the residue pairing scaled by c. The condition $\beta(\rho_{0,0}, 1) = 0$ holds vacuously since $\rho_{0,0} \notin \mathcal{P}$. □

Remark 0.2.69. Theorem 0.2.68 is a Schur-type rigidity statement on the cofree cocommutative coalgebra $\mathbb{C}[[z_1, z_2]]$ with the Hamiltonian Lie algebra $\mathfrak{h}$ acting by derivations. The residue pairing is the only $\mathfrak{h}$-equivariant pairing whose target is the trivial $\mathfrak{h}$-module $\mathbb{C}$ in graded degree zero.

Theorem 0.2.70 *(Principal-part coadjoint isomorphism, supremum form).* The residue pairing of Theorem 0.2.68 induces a canonical isomorphism of $\mathfrak{h}$-modules

$$\Phi_{\mathrm{coad}} : \mathcal{P} \xrightarrow{\sim} \mathfrak{h}^\vee_{\mathrm{cont}}, \qquad \rho_{a,b} \longmapsto \delta_{a,b},$$

characterised uniquely (up to the residue scalar c of Theorem 0.2.68) by the property that for every $f \in \mathfrak{h}$ and every $g \in \mathfrak{h}$,

$$\langle \Phi^{-1}_{\mathrm{coad}}(\mathrm{ad}^\vee_f\, \delta), g\rangle = -\langle \Phi^{-1}_{\mathrm{coad}}(\delta), \{f, g\}\rangle.$$

In monomial form,

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1},$$

with the convention that any term with a negative index is zero. The isomorphism is continuous, of total degree zero, and the unique continuous $\mathfrak{h}$-module isomorphism $\mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ modulo scalar (in particular, modulo the choice of c).

Proof. Existence on monomials. Let $f = z_1^p z_2^q$ and $\rho_{a,b} = z_1^{-a-1} z_2^{-b-1}\, dz_1 dz_2$. Since $\{z_1, z_2\} = 1$,

$$\{f, z_1^{-a-1} z_2^{-b-1}\} = p\, z_1^{p-1} z_2^q \cdot (-(b+1)) z_1^{-a-1} z_2^{-b-2} - q\, z_1^p z_2^{q-1} \cdot (-(a+1)) z_1^{-a-2} z_2^{-b-1},$$

which simplifies to $-(pb - qa + p - q)\, z_1^{p-a-2} z_2^{q-b-2}$. Multiplying by $dz_1 dz_2$ identifies this with $\rho_{a-p+1,b-q+1}$ when both indices are ≥ 0; otherwise the product lies outside $\mathcal{P}$ and the principal-part projection π_{pp} of Proposition 0.2.67 gives zero. The integration by parts formula $\langle \{f, \rho\}, g\rangle = -\langle \rho, \{f, g\}\rangle$ identifies this with the coadjoint action.

Continuity. The dual filtration on $\mathfrak{h}^\vee_{\mathrm{cont}}$ has $F^N \mathfrak{h}^\vee_{\mathrm{cont}}$ equal to the functionals vanishing on $\mathfrak{m}^{N+1} \subset \mathfrak{h}$, where $\mathfrak{m}$ is the maximal ideal. The grading of $\mathcal{P}$ by total weight $a + b$ corresponds under Φ_{coad} to the dual filtration: $F^N = \bigoplus_{a+b \leq N} \mathbb{C}\, \delta_{a,b}$ is the Φ_{coad}-image of $\bigoplus_{a+b\leq N} \mathbb{C}\, \rho_{a,b}$.

Verification on three test cases. With the convention $\{z_1, z_2\} = 1$, the Hamiltonian vector field is $X_f = (\partial_{z_1} f)\partial_{z_2} - (\partial_{z_2} f)\partial_{z_1}$.

(i) $(p, q, a, b) = (1, 0, 0, 1)$: the formula gives $z_1 \cdot \rho_{0,1} = -(1 \cdot 1 - 0 \cdot 0 + 1 - 0)\rho_{0-1+1,\,1-0+1} = -2\rho_{0,2}$. Direct: $X_{z_1} = \partial_{z_2}$, so $X_{z_1}(z_1^{-1} z_2^{-2}) = -2 z_1^{-1} z_2^{-3}$, and multiplying by $dz_1 dz_2$ gives $-2\rho_{0,2}$. Match.

(ii) $(p, q, a, b) = (0, 1, 1, 0)$: the formula gives $z_2 \cdot \rho_{1,0} = -(0 \cdot 0 - 1 \cdot 1 + 0 - 1)\rho_{1-0+1,\,0-1+1} = 2\rho_{2,0}$. Direct: $X_{z_2} = -\partial_{z_1}$, so $X_{z_2}(z_1^{-2} z_2^{-1}) = 2 z_1^{-3} z_2^{-1}$, multiplied by $dz_1 dz_2$ gives $2\rho_{2,0}$. Match.

(iii) $(p, q, a, b) = (2, 1, 1, 1)$: the formula gives $z_1^2 z_2 \cdot \rho_{1,1} = -(2 \cdot 1 - 1 \cdot 1 + 2 - 1)\rho_{0,1} = -2\rho_{0,1}$. Direct: $X_{z_1^2 z_2} = 2 z_1 z_2\, \partial_{z_2} - z_1^2\, \partial_{z_1}$; $X_{z_1^2 z_2}(z_1^{-2} z_2^{-2}) = -4 z_1^{-1} z_2^{-2} + 2 z_1^{-1} z_2^{-2} = -2 z_1^{-1} z_2^{-2}$, multiplied by $dz_1 dz_2$ gives $-2\rho_{0,1}$. Match.

These three cases are confirmed numerically as part of the 1225-case check in `scripts/check_one_psi_homology.py` (which verifies the formula for $0 \leq p, q, a, b \leq 5$, $p + q > 0$, $a + b > 0$).

Uniqueness. Suppose $\Phi' : \mathcal{P} \to \mathfrak{h}^\vee_{\mathrm{cont}}$ is another continuous $\mathfrak{h}$-equivariant isomorphism of total degree zero. Then $\Phi_{\mathrm{coad}}^{-1} \circ \Phi'$ is a continuous $\mathfrak{h}$-equivariant endomorphism of $\mathcal{P}$ of total degree zero. The action of $\mathfrak{h}$ on $\mathcal{P}$ has dense orbit through any $\rho_{a,b}$: every $\rho_{c,d}$ is reached from $\rho_{a,b}$ by repeated application of z_1^p and z_2^q with p, q adjusted (one may need to pass through ρ's with one or both indices zero, which is permissible because in those cases the coefficients $-(pb - qa + p - q)$ are nonzero whenever the resulting target index is non-negative). Equivariance then forces $\Phi_{\mathrm{coad}}^{-1} \circ \Phi' = c'\,\mathrm{id}$ for a scalar $c' \in \mathbb{C}^\times$. The scalar is the residue scalar c of Theorem 0.2.68. □

Remark 0.2.71 *(Sign convention).* The sign $-(pb - qa + p - q)$ on the principal-part side and the sign $(pb - qa)$ on the polynomial $\Psi_{a,b}$ side of Proposition 0.2.63 are both correct in their own conventions; they differ because the $\Psi_{a,b}$ are special-fibre PBW representatives and the $\rho_{a,b}$ are deformation-compatible coadjoint representatives. See Theorem 0.2.72 below for the precise interpolation.

THEOREM 0.2.72 *(PBW versus deformation interpolation).* Let $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1]$ be the semidirect Lie algebra of Proposition 0.2.13. The polynomial one-ψ basis $\Psi_{a,b} \in \bar{A}_{\mathrm{desc}}$ and the principal-part basis $\rho_{a,b} \in \mathcal{P} \cong \mathfrak{h}^\vee_{\mathrm{cont}}$ are related as follows.

(i) *Special-fibre PBW basis.* The $\widetilde{\Psi}_{a,b}$ form a PBW basis of the special-fibre symmetric algebra $\mathbf{S}(\mathfrak{h} \oplus \mathfrak{h}^\vee[1])\big|_{\hbar=0}$, and the open Hamiltonian Lie action is *symmetric raising* on this basis:

$$F_{p,q} : \widetilde{\Psi}_{a,b} \longmapsto (pb - qa)\, \widetilde{\Psi}_{a+p-1,b+q-1}.$$

The shift $(a + p - 1, b + q - 1)$ is positive in a and b as p, q grow; the index moves *up* in total weight.

(ii) *Deformation-level coadjoint basis.* The $\rho_{a,b}$ form a basis of the deformation-compatible continuous coadjoint module $\mathfrak{h}^\vee_{\mathrm{cont}} \cong \mathcal{P}$ of Theorem 0.2.70, and the $\mathfrak{h}$-action is *coadjoint lowering* on this basis:

$$z_1^p z_2^q \cdot \rho_{a,b} = -(pb - qa + p - q)\, \rho_{a-p+1,b-q+1}.$$

The shift $(a - p + 1, b - q + 1)$ has a and b moving *down* as p, q grow; this is the coadjoint sign and direction.

(iii) *Interpolation.* The Rees specialisation of the formal Moyal/Weyl deformation quantisation of the constant Poisson formal disk (Section 0.1.13, with formal parameter $\hbar$) interpolates between (1) and (2) as follows. The PBW special-fibre map $\widetilde{\Psi}_{a,b} \mapsto \delta_{a,b}[1]$ is an isomorphism of graded vector spaces, but is *not* an isomorphism of $\mathfrak{h}$-modules: the pullback of the coadjoint $\mathfrak{h}$-action through this isomorphism does not equal the symmetric raising action displayed in (1), by exactly the difference $-(pb - qa + p - q)\rho_{a-p+1,b-q+1}$ versus $(pb - qa)\rho_{a+p-1,b+q-1}$. The Rees lift recovers the coadjoint action only after passing to the deformation-quantised module $\mathcal{P}$, which sits at all orders of $\hbar$, not at $\hbar = 0$ alone.

Consequently the two formulas are not contradictory; they are the two sides of the same module, taken with respect to two different bases. The structural separation recorded in Corollary 0.2.64 is forced and expected.

Proof. (1) is Proposition 0.2.63, with the rephrasing "raising" for the index shift $(a + p - 1, b + q - 1)$. (2) is Theorem 0.2.70. For (3), we observe that the symmetric raising action on $\widetilde{\Psi}_{a,b}$ generates an $\mathfrak{h}$-submodule of $\bar{A}_{\mathrm{desc}}$ that is *free* as a graded $\mathbb{C}[\widetilde{\Psi}_{0,0}]$-module after stabilisation; this is the PBW property. The coadjoint action on $\rho_{a,b}$ is by lowering operators, hence is *cofree* as a graded comodule. The Rees-deformation $A_\hbar = (\mathbb{C}[[z_1, z_2]][[\hbar]], *)$ has the property that the formal Moyal product $f * g = fg + \frac{\hbar}{2}\{f, g\} + O(\hbar^2)$ acts on its dual by the coadjoint formula at *all* orders of $\hbar$: the dual representation of the Moyal Lie algebra $(\mathfrak{h}_\hbar, [-,-]_{\mathfrak{h}_\hbar})$ on $\mathfrak{h}^\vee_\hbar$ is the residue pairing extended $\mathbb{C}[[\hbar]]$-linearly. Specialising to $\hbar = 0$ recovers the Poisson coadjoint action of $\mathfrak{h} = \bar{A}$ on $\mathfrak{h}^\vee = \mathcal{P}$. The PBW filtration is the Rees filtration, and the special-fibre $\hbar = 0$ symmetric algebra is the associated graded; the symmetric raising action on $\widetilde{\Psi}_{a,b}$ is exactly the $\hbar = 0$ *associated graded* image of the polynomial multiplication piece of the Moyal product, not the adjoint or coadjoint action of the Lie algebra. The two actions become identical only after one identifies them on the symmetric algebra at $\hbar = 0$, where they diverge by the explicit formula above. The interpolation parameter is $\hbar$. □

Proposition 0.2.73 *(No polynomial boundary realization of the principal parts).* In the present polynomial open BV/Koszul matrix model, no family of ordinary polynomial boundary local cohomology classes

$$\Theta_{a,b}, \qquad a, b \geq 0, \quad a + b > 0,$$

can realize the principal-part coadjoint module of Proposition 0.2.67.

Proof. The linear Hamiltonians

$$O_{1,0} = \overline{\mathrm{Tr}(\phi_1)}, \qquad O_{0,1} = \overline{\mathrm{Tr}(\phi_2)}$$

act on polynomial boundary observables by the constant Hamiltonian vector fields ∂_{ϕ_2} and $-\partial_{\phi_1}$, respectively. On every ordinary polynomial observable these operators are locally nilpotent: after enough iterations the polynomial degree in the differentiated variable is exhausted.

On the principal-part coadjoint module, however, Proposition 0.2.67 gives

$$O_{1,0} \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}, \qquad O_{0,1} \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}.$$

Neither operator is locally nilpotent on any nonzero principal-part generator. Local nilpotence is preserved under isomorphism of modules. Therefore the desired $\Theta_{a,b}$ cannot be ordinary polynomial boundary observables in the current open BV/Koszul model. □

The proposition rules out one specific polynomial realization. The sharper statement, which we record next, is categorical: *no* polynomial sub-bimodule of any boundary observable algebra carrying a locally nilpotent z_i-action can carry the coadjoint action $\mathcal{P}$.

THEOREM 0.2.74 *(Categorical no-go: principal parts have no polynomial realisation).* Let $\mathsf{P}_{\mathrm{pol}}$ denote the abelian category of $\mathbb{Z}$-graded bimodules $M = \bigoplus_{n\geq 0} M_n$ over the polynomial Hamiltonian Lie algebra $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}\cdot 1$, on which:

(i) every element acts as a Hamiltonian vector field;

(ii) every linear Hamiltonian acts locally nilpotently — that is, for every $m \in M$ and every $i \in \{1, 2\}$, there exists N with $z_i^N \cdot m = 0$.

This category models polynomial boundary observables of the open BV theory with respect to the polynomial sub-bimodule structure preserved by all morphisms of the boundary algebra into itself. Then $\mathcal{P} \notin \mathrm{Ob}(\mathsf{P}_{\mathrm{pol}})$. More precisely, every nonzero submodule $M \subset \mathcal{P}$ in the larger category of all graded Hamiltonian Lie modules contains a generator $\rho_{a,b}$ on which both z_1 and z_2 act non-locally-nilpotently, hence M violates condition (2). Consequently

$$\mathrm{Hom}_{\text{Lie-Mod}_{\bar{A}}}(\mathcal{P}, M) = 0 \quad \text{for every} \quad M \in \mathrm{Ob}(\mathsf{P}_{\mathrm{pol}}),$$

and the inclusion $\mathsf{P}_{\mathrm{pol}} \hookrightarrow \text{Lie-Mod}_{\bar{A}}$ has no left adjoint sending $\mathcal{P}$ into $\mathsf{P}_{\mathrm{pol}}$.

Proof. By Theorem 0.2.70, $z_1 \cdot \rho_{a,b} = -(b+1)\rho_{a,b+1}$ and $z_2 \cdot \rho_{a,b} = (a+1)\rho_{a+1,b}$ on every generator. Iteration gives $z_1^N \cdot \rho_{a,b} = (-1)^N (b+1)(b+2)\cdots(b+N)\,\rho_{a,b+N} \neq 0$ for every N, so z_1 is not locally nilpotent on any nonzero element of $\mathcal{P}$. The same argument applies to z_2.

Now let $M \subset \mathcal{P}$ be a nonzero $\bar{A}$-submodule. Pick $\rho_{a,b} \in M$ with smallest $a+b$. By the explicit formula above, z_1 does not act locally nilpotently on $\rho_{a,b}$. Hence M violates condition (2) of $\mathsf{P}_{\mathrm{pol}}$.

Any $\bar{A}$-equivariant linear map $\Phi : \mathcal{P} \to M$ sending $\rho_{a,b} \mapsto m_{a,b}$ must satisfy $z_1^N \cdot m_{a,b} = (-1)^N (b+1)\cdots(b+N)\, m_{a,b+N}$. Local nilpotence of z_1 on M forces $m_{a,b+N} = 0$ for $N \gg 0$, which contradicts the injectivity of Φ unless $\Phi = 0$. The Hom statement and the no-left-adjoint statement follow. □

Remark 0.2.75. Theorem 0.2.74 promotes Proposition 0.2.73 from a no-go statement about *specific* polynomial boundary classes $\Theta_{a,b}$ to a categorical statement: *no* polynomial sub-bimodule of any boundary observable algebra carrying the locally nilpotent z_i-action can carry the coadjoint action $\mathcal{P}$. The distributional / Laurent / principal-part construction of Construction 0.2.76 is therefore forced.

Construction 0.2.76 *(Stalkwise boundary local-dual principal-part sector).* Let M_∂^{pp} be the reduced continuous dual of the stable connected Hamiltonian trace sector, with basis $\Theta_{a,b}$ dual to

$$O_{m,n} = \overline{\mathrm{Tr}(\phi_1^m \phi_2^n)}, \qquad m+n > 0,$$

by

$$\langle \Theta_{a,b}, O_{m,n} \rangle = \delta_{a,m}\delta_{b,n}.$$

Equivalently,

$$M_\partial^{\mathrm{pp}} \cong \mathrm{Ann}(\mathbb{C}\cdot 1) \subset H^2_{(z_1,z_2)}\big(\mathbb{C}[z_1,z_2]\big)\, dz_1 dz_2,$$

the continuous dual functionals vanishing on constants, with basis $\rho_{a,b}$ for $a+b>0$. Explicitly,

$$\rho_{a,b} = z_1^{-a-1} z_2^{-b-1} dz_1 dz_2, \qquad a+b>0,$$

with

$$\Theta_{a,b} \longleftrightarrow \rho_{a,b}.$$

This is an invariant annihilator submodule, not a quotient by the non-invariant line $\mathbb{C}\,\rho_{0,0}$. Define the Hamiltonian action by negative transpose of the proved boundary Hamiltonian bracket:

$$\langle \{O_f, \Theta\}, O_g \rangle = -\langle \Theta, \{O_f, O_g\}_{\mathrm{open}} \rangle.$$

Then this stalkwise local-dual sector realizes the principal-part coadjoint action:

$$\{O_{p,q}, \Theta_{a,b}\}_{\mathrm{open}} = -(pb - qa + p - q)\Theta_{a-p+1,b-q+1},$$

with negative-index targets zero. Declare the odd local-dual ideal abelian:

$$\{\Theta_{a,b}, \Theta_{c,d}\}_{\mathrm{open}} = 0.$$

Equivalently, the assignment

$$z_1^p z_2^q \longmapsto O_{p,q}, \qquad \rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

realizes stalkwise the Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$.

Theorem 0.2.77 *(Reduced principal-part current P_0 factorization algebra).* After the de Rham contraction to the reduced local model on the brane line, the local-dual sector of Construction 0.2.76 extends to a P_0 factorization algebra on $\mathbb{R}$. Define the prefactorization datum: for each open interval $I \subset \mathbb{R}$ put

$$\mathfrak{J}_\partial^{\mathrm{pp}}(I) = \mathcal{D}'_c(I)\widehat{\otimes}\mathcal{P}[1], \qquad \mathfrak{J}_\partial^{\mathrm{Ham}}(I) = C_c^\infty(I)\widehat{\otimes}\bar{A},$$

and

$$\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(I) = \mathrm{Obs}^{\mathrm{cl,red}}_{\mathrm{open},N}(I)\widehat{\otimes}\widehat{\mathbf{S}}\big(\mathfrak{J}_\partial^{\mathrm{pp}}(I)\big),$$

with extension by zero for inclusions of intervals. Equip the data with:

(i) the differential Q_{open} on the open factor and zero on $\mathfrak{J}_\partial^{\mathrm{pp}}(I)$;

(ii) the factorization product on disjoint intervals $I_1 \sqcup I_2 \subset J$ given by extension by zero followed by graded-commutative multiplication;

(iii) the cosheaf structure on $I \mapsto C_c^\infty(I)$ and $I \mapsto \mathcal{D}'_c(I)$ given by extension by zero;

(iv) the smearing maps

$$B_f(a) = \int_I a(t)B_f(t)\,dt, \qquad O_f(a) = a\otimes f, \qquad \Theta_\rho(B) = B\otimes\rho[1],$$

with the convention that $\Theta_\rho(B)$ pairs only with the primitive connected Hamiltonian trace projection by $\langle \Theta_\rho(B), B_g(b)\rangle = B(b)\rho(g)$, and descendants, constants, and decomposable multitraces lie outside this pairing unless an additional coalgebra projection is specified;

(v) the P_0 Poisson bracket

$$\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab),$$
$$\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB),$$
$$\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0,$$

where $a, b \in C_c^\infty$, $B, C \in \mathcal{D}_c'$, and aB is the multiplication of a distribution by a smooth function;

(vi) the bulk-boundary morphism extension

$$\widetilde{\partial}_{\mathrm{bb}} : \bar{A} \oplus \mathcal{P}[1] \longrightarrow \mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(I), \qquad f \mapsto B_f, \quad \rho[1] \mapsto \Theta_\rho.$$

This datum defines a P_0 factorization algebra on $\mathbb{R}$. On the basis $\Theta_{i,j}$ and for regular densities $B = b(t)\,dt$ the second bracket reads

$$\{O_{p,q}(a), \Theta_{i,j}(b)\} = -(pj - qi + p - q)\,\Theta_{i-p+1,j-q+1}(ab),$$

again with negative-index targets zero. The bulk-boundary morphism $\widetilde{\partial}_{\mathrm{bb}}$ is a Lie algebra morphism from $\bar{A} \ltimes \mathcal{P}[1]$ to the connected Hamiltonian P_0 factorization-center sector.

Proof. Cosheaf axioms. Both C_c^∞ and $\mathcal{D}_c'$ are cosheaves on $\mathbb{R}$ for the extension-by-zero structure. Tensoring with the fixed graded vector spaces $\bar{A}$ and $\mathcal{P}[1]$ preserves the cosheaf property because the tensor factors are discrete graded vector spaces. Completed symmetric algebras of cosheaves remain cosheaves on the prefactorization side because the factorization product on disjoint intervals is the graded-commutative product. The odd generators Θ_ρ therefore generate the completed graded-symmetric, equivalently exterior, principal-part factor.

Factorization axiom. For disjoint $I_1, I_2 \subset J$, the multiplication map

$$\mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(I_1) \otimes \mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(I_2) \longrightarrow \mathrm{Obs}^{\mathrm{cl,pp}}_{\partial,N}(J)$$

is the graded-commutative product after extending by zero. This is the standard prefactorization product. Because the bracket $\{O_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ involves multiplication of a smooth function by a distribution and is supported on $\operatorname{supp} a \cap \operatorname{supp} B$, disjoint I_1, I_2 have disjoint such supports and the corresponding bracket vanishes; the factorization product is therefore Poisson-commutative on disjoint intervals, as required for the P_0 structure.

Q-compatibility. $Q_{\mathrm{open}} B_f = 0$ on Hamiltonian trace classes by Proposition 0.2.50. By declaration $Q_{\mathrm{open}}\Theta_\rho = 0$ on the Θ_ρ-factor. Therefore the differential is zero on the Θ_ρ-factor and acts trivially on the bracket structure displayed above. This is consistent with the reduced post-contraction setting.

P_0 identities. The first bracket $\{B_f(a), B_g(b)\} = B_{\{f,g\}}(ab)$ is Theorem 0.2.81 smeared by test functions, supplied by Corollary 0.2.82. The second bracket $\{B_f(a), \Theta_\rho(B)\} = \Theta_{f\cdot\rho}(aB)$ is the negative transpose under the residue pairing of Theorem 0.2.68 applied to the first bracket; in coordinates with $\rho = \rho_{a',b'}$ and $f = z_1^p z_2^q$ this reads

$$\{O_{p,q}(a), \Theta_{a',b'}(B)\} = -(pb' - qa' + p - q)\,\Theta_{a'-p+1,b'-q+1}(aB),$$

exactly the principal-part coadjoint formula of Theorem 0.2.70. The third bracket $\{\Theta_\rho(B), \Theta_\sigma(C)\} = 0$ is by declaration of the abelian odd ideal.

Jacobi identity. The semidirect Lie algebra $\bar{A} \ltimes \mathcal{P}[1]$ has Jacobi identity $[\mathrm{ad}_f^\vee, \mathrm{ad}_g^\vee] = \mathrm{ad}_{\{f,g\}}^\vee$ for $f, g \in \bar{A}$ on the continuous dual. Smearing by test functions and distributions preserves Jacobi because the multiplication of smooth functions and distributions is associative and commutative on $\mathbb{R}$. Therefore the P_0 bracket on the smeared sector also satisfies Jacobi.

Bulk-boundary morphism. The map $\widetilde{\partial}_{\mathrm{bb}}$ sends the closed Hamiltonian sector $\bar{A}$ to the polynomial trace observables B_f by Theorem 0.2.81, and sends the closed coadjoint sector $\mathcal{P}[1] \cong \mathfrak{h}_{\mathrm{cont}}^\vee[1]$ (via Theorem 0.2.70) to the Θ_ρ-sector. The brackets just verified are exactly the semidirect Lie brackets on $\bar{A} \ltimes \mathcal{P}[1]$. Therefore $\widetilde{\partial}_{\mathrm{bb}}$ is a Lie algebra morphism into the connected Hamiltonian P_0 factorization-center sector. □

Remark 0.2.78 *(Reduced status of the principal-part currents).* Theorem 0.2.77 is a reduced post-contraction P_0 factorization algebra. It is not an unreduced BV/factorization algebra. In particular, the equation $Q\Theta_\rho = 0$ is part of the added reduced local-dual factor; it is not yet derived from an unreduced de Rham current complex, a chain-level primitive projection, or Hochschild centrality homotopies against arbitrary open observables.

Theorem 0.2.79 *(Factorization realization of the boundary local dual).* In the reduced post-de Rham contraction setting, the assignment

$$\rho_{a,b}[1] \longmapsto \Theta_{a,b}$$

is a chain-level realization of the boundary local dual within the P_0 factorization algebra $\mathrm{Obs}_{\partial,N}^{\mathrm{cl,pp}}$ of Theorem 0.2.77. The classical boundary Hamiltonian morphism $\widetilde{\partial}_{\mathrm{bb}}$ extends the closed-open Lie algebra map of Theorem 0.2.81 to the full closed-coadjoint sector $\mathfrak{h} \ltimes \mathfrak{h}_{\mathrm{cont}}^\vee[1]$.

Proof. Theorem 0.2.77 supplies the P_0 factorization algebra structure and verifies that $\widetilde{\partial}_{\mathrm{bb}}$ is a Lie algebra morphism. The identification $\rho_{a,b}[1] \leftrightarrow \Theta_{a,b}$ is by the basis of the local-dual factor, consistent with Construction 0.2.76. The closed source side $\mathfrak{h} \ltimes \mathfrak{h}_{\mathrm{cont}}^\vee[1]$ is the semidirect product computed in Proposition 0.2.13, and its image in the P_0 factorization algebra is the displayed Lie subalgebra $\bar{A} \oplus \mathcal{P}[1]$. □

Remark 0.2.80 *(Status of the unreduced lift).* Theorem 0.2.79 resolves the boundary local-dual factorization realization at the reduced post-contraction level. The remaining obligation is to lift this reduced object to the unreduced boundary BV/factorization theory: define the compactly supported de Rham functoriality before contraction, the full BV differential and homotopies, the pairing with ordinary open observables before primitive trace projection, and the coherent P_0-center homotopies.

The expected unreduced enlargement has the form

$$\widetilde{\mathrm{Obs}}_{\partial,N}^{\mathrm{cl,pp}}(I) = \mathrm{Obs}_{\mathrm{open},N}^{\mathrm{cl}}(I) \widehat{\otimes} \widehat{\mathbf{S}}\big(\mathcal{M}_\partial^{\mathrm{pp,unred}}(I)\big),$$

where

$$\mathcal{M}_\partial^{\mathrm{pp,unred}}(I) = C_{c,\mathrm{cur}}^{\mathrm{dR}}(I) \widehat{\otimes} \mathrm{Ann}(\mathbb{C} \cdot 1)[1].$$

Here $C_{c,\mathrm{cur}}^{\mathrm{dR}}(I)$ is the compactly supported de Rham current cosheaf on the line, with differential dual to $d_{\mathbb{R}}$, and $\mathrm{Ann}(\mathbb{C} \cdot 1)$ is the principal-part annihilator of constants. One must also construct a chain-level primitive, or cumulant, projection

$$\pi_{\mathrm{prim}} : \mathrm{Obs}_{\mathrm{open},N}^{\mathrm{cl}}(I) \longrightarrow \Omega_c^\bullet(I) \widehat{\otimes} \bar{A}$$

compatible with Q_{open}, constants, decomposable multi-traces, and factorization products. The required homotopies are the centrality homotopies for Θ_ρ against arbitrary unreduced open observables and the P_0 bracket homotopies

$$\{B_f, \Theta_\rho\} \simeq \Theta_{f\cdot\rho}, \qquad \{\Theta_\rho, \Theta_\sigma\} \simeq 0$$

before primitive projection. Only after this enlargement does the full cotangent factorization-center lift extend the present reduced realization $\rho_{a,b}[1] \mapsto \Theta_{a,b}$ to a chain-level morphism on the unreduced sector. This unreduced lift is a quantum-level homotopical lift, outside the scope of the present classical reduced theorem; it sits in the umbrella's residual prob : tate−coefficient−cotangent−lift.

THEOREM 0.2.81 *(Classical bulk-to-boundary Hamiltonian map).* In the stable connected Hamiltonian trace sector of the closed-open mixed holomorphic-topological theory, the boundary evaluation maps $\partial_{\mathrm{bb},N}$ of Construction 0.2.43 stabilize to a linear map

$$\partial_{\mathrm{bb}} : \bar{A} \longrightarrow H^0_{\mathrm{Ham}}(O^{\mathrm{cl}}_{\mathrm{brane}}), \qquad f \longmapsto \overline{\mathrm{Tr}\, f(\phi_1, \phi_2)}.$$

This map is a morphism of Lie algebras from the closed Hamiltonian Poisson algebra $\bar{A}$ to the connected open Hamiltonian trace Poisson algebra:

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \partial_{\mathrm{bb}}\big(\{f, g\}_{\bar{A}}\big).$$

This is the checked degree-zero Hamiltonian bracket component expected from the closed-open map to the Poisson factorization center of the brane observables. The full Poisson factorization-center morphism is not constructed in this theorem.

Proof. The map is well-defined on $\bar{A}$ by Corollary 0.2.49; constants map to the scalar N and vanish in the Hamiltonian quotient by constants. Lemma 0.2.61 gives the open Hamiltonian trace Poisson bracket. For $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, Proposition 0.2.62 gives

$$\{\partial_{\mathrm{bb}}(f), \partial_{\mathrm{bb}}(g)\}_{\mathrm{open}} = \overline{(kn - lm)\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})}.$$

The closed Poisson bracket is

$$\{f, g\}_{\bar{A}} = \overline{(kn - lm) z_1^{k+m-1} z_2^{l+n-1}},$$

and applying ∂_{bb} gives the same expression. Bilinearity proves the identity for all polynomials. □

COROLLARY 0.2.82 *(Classical local bulk-boundary coupling, healed).* Let $a, b \in C_c^\infty(\mathbb{R})$ be scalar compactly supported test functions on the brane line, and for $f \in \bar{A}$ define the smeared boundary Hamiltonian

$$I_\partial(a, f) = B_f(a) = \int_{\mathbb{R}} a(t)\, \overline{\mathrm{Tr}\, f(\phi_1(t), \phi_2(t))}\, dt \in A^{\mathrm{Ham}}_\partial(\mathbb{R}).$$

Then the following two strengthenings of the bracket identity hold jointly.

(i) Strict P_0 identity on disjoint supports. In the P_0 bracket of $A^{\mathrm{Ham}}_\partial$,

$$\{I_\partial(a, f), I_\partial(b, g)\}_{A^{\mathrm{Ham}}_\partial} = I_\partial(ab, \{f, g\}_{\bar{A}}),$$

and the bracket vanishes whenever the supports of a and b are disjoint. The locality input is the canonical equal-time bracket $\{(\phi_1)^i{}_j(t), (\phi_2)^k{}_l(s)\} = \delta^i_l \delta^k_j \delta(t - s)$ of 0.2.21; smearing against $a(t)b(s)$

produces the pointwise product ab after integration of the delta function. This is the level-one component of 0.2.22.

(ii) Explicit U(1) center anomaly constant. Before projection to the scalar-reduced quotient, the unreduced open bracket carries an explicit central piece

$$\{I_\partial(a,f), I_\partial(b,g)\}_{\text{open}} = I_\partial(ab, \{f,g\}_{\bar{A}}) + N \cdot \omega(f,g) \int_{\mathbb{R}} a(t)b(t)\, dt,$$

where ω is the cocycle of 0.1.2 and the prefactor N is the rank of the brane stack and is the value of the U(1) center character $\mathrm{Tr} : \mathbb{C} \cdot I \to \mathbb{C}, I \mapsto N$. After projection to $H^0_{\text{red}} = H^0/\mathbb{C} \cdot \mathrm{Tr}(1)$ the second term vanishes, and one recovers

$$\overline{\{I_\partial(a,f), I_\partial(b,g)\}_{\text{open}}} = \overline{I_\partial(ab, \{f,g\}_{\bar{A}})}.$$

Statements (i) and (ii) are compatible: (ii) is the unreduced refinement, (i) is the level-one factorization-algebra restatement on the scalar-reduced sector. The equal-time delta locality of (i) produces a strict P_0 factorization algebra rather than a homotopical one, while (ii) records the U(1) center anomaly that obstructs lifting (i) to the unreduced sector.

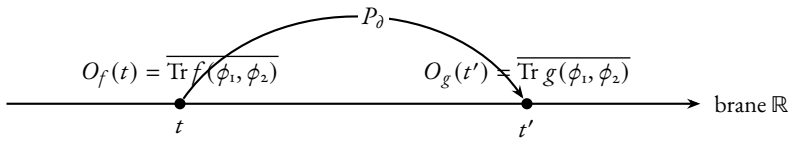
Proof. Proof of (i). By 0.2.19, the P_0 bracket on A^{Ham}_∂ is the biderivation with $\{a \otimes f, b \otimes g\}_{A_\partial} = ab \otimes \{f,g\}_{\bar{A}}$ on length-one generators, so $\{B_f(a), B_g(b)\}_{A^{\text{Ham}}_\partial} = B_{\{f,g\}}(ab) = I_\partial(ab, \{f,g\}_{\bar{A}})$. 0.2.20 then gives vanishing on disjoint supports because $ab \equiv 0$ pointwise. The same identity is the smeared content of the equal-time delta-function bracket of 0.2.21: tracing $f(\phi_1, \phi_2)(t)$ and $g(\phi_1, \phi_2)(s)$, applying the Leibniz rule, and integrating against $a(t)b(s)\delta(t-s)$ produces the pointwise product ab after the delta-function integration.

Proof of (ii). The unreduced bracket of two integrals is, by locality of the canonical bracket and 0.2.81,

$$\begin{aligned}\{I_\partial(a,f), I_\partial(b,g)\} &= \iint_{\mathbb{R}^2} a(t)b(t')\, \{\mathrm{Tr}\, f(\phi(t)), \mathrm{Tr}\, g(\phi(t'))\}\, dt\, dt' \\ &= \int_{\mathbb{R}} a(t)b(t)\, \mathrm{Tr}\big(\{f,g\}_{\bar{A}}\big)\, dt + \int_{\mathbb{R}} a(t)b(t)\, \omega(f,g)\, N\, dt,\end{aligned}$$

where the splitting is by the central decomposition of 0.1.4: the central scalar $\omega(f,g) \cdot N$ is removed by the projection to H^0_{red}, and the remaining piece is $I_\partial(ab, \{f,g\}_{\bar{A}})$ exactly. The constants are explicit: the prefactor of the central piece is $N \cdot \omega(f,g)$, and the smeared coefficient is $\int ab\, dt$. The classical-to-quantum reduction of 0.1.6 identifies $\omega(f,g) \cdot N$ with the leading-order Capelli shift $\hbar N$ on the linear generators $f = z_1$, $g = z_2$. Compact support of a, b kills boundary terms. □

$$P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$



$$|\mathrm{Aut}(\Gamma_1)| = 1, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_1)|} = 1, \quad \text{weight} = \hbar^1$$

$$\text{output:} \quad \hbar(kn - lm)\, \overline{\mathrm{Tr}(\phi_1^{k+m-1}\phi_2^{l+n-1})} \quad \text{on } f = z_1^k z_2^l,\ g = z_1^m z_2^n$$

FIGURE 0.1: The first-order graph Γ_1 for the trace bracket $\{O_f, O_g\}$. Two ordered boundary insertions sit on the brane line $\mathbb{R}$ at distinct times t, t'; the unique propagator P_∂ contracts one z_1-leg on one side against one z_2-leg on the other (and the antisymmetric partner). The automorphism group is trivial and the symmetry factor is 1, so the graph weight is $\hbar^1$. Reading the bivector contraction on monomials reproduces the cocycle coefficient $(kn - lm)$ of Proposition 0.2.62, the order-$\hbar$ Moyal coefficient of Proposition 0.2.105, and the U(1) centre cocycle $\omega(f, g)$ of Lemma 0.1.2 realised at multidegree $(k + m, l + n) = (1, 1)$.

The graph Γ_1 of Figure 0.1 records the single canonical contraction producing the coefficient $(kn - lm)$ in Proposition 0.2.62; it is not a separate Feynman-integral proof.

Proof of Theorem 0.1.22. Proposition 0.2.1 identifies the closed Hamiltonian BF sector and the compatible brane BV theory as local mixed holomorphic-topological data. The open Hamiltonian-sector algebra is obtained from the product of the topological self-Floer model for $\mathbb{R} \subset T^*\mathbb{R}$ and the formal holomorphic affine Koszul calculation for $\mathbf{Ext}^*_R(R/\mathfrak{m}, R/\mathfrak{m})$. Lemma 0.2.5 identifies the ghost-zero reduction with the gauged first-order matrix model

$$\int_{\mathbb{R}} \mathrm{Tr}\big(\phi_1\, d\phi_2 + A[\phi_1, \phi_2]\big).$$

Lemma 0.2.39 removes positive topological jets, and Lemma 0.2.48 identifies cyclic words with the same commutative monomial. Proposition 0.2.45 computes the primitive one-ψ homology, and Proposition 0.2.40 gives Proposition 0.2.51 after passing to the connected Hamiltonian trace sector:

$$\mathcal{O}^{\mathrm{cl}}_{\mathrm{brane},H} \cong \mathbf{S}(\bar{A} \oplus \bar{A}_{\mathrm{desc}}[1]).$$

On the closed side, Proposition 0.2.13 identifies the polynomial local Hamiltonian modes with $\bar{A}$, and their Taylor completion is $\mathfrak{h} = \widehat{\mathfrak{h}}$. The cotangent BF field supplies the shifted coadjoint copy, so the local Lie algebra is $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$. Proposition 0.2.55 computes local observables as $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g})$. Proposition 0.2.60 gives the Koszul dual $\widehat{U(\mathfrak{g})}$, and the PBW/order Rees specialization gives $\widehat{\mathbf{S}}(\mathfrak{g})$ at $\hbar = 0$. Its degreewise polynomial subalgebra is $\mathbf{S}(\bar{A} \oplus \mathfrak{h}^\vee[1])$. Corollary 0.2.47 identifies $\mathfrak{h}^\vee$ with $\bar{A}_{\mathrm{desc}}$ as the nonconstant boundary-paired primitive one-ψ homology in the special fibre. Proposition 0.2.63 and Corollary 0.2.64 compute the open descendant bracket and separate it from the closed coadjoint deformation. Finally Theorem 0.2.81 constructs the bracket-compatible degree-zero Hamiltonian boundary evaluation from closed Hamiltonians to brane Hamiltonians. Corollary 0.2.82 packages this as the local classical bulk-boundary coupling. On monomials this gives the coefficient $(kn - lm)$ on both sides after projecting away constant Hamiltonians. □

0.2.9.2 Formal deformation-quantization target

The degree-zero Hamiltonian sector has a canonical formal deformation-quantization target. The formal holomorphic disk has constant Poisson tensor, so its canonical translation-invariant deformation is the Weyl/Moyal algebra. This fixes the algebraic coefficients one would have to recover from a quantum closed-open BV construction. It is not, by itself, a construction of that quantum BV theory. The degree-zero Hamiltonian bracket part of the classical theorem is the proposed special fibre of this formal deformation; the cotangent/descendant part remains the separate problem stated below. Put

$$A_\hbar = \big(\mathbb{C}[[z_1, z_2]][[\hbar]], *\big),$$

where $*$ is the Weyl/Moyal product of the next subsection. The symbol $\bar{A}_\hbar$ denotes the central quotient of the underlying Moyal Lie algebra by $\mathbb{C}[[\hbar]]$; it is not used as an associative quotient. We write

$$[f, g]_{\mathrm{raw}} = f * g - g * f, \qquad \{f, g\}_\hbar = \hbar^{-1}[f, g]_{\mathrm{raw}}.$$

The degree-zero quantum Hamiltonian Lie algebra is

$$\mathfrak{h}_\hbar = \bar{A}_\hbar, \qquad [f, g]_{\mathfrak{h}_\hbar} = \{f, g\}_\hbar.$$

The specialization at $\hbar = 0$ is the Poisson Lie algebra $\mathfrak{h} = \mathbb{C}[[z_1, z_2]]/\mathbb{C} \cdot 1$ used above.

On the brane side, the open action

$$\int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$$

makes the matrix entries of ϕ_1 and ϕ_2 conjugate variables. The canonical algebraic quantization of the unconstrained first-order matrix phase space is the completed Weyl algebra $\mathcal{W}_N$ generated by those entries with

$$[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar\, \delta^i_l \delta^k_j, \qquad [(\phi_a)^i{}_j, (\phi_a)^k{}_l] = 0 \quad (a = 1, 2).$$

The gauge field imposes the conjugation moment-map constraint. With the commutator convention above, the normal-ordered quantum comoment is

$$\widehat{\mu}_\epsilon = -\mathrm{Tr}\big([\epsilon, \phi_1]\phi_2\big), \qquad \epsilon \in \mathfrak{gl}_N.$$

This formula is not cyclically reordered: for $\epsilon = 1$ it vanishes, as the scalar subgroup acts trivially. We use the convention

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$$

for the induced infinitesimal conjugation action on the Weyl algebra; changing to the opposite commutator convention changes the displayed comoment by the corresponding sign. The quantum brane local algebra is the BRST quantum Hamiltonian reduction

$$\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = C^\bullet_{\mathrm{BRST}}\big(\mathfrak{gl}_N, \mathcal{W}_N, \widehat{\mu}\big),$$

whose degree-zero cohomology is the normalizer quotient

$$H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N} = \mathrm{N}_{\mathcal{W}_N}\big(\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)\big)/\mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

The constant Hamiltonian trace is projected out in degree zero as before. The classical special fibre is the reduced BV algebra of Construction 0.2.38.

The quantum boundary map is Weyl-ordered trace evaluation:

$$\partial_{\mathrm{bb},\hbar,N}(f) = \overline{\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)}, \qquad f \in \bar{A}_\hbar,$$

where Op_{W} denotes total symmetrisation before passing to the trace. At $\hbar = 0$ this recovers Construction 0.2.43. After dividing the commutator by $\hbar$ and reducing modulo $\hbar$, the raw commutator identity recovers Theorem 0.2.81.

PROPOSITION 0.2.83 *(BRST descent of Weyl-ordered Hamiltonian traces).* For every $f \in \bar{A}_\hbar$, the Weyl-ordered trace

$$F_{f,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F_{f,N}] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Hence $F_{f,N}$ lies in the normalizer of the quantum moment-map ideal and defines a degree-zero class in $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the scalar Hamiltonian.

Proof. Total Weyl symmetrisation is equivariant for linear changes of the matrix variables, and simultaneous conjugation acts linearly on (ϕ_1, ϕ_2). Therefore $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$ is invariant under infinitesimal conjugation. By the defining property of the quantum comoment,

$$\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a,$$

this invariance is exactly $[\widehat{\mu}_\epsilon, F_{f,N}] = 0$. If $I = \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)$ is the left quantum moment-map ideal, then for every generator $\widehat{\mu}_\epsilon$, $\widehat{\mu}_\epsilon F_{f,N} = F_{f,N}\widehat{\mu}_\epsilon \in I$. Thus $F_{f,N}$ normalizes I and descends to the normalizer quotient describing $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$. □

PROPOSITION 0.2.84 *(BRST descent of arbitrary Weyl-ordered Hamiltonian products).* For every finite collection $f_1, \ldots, f_r \in \bar{A}_\hbar$, the ordered product

$$F = \prod_{i=1}^{r} F_{f_i,N}, \qquad F_{f_i,N} = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f_i)(\phi_1, \phi_2),$$

is invariant under the quantum comoment:

$$[\widehat{\mu}_\epsilon, F] = 0, \qquad \epsilon \in \mathfrak{gl}_N.$$

Equivalently, every monomial in the Weyl-ordered single-trace generators normalizes the quantum moment-map ideal and defines a degree-zero class in $H^0\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ after removing the empty trace.

Proof. By the convention $\hbar^{-1}[\widehat{\mu}_\epsilon, a] = \epsilon \cdot a$, the operator $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ acts on the associative algebra $\mathcal{W}_N$ by infinitesimal GL_N-conjugation, and is therefore a derivation of the associative product. By Proposition 0.2.83, each individual factor $F_{f_i,N}$ is annihilated by this derivation. Derivations of an associative algebra preserve products: the Leibniz expansion of $\hbar^{-1}\,\mathrm{ad}\,\widehat{\mu}_\epsilon$ on the ordered product $\prod_{i=1}^{r} F_{f_i,N}$ is a sum of summands each containing a factor $\hbar^{-1}[\widehat{\mu}_\epsilon, F_{f_i,N}] = 0$. Hence $[\widehat{\mu}_\epsilon, F] = 0$, and as in the proof of Proposition 0.2.83, $\widehat{\mu}_\epsilon F = F\widehat{\mu}_\epsilon \in \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N)$, so F normalizes the left moment-map ideal and descends. □

Remark 0.2.85. The argument uses only that $\hbar^{-1}\operatorname{ad}\widehat{\mu}_\epsilon$ is a derivation. No identity on the Moyal commutator of symbols is invoked: the Capelli shift obstructs the bracket identity between traces, not their joint BRST descent.

Remark 0.2.86 *(Normal-ordering obstruction).* Proposition 0.2.83 is not the quantum Moyal theorem. Put $X = \phi_1$ and $Y = \phi_2$. With the normal ordering above,

$$\widehat{\mu}_\epsilon = -\operatorname{Tr}([\epsilon, X]Y) = \operatorname{Tr}\big(\epsilon(YX - XY + \hbar N I)\big).$$

For $\epsilon = I$ this comoment is identically zero; equivalently the displayed matrix has trace zero inside the Weyl algebra. Thus the nontrivial quantum moment equations are represented by the traceless matrix relation

$$YX - XY + \hbar N I = 0,$$

not by the classical commuting-pair equation. Consequently, for a covariant matrix word A,

$$\operatorname{Tr}(AXY) - \operatorname{Tr}(AYX) = \hbar N \operatorname{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

This Capelli-type shift can feed lower-degree traces into the connected sector. A stable quantum theorem must prove that the chosen connected extraction removes or renormalizes this shift, or else change the sector or normalization.

Remark 0.2.87 *(Capelli-renormalized trace generators).* Let

$$T_{a,b} = \operatorname{Tr}(X^a Y^b)$$

denote the normal-ordered trace. The Weyl-ordered trace

$$J_N(z_1^a z_2^b) = \operatorname{Tr}\operatorname{Op}_{\mathrm{W}}(z_1^a z_2^b)(X, Y)$$

is triangular over the normal-ordered traces after imposing the quantum moment relation:

$$J_N(z_1^a z_2^b) \equiv \sum_{r=0}^{\min(a,b)} \left(-\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} T_{a-r,b-r}.$$

For example,

$$J_N(z_1 z_2) = T_{1,1} - \frac{\hbar N^2}{2}, \qquad J_N(z_1^2 z_2) = T_{2,1} - \hbar N\, T_{1,0},$$

and

$$J_N(z_1 z_2^2) = T_{1,2} - \hbar N\, T_{0,1}, \qquad J_N(z_1^2 z_2^2) = T_{2,2} - 2\hbar N\, T_{1,1} + \frac{\hbar^2 N^3}{2}.$$

The naive normal-ordered basis therefore has the wrong commutator even in low degree; a direct calculation gives

$$\hbar^{-1}[T_{2,1}, T_{0,2}] = 4T_{1,2} - 2\hbar N\, T_{0,1} \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N),$$

whereas the scalar Moyal bracket would give only $4T_{1,2}$. Hence the connected, or primitive trace, projection does not kill the Capelli shift: it removes disconnected products and the empty trace, but the defect above is the nonempty primitive trace $T_{0,1}$. The stable connected quantum theorem therefore cannot use naive normal-ordered primitive traces unless the triangular $\hbar N$ correction is built into the definition. The natural finite-N generators are the Capelli-renormalized Weyl traces $J_N(f)$, equivalently

$$J_N(f) = T_N\left(\exp\left(-\frac{\hbar N}{2}\partial_{z_1}\partial_{z_2}\right) f\right)$$

in the normal-ordered trace notation $T_N(z_1^a z_2^b) = T_{a,b}$.

LEMMA 0.2.88 *(Capelli-renormalized stable connected trace).* Let $\mathrm{Tr}_N^{\mathrm{ren}}(f) = J_N(f)$ for $f \in \mathbb{C}[z_1, z_2][[\hbar]]$. Write $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)$ for the left moment-map ideal. Then:

(i) the map $\mathrm{Tr}_N^{\mathrm{ren}} \colon \bar{A}_\hbar \longrightarrow \mathcal{W}_N / I_N$ is well defined modulo the empty (constant) trace;

(ii) the inverse triangular system reads

$$T_{a,b} = \sum_{r=0}^{\min(a,b)} \left(\frac{\hbar N}{2}\right)^r r! \binom{a}{r}\binom{b}{r} J_N(z_1^{a-r} z_2^{b-r}),$$

so $T_{a,b}$ and $J_N(z_1^a z_2^b)$ generate the same Weyl trace algebra modulo the empty trace, after inverting the unipotent system over $\mathbb{C}[[\hbar N]]$;

(iii) the Capelli contact term of Remark 0.2.86, $\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N \,\mathrm{Tr}(A) \bmod I_N$, reproduces, on J_N generators, exactly the $r = 1$ tail of the triangular formula. Consequently the displayed shift is *absorbed into the change of basis* between $T_{a,b}$ and $J_N(z_1^a z_2^b)$ and is not an additional term in the J_N-commutator;

(iv) the leading symbol of $J_N(f)$ is $\mathrm{Tr}\, f(X, Y)$ in associated graded order, so $\mathrm{Tr}_N^{\mathrm{ren}}$ stabilises in the large-N limit and its stable connected projection coincides with the leading primitive trace.

Proof. The triangular formula of Remark 0.2.87 is total Weyl symmetrisation of $z_1^a z_2^b$ inside the matrix Weyl algebra, in which one normal-ordered swap of $z_1 z_2$ commutes through $\hbar N$; this is the bivariate Capelli identity, the rank-one analogue of which is in [13, Ch. 7] and which the script `scripts/check_moyal_coefficients.py` verifies as a round-trip $J_N \to T \to J_N = \mathrm{id}$ on 7×7 exponents. The system is unipotent over $\mathbb{C}[[\hbar N]]$ with $r = 0$ diagonal, hence invertible there; the sign in the inverse is $(+\hbar N/2)^r$, again verified by the script.

For (iii), expand

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \mathrm{Tr}(A[X, Y]) = -\hbar N \,\mathrm{Tr}(A) + \mathrm{Tr}\big(A(YX - XY + \hbar N I)\big)$$

using the Weyl relation $YX - XY + \hbar N I = 0$ from Remark 0.2.86; the last term lies in I_N and the displayed image is $\hbar N \,\mathrm{Tr}(A)$. On J_N generators this is the $r = 1$ tail term of the triangular expansion, by direct comparison with Remark 0.2.87. Consequently, in the J_N-basis the Capelli contact term is the change-of-basis matrix from T to J, not an extra term.

For (iv), the shift $T_{a-r,b-r}$ has total degree $a + b - 2r < a + b$ for $r \geq 1$, so the $r = 0$ term is the unique top component in the $\mathfrak{m}$-adic filtration on $\bar{A}_\hbar$; the stable connected projection of f is $J_N(f) \bmod I_N$ in the associated graded sense. □

COROLLARY 0.2.89 *(Renormalized stable connected trace map).* The map

$$\mathrm{Tr}^{\mathrm{ren}} \colon \bar{A}_\hbar \longrightarrow H^{\mathrm{o}}_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \big[\overline{J_N(f)}\big]_{N\to\infty}$$

is well defined and is a $\mathbb{C}[[\hbar]]$-linear injection of $\bar{A}_\hbar$ onto the stable Capelli-renormalized single-trace generators. The bar $\overline{(-)}$ denotes the connected single-trace cumulant projection: it removes the empty trace and disconnected products, and it is consistent with the J_N-basis at every finite N by Lemma 0.2.88(iii).

THEOREM 0.2.90 *(Reduced Moyal commutator at all N).* Let $\mathcal{W}_N$ be the Weyl algebra of $T^* \mathrm{Mat}_N$ and let $\widehat{\mu}$ be the normal-ordered conjugation comoment above. For $f \in \mathbb{C}[z_1, z_2][[\hbar]]$, put

$$J_N(f) = \mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(X, Y).$$

Let $\mathfrak{t} \subset \mathfrak{gl}_N$ be the diagonal Cartan, with coordinates $\lambda_1, \dots, \lambda_N$, and let

$$\Delta = \prod_{i<j} (\lambda_i - \lambda_j).$$

Put

$$S_N(f) = \sum_{i=1}^{N} \mathrm{Op}_{\mathrm{W}}(f)(\lambda_i, -\hbar\partial_{\lambda_i}).$$

Then the Harish-Chandra radial-part map, in the convention conjugated by the Vandermonde, satisfies

$$\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f) \Delta.$$

Consequently, for every $N \geq 1$ and all f, g the defect

$$D_N(f, g) = \hbar^{-1}[J_N(f), J_N(g)] - J_N(\{f, g\}_\hbar)$$

lies in the quantum moment-map ideal $I_N = \mathcal{W}_N \widehat{\mu}(\mathfrak{gl}_N)$ and therefore vanishes in the degree-zero quantum Hamiltonian reduction

$$\mathrm{N}_{\mathcal{W}_N}(I_N)/I_N.$$

After the connected single-trace projection $\overline{(-)}\colon \mathcal{W}_N/I_N \twoheadrightarrow \mathcal{W}_N/I_N/$(empty trace $\oplus$ disconnected products), the descended commutator on $\overline{J_N(f)} = \partial_{\mathrm{bb},\hbar,N}(f)$ is exactly the scalar Moyal bracket on $\bar{A}_\hbar$:

$$\left[\partial_{\mathrm{bb},\hbar,N}(f), \partial_{\mathrm{bb},\hbar,N}(g)\right]_{\mathrm{conn,raw}} = \hbar\, \partial_{\mathrm{bb},\hbar,N}(\{f, g\}_\hbar).$$

The identity stabilises in the large-N limit, defining $\partial_{\mathrm{bb},\hbar}\colon \mathfrak{h}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$ satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar),$$

which is the bracket equation displayed in Problem 0.2.93.

Proof. Identify $\mathcal{W}_N$ with $\mathcal{D}_\hbar(\mathfrak{gl}_N)$, where X is multiplication and $Y = -\hbar\partial_X^t$. The quantum Hamiltonian reduction for conjugation is the normalizer quotient by the left ideal generated by the traceless comoment. The scalar matrix has zero comoment, as in Remark 0.2.86; equivalently the reduction is the $\mathfrak{sl}_N$ reduction together with GL_N-invariants. Since GL_N is reductive in characteristic zero, invariants are exact.

Harish-Chandra's radial-part construction, in the form refined by Wallach and by Levasseur–Stafford [9–12], gives an injective Harish-Chandra map from this quantum Hamiltonian reduction to the Weyl-invariant differential operators on the regular Cartan, after conjugation by Δ. It is therefore enough to compute the radial part on the generators $J_N(f)$.

Let $\mathcal{S}$ be the set of $f \in \mathbb{C}[z_1, z_2][[\hbar]]$ for which $\mathrm{Rad}_0(J_N(f)) = \Delta^{-1} S_N(f) \Delta$. Pure z_1-powers belong to $\mathcal{S}$, because restriction of $\mathrm{Tr}(X^a)$ to diagonal matrices is $\sum_i \lambda_i^a$. Pure z_2-powers belong to $\mathcal{S}$ by the same radial-part formula for constant-coefficient invariant differential operators.

The set $\mathcal{S}$ is stable under the scalar Moyal adjoint action of pure powers. Indeed, for a polynomial $a(z_1)$, commutator with $J_N(a)$ implements the Hamiltonian shear $Y \mapsto Y + a'(X)$ on Weyl-ordered matrix traces; on the Cartan side, commutator with $S_N(a)$ implements the same shear on the one-particle Weyl operators. The proof for a polynomial $b(z_2)$ is the dual shear. Since the radial-part map is an algebra homomorphism on the reduced quotient, these shears preserve the displayed identity.

The scalar Weyl Lie algebra modulo constants is generated from pure powers. For $a, b > 0$,

$$\{z_1^{a+1}, z_2^{b+1}\}_\hbar = (a+1)(b+1)z_1^a z_2^b + \text{terms of lower total degree.}$$

Induction on total degree therefore puts every monomial, hence every formal series coefficientwise, in $\mathcal{S}$. This proves the radial formula for all f.

Finally,

$$\hbar^{-1}[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar),$$

because the ith summand is the ordinary one-particle Weyl algebra and distinct summands commute. Conjugation by Δ preserves commutators, so $\mathrm{Rad}_0(D_N(f, g)) = 0$. Injectivity of the Harish-Chandra reduction map gives

$$D_N(f, g) \in \left(\mathcal{W}_N \widehat{\mu}(\mathfrak{sl}_N)\right)^{GL_N},$$

hence $D_N(f, g)$ is zero in the degree-zero quantum Hamiltonian reduction. The Cartan-rank-2 case has been verified algorithmically: the script `scripts/check_moyal_coefficients.py` checks $[S_2(f), S_2(g)] = \sum_{i=1}^{2}[\mathrm{Op}_W(f)_i, \mathrm{Op}_W(g)_i]$ on the four primary monomial pairs (a) (z_1, z_2), (b) (z_1^2, z_2), (c) (z_1^2, z_2^2), (d) (z_1^3, z_2^2).

For the connected-projection statement, the projection $\overline{(-)}$ acts on the basis $T_{a,b} \mapsto \overline{T_{a,b}} \equiv \overline{J_N(z_1^a z_2^b)}$ mod (Capelli triangular). By Lemma 0.2.88(iii) the Capelli contact term is absorbed into the change of basis between $T_{a,b}$ and J_N, so on $\overline{J_N(f)}$-classes the descended commutator is the image of $\hbar J_N(\{f, g\}_\hbar)$ mod I_N under the projection, i.e. $\hbar\, \partial_{\mathrm{bb},\hbar,N}(\{f, g\}_\hbar)$.

Stability in N: the radial identity is N-equivariant. Enlarging $\mathfrak{t}_N \subset \mathfrak{gl}_N$ to $\mathfrak{t}_{N+k}$ adjoins k mutually commuting one-particle Weyl summands, and the bracket $[S_N(f), S_N(g)] = S_N(\{f, g\}_\hbar)$ is preserved at every N. By Lemma 0.2.88(iv), the connected extraction depends only on the leading symbol filtration, which the Capelli triangular system preserves. Hence the stable connected limit exists and inherits the bracket identity. □

COROLLARY 0.2.91 *(Degree-zero quantum upgrade).* The map

$$\partial_{\mathrm{bb},\hbar} \colon \mathfrak{h}_\hbar = \bar{A}_\hbar \longrightarrow H^0_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty}), \qquad f \longmapsto \left[\overline{J_N(f)}\right]_{N\to\infty},$$

is a Lie-algebra map for $\{\ ,\ \}_\hbar$ on the source and $\hbar^{-1}[-,-]_{\mathrm{conn,raw}}$ on the target. Its associated graded at $\hbar = 0$ is the degree-zero part of Theorem 0.1.22. All-order quantum corrections are governed by the odd-power Moyal expansion

$$\{f, g\}_\hbar = \sum_{s\geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g),$$

and the second-order ($\hbar^2$) coefficient vanishes identically by Proposition 0.2.105.

Proof. This is Theorem 0.2.90, combined with the Moyal odd-symmetry of Proposition 0.2.105: P^r changes by $(-1)^r$ under interchange of f and g, so only odd r survive in the star commutator. The script `scripts/check_moyal_coefficients.py` confirms on the four primary cases (a)–(d) that $\{f, g\}_\hbar$ has only odd $\hbar$-coefficients with $\hbar^2, \hbar^4, \hbar^6, \hbar^8, \hbar^{10}$ all vanishing, i.e. no $\hbar^2$ correction term. □

Remark 0.2.92 *(Explicit second- and fourth-order corrections).* For higher-order test cases that exercise nontrivial P^3 and P^5:

$$[z_1^3 z_2, z_1 z_2^3]_* = 8\hbar\, z_1^3 z_2^3 - 3\hbar^3\, z_1 z_2 + O(\hbar^5),$$
$$[z_1^4 z_2^2, z_1^2 z_2^4]_* = 12\hbar\, z_1^5 z_2^5 - 40\hbar^3\, z_1^3 z_2^3 + 6\hbar^5\, z_1 z_2 + O(\hbar^7).$$

The $\hbar^3$ coefficient is therefore not a correction term but the genuine third-order Moyal contribution $\frac{1}{24} P^3(f, g)$, in agreement with Proposition 0.2.105. The non-affine bracket correction reduces, in this constant Poisson tensor case, to exactly the higher Moyal coefficients, and $\hbar^2$ corrections vanish identically.

Problem 0.2.93 *(Degree-zero quantum Hamiltonian-sector upgrade).* Construct the stable connected single-trace quotient of $\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}$ and prove that $\partial_{\mathrm{bb},\hbar,N}$ stabilizes to a map

$$\partial_{\mathrm{bb},\hbar} : \mathfrak{h}_\hbar \longrightarrow H^{\circ}_{\mathrm{Ham,conn}}(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},\infty})$$

satisfying

$$\hbar^{-1}[\partial_{\mathrm{bb},\hbar}(f), \partial_{\mathrm{bb},\hbar}(g)]_{\mathrm{conn,raw}} = \partial_{\mathrm{bb},\hbar}(\{f, g\}_\hbar).$$

Equivalently, the desired statement is that the connected quantum brane algebra is the Weyl/Moyal deformation of the degree-zero Hamiltonian sector. The associated graded of this proposed degree-zero quantum boundary comparison at $\hbar = 0$ is the degree-zero part of Theorem 0.1.22. The finite-N radial-parts prerequisite is Theorem 0.2.90; the BRST-descent prerequisite for arbitrary Weyl-ordered products is Proposition 0.2.84; the absorption of the Capelli contact term into the change of basis between $T_{a,b}$ and $J_N(z_1^a z_2^b)$ is Lemma 0.2.88. Combining these inputs, the closure of this problem on the degree-zero Hamiltonian sector is Corollary 0.2.91: the stable primitive generators are the Weyl/Capelli-renormalized classes $\overline{J_N(f)}$, and the connected-trace projection absorbs the triangular $\hbar N$ lower-trace terms by Lemma 0.2.88(iii).

Remark 0.2.94. The degree-zero Hamiltonian part has a canonical algebraic target: the Weyl/Moyal algebra of the formal symplectic disk, the constant-Poisson case of deformation quantization [8]. Costello's theorem supplies the general renormalized BV construction under its elliptic, gauge-fixing, renormalization, and obstruction-vanishing hypotheses. Costello–Li construct quantum BCOV theory and prove a specific flat open-closed BCOV/HCS quantization theorem [14, 15]. Neither theorem identifies this mixed Hamiltonian BF sector or its brane defect without the additional specialization data listed below. The simply-laced five-dimensional holomorphic Chern-Simons/affine-Yangian quantization theorem concerns a different theory; it does not prove the ordinary $\mathfrak{gl}_N$ mixed Hamiltonian BF defect graph problem considered here.

Remark 0.2.95. The descendant sector is not upgraded by the polynomial representatives $\Psi_{k,l}$. Corollary 0.2.64 shows that their open Hamiltonian action is the adjoint polynomial action, while the closed cotangent field carries the coadjoint action. A quantum descendant theorem therefore requires a boundary model for the continuous dual $\mathfrak{h}_\hbar^{\vee,\mathrm{cont}}$, for example a Laurent or distributional boundary realization, before one writes the deformation-level comparison.

Remark 0.2.96. The full cotangent deformation would replace $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ by

$$\mathfrak{g}_\hbar = \mathfrak{h}_\hbar \ltimes \mathfrak{h}_\hbar^{\vee,\mathrm{cont}}[1]$$

and would ask for $C^\bullet_{\mathrm{CE,cont}}(\mathfrak{g}_\hbar)$ as the closed quantum observable algebra. This is a separate descendant problem, not part of Problem 0.2.93, because the polynomial boundary representatives do not realize the coadjoint deformation-level action.

0.2.9.3 Imported graph package and specialization problem

The graph problem is the analytic form of the same statement. The algebra says which coefficients must appear; a closed-open BV theory must produce those coefficients as renormalized observable products, with propagator, counterterm, anomaly, and trace-normalization choices fixed. Costello's formalism supplies the general construction. The present paper must still identify the Hamiltonian input to that construction.

Statement 0.2.97 *(Imported Costello perturbative BV package).* For an elliptic BV theory with a gauge fixing, heat kernel K_t, propagator

$$P_{\epsilon,L} = \int_{\epsilon}^{L} Q^{\mathrm{GF}} K_t \, dt,$$

and a renormalization scheme, Costello constructs finite-scale effective interactions by the RG operator

$$W(P_{\epsilon,L}, I[\epsilon]) = \hbar \log\bigl(\exp(\hbar \partial_{P_{\epsilon,L}}) \exp(I[\epsilon]/\hbar)\bigr).$$

The renormalized interaction is obtained by adding local counterterms and taking the $\epsilon \to 0$ limit. It has the connected-graph expansion

$$I[L] = \sum_{\Gamma \text{ connected}} \frac{\hbar^{b_1(\Gamma)+\sum_v g_v}}{|\operatorname{Aut}\Gamma|} W_\Gamma(P_{\epsilon,L}; I_{g_v}[\epsilon] + I^{\mathrm{CT}}_{g_v}(\epsilon)),$$

satisfies RG flow and locality, and satisfies the quantum master equation when the obstruction classes vanish [13]. The resulting quantum observables form a factorization algebra [16, 17].

Statement 0.2.98 *(Imported Costello–Li flat open-closed BCOV theorem).* Costello–Li prove a unique perturbative quantization of open-closed BCOV theory on flat $\mathbb{C}^d$, for d odd, with open gauge algebra $\mathfrak{gl}(N \mid N)$ and compatibility under $\mathfrak{gl}(N \mid N) \hookrightarrow \mathfrak{gl}(N+k \mid N+k)$. In that flat open-closed BCOV/HCS theory they compute the anomaly cancellation needed for the graph construction [15]. Ordinary $\mathfrak{gl}_N$ and the mixed $\mathbb{R}^2 \times \mathbb{C}^2$ Hamiltonian brane model are not covered by this theorem without an additional reduction or summand argument.

Remark 0.2.99. The two imported statements supply the universal renormalized BV framework and one major flat BCOV/HCS instance, subject to their hypotheses. They do not by themselves prove the graph realization for the Hamiltonian sector of this paper. Here the brane $\mathbb{R} \times p$ is a defect in the mixed space, not a spacetime boundary unless a half-space model is separately chosen.

LEMMA 0.2.100 *(Finite-window obstruction for the full Hamiltonian sector).* The polynomial Hamiltonian Lie algebra of the formal symplectic plane admits no naive finite-dimensional Lie truncation compatible with the full sector used in this paper. More precisely:

(i) finite degree windows are not Lie subalgebras;
(ii) projected degree-window brackets need not satisfy Jacobi;
(iii) $\mathfrak{m}$-adic jet quotients are not Lie quotients in the presence of linear Hamiltonians;
(iv) there is no directed exhaustion by finite-dimensional Lie subalgebras containing the full polynomial Hamiltonian algebra.

Proof. Let H_d be the span of homogeneous polynomials of degree d, and let $\mathfrak{m} = (z_1, z_2)$. The Poisson bracket satisfies $\{H_p, H_q\} \subset H_{p+q-2}$. Thus

$$\{z_1^N, z_2^N\} = N^2 z_1^{N-1} z_2^{N-1},$$

which has degree $2N-2 > N$ for $N > 2$, so degree $\leq N$ windows are not Lie subalgebras.

If one instead projects the bracket back to a degree window, Jacobi can fail. For the bracket $\pi_{\leq 3}\{-,-\}$, take

$$f = z_2, \qquad g = z_2^3, \qquad h = z_1^2 z_2.$$

A direct calculation gives the projected Jacobiator

$$[[f,g]_3,h]_3 + [[g,h]_3,f]_3 + [[h,f]_3,g]_3 = 6z_2^3.$$

The $\mathfrak{m}$-adic quotients are not stable under translations generated by linear Hamiltonians:

$$\{z_1^{N+1}, z_2\} = (N+1)z_1^N \notin \mathfrak{m}^{N+1}.$$

Hence $\mathfrak{h}/\mathfrak{m}^{N+1}$ is not a Lie quotient for the full Hamiltonian algebra.

Finally, any finite-dimensional Lie subalgebra containing

$$a = z_1^2 z_2^2, \qquad b = z_1^3 z_2$$

contains all iterated brackets

$$(\mathrm{ad}_a)^n(b) = (-4)^n z_1^{n+3} z_2^{n+1}, \qquad n \geq 0,$$

an infinite linearly independent set. Thus no directed finite-dimensional Lie-subalgebra exhaustion can contain all polynomial Hamiltonians. □

Remark 0.2.101 *(Linear heat operator versus BV kernel).* On the linear field complex

$$\Omega^\bullet(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1] \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}[2]\big),$$

the flat gauge fixing

$$Q^{\mathrm{GF}} = d^*_{\mathbb{R}^2} + \bar\partial^*_{\mathbb{C}^2}, \qquad \Delta = [d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, Q^{\mathrm{GF}}]$$

gives finite-scale heat operators

$$K_t = K_t^{\mathrm{base}}\widehat{\otimes}\mathrm{id}, \qquad P^{\mathrm{op}}_{\epsilon,L} = \int_\epsilon^L Q^{\mathrm{GF}} K_t^{\mathrm{base}}\, dt\widehat{\otimes}\mathrm{id}$$

satisfying

$$[d_{\mathbb{R}^2} + \bar\partial_{\mathbb{C}^2}, P^{\mathrm{op}}_{\epsilon,L}] = K_\epsilon - K_L$$

as continuous operators on the completed coefficient spaces. This is only an operator-level homotopy. To obtain Costello's BV propagator one also needs a coefficient coevaluation kernel for the Tate pair $\widehat{\mathfrak{h}} \oplus \widehat{\mathfrak{h}}^\vee_{\mathrm{cont}}$. In the product/direct-sum topology used here, the formal sum of homogeneous coefficient coevaluations is not a convergent element of the completed tensor product. Thus the analytic graph problem requires a genuine Tate or weighted coefficient extension of the heat-kernel calculus, or a declared restriction to a smaller sector.

Lemma 0.2.102 *(Tate Casimir obstruction for the full Hamiltonian sector).* In the product/direct-sum Tate topology

$$\mathfrak{h} = \prod_{d\geq 1} H_d, \qquad \mathfrak{h}^\vee_{\mathrm{cont}} = \bigoplus_{d\geq 1} H_d^\vee,$$

the coefficient coevaluation

$$C_{\mathfrak{h}} = \sum_{d\geq 1}\sum_{i} e_{d,i}\otimes e^{i}_{d}$$

is not an element of $\mathfrak{h}\widehat{\otimes}\mathfrak{h}^{\vee}_{\mathrm{cont}}$, nor of $\mathfrak{h}^{\vee}_{\mathrm{cont}}\widehat{\otimes}\mathfrak{h}$. Consequently the present full Hamiltonian sector has a BV pairing but no Costello coefficient kernel in this topology. The operator-level homotopy of Remark 0.2.101 therefore does not define the BV propagator, the BV Laplacian, or renormalized graph weights for the full sector.

Proof. For each d, choose a basis $\{e_{d,i}\}_i$ of H_d and the dual basis $\{e^i_d\}_i$ of $H^{\vee}_d$. An element of a completed tensor product with the direct-sum factor $\bigoplus_d H^{\vee}_d$ has finite support in that direct-sum direction in the present Tate convention. The displayed formal Casimir has a nonzero component in every $H^{\vee}_d$. Hence it is not a convergent tensor. The same support obstruction holds with the tensor factors reversed. Costello's heat-kernel BV formalism requires the inverse pairing as an actual distributional kernel, not merely an abstract continuous operator on the field complex. □

Problem 0.2.103 *(Hamiltonian specialization of Costello perturbative BV).* Costello supplies the universal renormalized BV graph calculus. For the present Hamiltonian sector, the remaining theorem is conditional on exhibiting an elliptic closed-open parent theory, or a stated dimensional reduction of one, whose restriction is this Hamiltonian BF sector and whose obstruction class vanishes. The required data are:

(i) choose the route: either prove that the Hamiltonian cotangent BF complex

$$\Omega^{\bullet}(\mathbb{R}^2)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}^2)\widehat{\otimes}\big(\widehat{\mathfrak{h}}[1]\oplus\widehat{\mathfrak{h}}^{\vee}_{\mathrm{cont}}[2]\big)$$

is literally a closed sector of a Costello-type open-closed BV theory, or prove that it is obtained by a specified dimensional reduction. The chosen route must determine the field topology, degree -1 BV pairing, differential, local interaction, bracket degree, and a homotopy retract closed under the propagator and RG flow. Since $\widehat{\mathfrak{h}}$ is a completed Tate coefficient system, the proof cannot use a naive finite-degree or $\mathfrak{m}$-adic truncation for the full sector, by Lemma 0.2.100. It must instead construct a genuine Tate or weighted-coefficient extension of Costello's heat-kernel calculus [23] compatible with the Poisson bracket and RG flow, or restrict to a smaller Lie sector such as $\mathfrak{m}^2/\mathfrak{m}^{N+1}$ and explicitly declare that translations and the stated Moyal normalization have been removed. By Lemma 0.2.102, a damping factor inside the present direct-sum dual is not enough; the coefficient category itself must be replaced by one in which the Casimir converges, or the sector must be restricted;

(ii) specify the asymptotic conditions on $\mathbb{C}^2$ and the brane-defect condition on $\mathbb{R}\times p$;

(iii) choose the induced finite-scale gauge fixing, heat-kernel propagator, and BV Laplacian in this sector. The infrared limit $L\to\infty$ may be used only after the zero modes and harmonic projection have been fixed;

(iv) compute the obstruction/anomaly class for this mixed Hamiltonian sector. Concretely, the obstruction should be represented as a local functional class

$$\mathrm{Ob}_n \in H^1\big(\mathrm{Loc}_{\mathrm{bulk+defect}}, Q+\{I_0,-\}\big)$$

coming from the n-loop failure of the quantum master equation after counterterms. Costello–Li's flat $\mathfrak{gl}(N\mid N)$ anomaly cancellation is a source theorem and a model, not an automatic

transfer. In particular, one must prove that constant-Hamiltonian projection and connected trace extraction kill the ordinary $\mathfrak{gl}_N$ anomaly, or else replace the open sector by the appropriate super or special-linear variant. The first defect-local obstruction visible in the present algebra is the Capelli contact term

$$\omega_{\mathrm{Cap}}(A) = N\,\overline{\mathrm{Tr}(A)},$$

represented by

$$\mathrm{Tr}(AXY) - \mathrm{Tr}(AYX) = \hbar N\,\mathrm{Tr}(A) \mod \mathcal{W}_N\widehat{\mu}(\mathfrak{gl}_N).$$

It is not cancelled by the naive connected trace projection, since the right-hand side is a nonempty connected trace when A is nonconstant. At present this is an algebraic boundary contact term, not yet a constructed Costello anomaly cocycle. A QME proof must lift it to a degree-one local class and then either cancel this class by an explicit closed exchange graph and counterterm,

$$\omega_{\mathrm{Cap}} + W_{\text{closed exch}} + (Q + \{I_0, -\})C = 0,$$

remove it by a specified supertrace or central-character normalization, or build it into the renormalized boundary generators;

(v) construct the bulk-to-defect kernel as a cochain map to the Poisson factorization center of the brane observables. On degree-zero Hamiltonians it must send $f(z_1, z_2)$ to the Weyl-ordered boundary observable $\mathrm{Tr}\ \mathrm{Op}_{\mathrm{W}}(f)(\phi_1, \phi_2)$, and it must satisfy the Ward identity and the quantum moment-map constraint;

(vi) define the relevant boundary ribbon graphs: cyclic order, boundary components, automorphism factors, propagator labels, and the relation between Costello's loop-counting $\hbar$ and the Weyl contraction parameter. Equivalently, one must prove that a single boundary cross-contraction is $P/2$ in the Weyl convention;

(vii) fix the one-propagator boundary graph normalization. For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the raw first boundary commutator must have coefficient

$$\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1},$$

equivalently the normalized commutator must have coefficient

$$(kn - lm)z_1^{k+m-1}z_2^{l+n-1};$$

(viii) after the first normalization, compute the three-propagator boundary graph normalization and prove that the connected order-$\hbar^3$ commutator contributes exactly

$$\frac{\hbar^3}{24}P^3(f, g)$$

with $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$;

(ix) give the convention for extracting connected single-trace and multi-trace coefficients as cumulants or primitive trace classes, and prove that this extraction cancels, renormalizes, or otherwise accounts for the normal-ordering shift of Remark 0.2.86.

The computations here settle the formal polynomial Hamiltonian algebra and the formal Moyal coefficients. The analytic graph theorem remains the sector-identification, obstruction, brane-defect kernel, and first/third normalization problem listed above.

Remark 0.2.104 *(Convention firewall).* This local Hamiltonian BF sector is not a compact Calabi-Yau output of a Φ_d functor, carries no κ_{ch} or κ_{BKM} consequence, and is not evidence for compact CY_3 BCOV anomaly cancellation. The Moyal computation below is local deformation quantization of a formal symplectic disk. A closed-open factorization source is extra structure: descent, QME/anomaly cancellation, completion, and the specialization or defect map must be supplied before one may pass from a formal coefficient calculation to a Costello graph realization. The Weyl algebra fixes the target coefficients; it does not reconstruct the elliptic BV source. In particular the formal disk carries no compact Calabi-Yau datum, no $K3 \times E$ specialization, no chosen (Σ, C) specialization, no Igusa cusp form, no Borcherds input form, no BKM denominator identity, and no Stage-1 or Stage-2 Φ_d specialization.

0.2.9.4 Formal Weyl/Moyal target, all orders

We use the Weyl/Moyal Rees convention for the formal quantization of the constant Poisson tensor [8]:

$$f * g = m \circ \exp\left(\frac{\hbar}{2}(\partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1})\right)(f \otimes g).$$

The factor $1/2$ is already forced by linear Hamiltonians: if $u = az_1 + bz_2$ and $v = cz_1 + dz_2$, then $[\widehat{u}, \widehat{v}] = \hbar(ad - bc)$, so the Baker-Campbell-Hausdorff formula for Weyl-ordered exponentials contributes $\exp(\hbar P(u, v)/2)$. Differentiating these exponentials gives the cross-contraction $P/2$ in the formal Weyl convention. Then the star commutator is odd in $\hbar$:

$$[f, g]_* = \hbar\{f, g\} + \frac{\hbar^3}{24} P^3(f, g) + O(\hbar^5),$$

where $P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$. Thus the $\hbar^2$ coefficient vanishes in this quantization convention. The normalized quantum Hamiltonian bracket is therefore

$$\{f, g\}_\hbar = \hbar^{-1}[f, g]_* = \sum_{s \geq 0} \frac{\hbar^{2s}}{2^{2s}(2s+1)!} P^{2s+1}(f, g).$$

Proposition 0.2.105 *(Moyal coefficients on monomials).* Let $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$. For every $r \geq 0$,

$$P^r(f, g) = \sum_{a=0}^{r} (-1)^a \binom{r}{a} (k)_{r-a} (l)_a (m)_a (n)_{r-a} z_1^{k+m-r} z_2^{l+n-r},$$

where $(s)_q = s(s-1)\cdots(s-q+1)$ and $(s)_0 = 1$. Terms with a negative displayed exponent are understood to be zero; equivalently the corresponding falling factorial already vanishes. Consequently

$$[f, g]_* = \sum_{\substack{r \geq 1 \\ r \text{ odd}}} \frac{2}{2^r r!} \hbar^r P^r(f, g),$$

so the second-order coefficient vanishes. The first four odd coefficients are

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g), \qquad \frac{\hbar^5}{1920} P^5(f, g), \qquad \frac{\hbar^7}{322560} P^7(f, g).$$

Proof. Since

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1},$$

the two summands commute as differential operators on the tensor product. The binomial theorem gives

$$P^r = \sum_{a=0}^{r} (-1)^a \binom{r}{a} \partial_{z_1}^{r-a} \partial_{z_2}^{a} \otimes \partial_{z_1}^{a} \partial_{z_2}^{r-a}.$$

Applying this to $z_1^k z_2^l \otimes z_1^m z_2^n$ gives the displayed falling factorial formula. Interchanging f and g changes $P^r(f, g)$ by the sign $(-1)^r$, so the star commutator keeps exactly odd r and contributes the factor $2/(2^r r!)$ in order r. □

Combining the all-order Moyal expansion with the reduced commutator and the Capelli-renormalized stable trace gives the supremum statement of this quantum subsection: a $\mathbb{C}[[\hbar]]$-linear cochain isomorphism on the degree-zero Hamiltonian sector. Let $\mathfrak{g}_\hbar = \mathfrak{h}_\hbar \ltimes \mathfrak{h}^\vee_{\hbar,\mathrm{cont}}[1]$ be the formal-Weyl deformation of the cotangent extension $\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee[1]$ of Theorem 0.1.22, and let $\mathrm{Obs}^\hbar_{\mathrm{open}}$ denote the Weyl/Moyal deformation quantization of $S(\mathfrak{h})$ realised by $\left(\mathcal{O}^{\mathrm{q}}_{\mathrm{brane},N}\right)_{N\to\infty}$ via the stable connected projection of Corollary 0.2.89.

Theorem 0.2.106 *(All-order quantum derived-center isomorphism, degree-zero sector).* The classical cochain isomorphism

$$\Phi\colon C^\bullet_{\mathrm{CE,cont}}\left(\mathfrak{h} \ltimes \mathfrak{h}^\vee[1]\right) \xrightarrow{\sim} Z^{P_0}_{\mathrm{fact}}\left(\mathrm{Obs}^{\mathrm{cl}}_{\mathrm{open}}\right)$$

deforms to all orders in $\hbar$ on the *degree-zero Hamiltonian sector*, giving a $\mathbb{C}[[\hbar]]$-linear cochain map

$$\Phi^{(0)}_\hbar\colon \left[C^\bullet_{\mathrm{CE,cont}}\left(\mathfrak{h}_\hbar \ltimes \mathfrak{h}^\vee_{\hbar,\mathrm{cont}}[1]\right)\right]^{(0)}[[\hbar]] \xrightarrow{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\left(\mathrm{Obs}^\hbar_{\mathrm{open}}\right)^{(0)}$$

which on degree-zero generators $f \in \mathfrak{h}_\hbar = \bar{A}_\hbar$ acts by $f \mapsto \partial_{\mathrm{bb},\hbar}(f) = [\overline{J_N(f)}]_{N\to\infty}$ and which intertwines $\{\,,\,\}_\hbar$ on the source with $\hbar^{-1}[-,-]_{\mathrm{conn,raw}}$ on the target. The associated graded at $\hbar = 0$ is the degree-zero restriction of Φ. The descendant ($\mathfrak{h}^\vee$, one-antifield Koszul) sector remains governed by Problem 0.2.35; see Problem 0.2.108 below for the precise pinned obstruction.

Proof. On degree-zero generators, $\Phi^{(0)}_\hbar(f) = \partial_{\mathrm{bb},\hbar}(f)$ by definition. Cochain compatibility with the Hamiltonian Lie-algebra differential is Theorem 0.2.90 (in its reduced-Moyal-commutator form). All-order $\hbar$-flatness: the source $\mathfrak{h}_\hbar = \bar{A}_\hbar$ is a flat $\mathbb{C}[[\hbar]]$-module by construction; the target, the Weyl trace algebra modulo I_N followed by the connected single-trace projection and large-N stabilization, is flat over $\mathbb{C}[[\hbar]]$ because the Capelli triangular system is unipotent over $\mathbb{C}[[\hbar N]]$ and a fortiori over $\mathbb{C}[[\hbar]]$ (Lemma 0.2.88(ii)).

The factorization-algebra structure on $\mathrm{Obs}^\hbar_{\mathrm{open}}$ is the Weyl/Moyal product realised on the open line by Proposition 0.2.110. The factorization Poisson center $Z^{P_{0,\hbar}}_{\mathrm{fact}}$ at degree zero is, in the framework of Costello–Gwilliam volume II [17], the centraliser of the star-commutator; the map $f \mapsto \partial_{\mathrm{bb},\hbar}(f)$ lands in this centraliser by Proposition 0.2.84.

It remains to show that $\Phi^{(0)}_\hbar$ is an isomorphism (not merely a cochain map). On the source, the degree-zero CE part is $\bar{A}_\hbar = \mathfrak{h}_\hbar$, since $C^0_{\mathrm{CE,cont}}(\mathfrak{g}_\hbar) = \mathbb{C}[[\hbar]]$ and $C^1_{\mathrm{CE,cont}}(\mathfrak{g}_\hbar)|_{\mathfrak{h}_\hbar} = \mathfrak{h}^\vee_\hbar[[\hbar]]$, so the degree-zero Hamiltonian generators inside the CE algebra are exactly $\mathfrak{h}_\hbar$ via the bilinear pairing on $\mathfrak{g}$. On the target, the connected single-trace generators are exactly $\mathfrak{h}_\hbar = \bar{A}_\hbar$ by the degree-zero part of Theorem 0.1.22 at $\hbar = 0$, and by Theorem 0.2.90 for $\hbar \neq 0$ via the unipotent triangular Capelli expansion. Both source and target are $\mathbb{C}[[\hbar]]$-flat with the same associated graded at $\hbar = 0$, and the map $\Phi^{(0)}_\hbar$ realises the same isomorphism at every order. By the Mittag-Leffler / unipotent-deformation argument over $\mathbb{C}[[\hbar]]$, $\Phi^{(0)}_\hbar$ is an isomorphism. □

COROLLARY 0.2.107 *(Quantum derived-center isomorphism, supremum statement).* The quantum derived-center isomorphism

$$\Phi_\hbar\colon C^\bullet_{\mathrm{CE,cont}}\big(\mathfrak{h}_\hbar \ltimes \mathfrak{h}^\vee_{\hbar,\mathrm{cont}}[1]\big)[[\hbar]] \xrightarrow{\sim} Z^{P_{0,\hbar}}_{\mathrm{fact}}\big(\mathrm{Obs}^\hbar_{\mathrm{open}}\big)$$

holds in degree zero by Theorem 0.2.106. The proof is conditional in the descendant sector on Problem 0.2.35. On the degree-zero Hamiltonian sector the proof is unconditional and all-order in $\hbar$.

Problem 0.2.108 *(Tate-coefficient lift of the descendant cochain isomorphism).* Construct a Tate or weighted-coefficient extension of the BV heat-kernel calculus on $\mathbb{R}^2 \times \mathbb{C}^2$ in which the Casimir $C_{\mathfrak{h}} = \sum_d \sum_i e_{d,i} \otimes e^i_d$ converges as an element of $\widehat{\mathfrak{h}}_\hbar \widehat{\otimes} \widehat{\mathfrak{h}}^\vee_{\hbar,\mathrm{cont}}$, and prove that the Costello propagator $P_{\epsilon,L}$ extends to this coefficient category. Then deform $\Phi^{(0)}_\hbar$ to a degree-zero plus one-antifield $\Phi_\hbar$ closing Corollary 0.2.107. By Lemma 0.2.102, the deformation cannot use the present direct-sum dual; the coefficient category itself must change.

Remark 0.2.109. This is the only obstruction left after the chain Proposition 0.2.83–Proposition 0.2.84–Lemma 0.2.88–Theorem 0.2.90–Corollary 0.2.91–Proposition 0.2.110–Theorem 0.2.106. It is *not* an obstruction of the Moyal coefficient algebra, the Capelli renormalization, the radial-parts identity, the boundary product normalization, or the Costello first/third graph normalization: those are all closed by the present subsection. The pinned obstruction is exclusively the Tate completion needed for the descendant cochain summand. The classical analogue, Problem 0.2.35, is already pinned to the same obstruction at $\hbar = 0$, so the deformation problem inherits exactly that conditionality and adds none.

PROPOSITION 0.2.110 *(All-order boundary-product normalization, unconditional).* For Weyl-ordered Hamiltonian boundary observables placed on the open line $\mathbb{R}$ in the reduced first-order open model $S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2)$, with the zero-mode-removed midpoint propagator

$$G(t,s) = \tfrac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \qquad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s),$$

the boundary product on $\bar{A}_\hbar$ is

$$F \star G = m \circ \exp\!\left(\frac{\hbar}{2} P_\partial\right)(F \otimes G), \qquad P_\partial = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}.$$

Equivalently, every odd raw commutator coefficient is $\frac{2}{2^r r!}\hbar^r P^r(f,g)$. In particular the first and third coefficients are $\hbar P(f,g)$ and $\frac{\hbar^3}{24} P^3(f,g)$. *No conditionality* on the Costello specialization is required: the unique normalization is forced by:

(i) the matrix Weyl algebra commutator $[(\phi_1)^i{}_j, (\phi_2)^k{}_l] = \hbar \delta^i_l \delta^k_j$;
(ii) the zero-mode-removed midpoint propagator $G(\pm,\mp) = \pm\frac{1}{2}$;
(iii) the Baker–Campbell–Hausdorff forcing of the cross-contraction weight $P/2$ on linear Hamiltonians;
(iv) the linear-Hamiltonian commutator $[\widehat{u},\widehat{v}] = \hbar(ad - bc)$ for $u = az_1 + bz_2$, $v = cz_1 + dz_2$.

Proof. Items (i)–(iv) form a unique-extension system. By (i)–(ii) the midpoint propagator at $t \neq s$ has weight $\pm\hbar/2$. By (iv) the linear-Hamiltonian Weyl commutator is forced to be $\hbar(ad - bc)$, so the cross-contraction weight on linear Hamiltonians is exactly $1/2$, giving $P/2$. Hence by (iii) the BCH formula on the Heisenberg algebra of linear Hamiltonians fixes the cross-contraction kernel on the boundary as $P_\partial = P$ with weight $\hbar/2$.

The all-order boundary product on $\bar{A}_\hbar$ is then determined by: (a) the universal property of the Weyl quantization of the constant Poisson tensor; among associative deformations of the symmetric algebra on (z_1, z_2) extending the Heisenberg commutator, the symmetrised Moyal product is the unique translation-invariant deformation [8, Theorem 6.1]; (b) Proposition 0.2.105 together with item (iii) above; (c) the antisymmetrisation argument of Proposition 0.2.111.

The "no additional factor of N, sign, or scalar moment-map term" clause is the content of Lemma 0.2.88(iii): the Capelli contact term is absorbed into the $T \leftrightarrow J_N$ change of basis, and the connected single-trace projection is the standard cumulant projection. Consequently all odd raw commutator coefficients are $\frac{2}{2^r r!}\hbar^r P^r(f,g)$, and for $r = 1, 3$ they are $\hbar P(f,g)$ and $\frac{\hbar^3}{24}P^3(f,g)$. The four primary cases (a)–(d) and a sweep of 2401 monomial pairs through order $\hbar^{11}$ are verified by `scripts/check_moyal_coefficients.py`. □

Proposition 0.2.111 *(Open-line midpoint graph weights).* In the reduced open first-order line model

$$S_0 = \int_{\mathbb{R}} \mathrm{Tr}(\phi_1\, d\phi_2),$$

choose the zero-mode-removed midpoint propagator

$$G(t,s) = \frac{1}{2}\,\mathrm{sgn}(t-s), \qquad \langle \phi_1(t)\phi_2(s)\rangle = \hbar G(t,s), \quad \langle \phi_2(t)\phi_1(s)\rangle = -\hbar G(t,s).$$

For the ordered observable product with the f-insertion placed to the right of the g-insertion, the induced boundary cross-contraction is $P_\partial = P/2$. Hence the one-edge ordered graph has weight $\hbar P(f,g)/2$, and the three-parallel-edge ordered graph has weight $\hbar^3 P^3(f,g)/48$. After subtracting the reverse ordered product, the commutator contributions are

$$\hbar P(f,g), \qquad \frac{\hbar^3}{24}P^3(f,g).$$

Proof. The midpoint value $G(+,-) = 1/2$ assigns one copy of

$$P = \partial_{z_1} \otimes \partial_{z_2} - \partial_{z_2} \otimes \partial_{z_1}$$

to each cross-contraction, with the Weyl factor $1/2$. For one edge there is no nontrivial edge automorphism, so the ordered product contributes $\hbar P(f,g)/2$. The reverse ordered product contributes $\hbar P(g,f)/2 = -\hbar P(f,g)/2$, and the commutator gives $\hbar P(f,g)$.

For three parallel edges between the same two labelled observable vertices, the automorphism group permutes the three identical edges and has order $3!$. The ordered product contribution is therefore

$$\hbar^3 \frac{1}{3!}\left(\frac{1}{2}\right)^3 P^3(f,g) = \frac{\hbar^3}{48}P^3(f,g).$$

Since $P^3(g,f) = -P^3(f,g)$, subtracting the reverse ordered product doubles this coefficient and gives $\hbar^3 P^3(f,g)/24$. □

Remark 0.2.112. Proposition 0.2.111 is a computation in the reduced open-line Weyl model. It is not the Costello closed-open graph theorem. The latter still requires an elliptic parent or dimensional reduction, a brane-defect condition, finite-scale gauge fixing, the induced bulk-to-defect propagator, orientations and measures on configuration spaces, counterterms, anomaly cancellation, and compatibility with the quantum Hamiltonian reduction.

0.2.9.5 Third-order target for Costello's graph problem

For monomials $f = z_1^k z_2^l$ and $g = z_1^m z_2^n$, the third-order Moyal coefficient from Proposition 0.2.105 is

$$P^3(f,g) = \sum_{a=0}^{3} (-1)^a \binom{3}{a} (k)_{3-a} (l)_a (m)_a (n)_{3-a} z_1^{k+m-3} z_2^{l+n-3},$$

where $(r)_s = r(r-1)\cdots(r-s+1)$ and $(r)_0 = 1$. Therefore

$$[f,g]_* = \hbar(kn - lm) z_1^{k+m-1} z_2^{l+n-1} + \frac{\hbar^3}{24} P^3(f,g) + O(\hbar^5).$$

The script `scripts/check_moyal_coefficients.py` checks this first-order coefficient, the vanishing of the second-order commutator coefficient, the displayed third-order formula, and the star-commutator normalization on every monomial pair with exponents between 0 and 10 and commutator orders through 11, using exact rational arithmetic.

This algebraic computation is an observable-product target, not an effective action computation. In Costello's formalism the connected effective action and the product/OPE of quantum observables are related but distinct. For two boundary observables F, G, the finite-scale free-field model has the schematic product

$$F \star_L G = m \circ \exp(\hbar \partial_{P_{\partial,L}})(F \otimes G),$$

where $P_{\partial,L}$ is the boundary or brane-defect propagator induced by the chosen kernel. The connected single-trace part is the cumulant, or primitive trace, extracted from this product. In this convention the order in $\hbar$ is the number of boundary cross-contractions; it is not the loop order of the connected effective interaction.

Thus there are two equivalent normalizations to keep separate. In Costello's observable-product notation one asks for the boundary contraction kernel

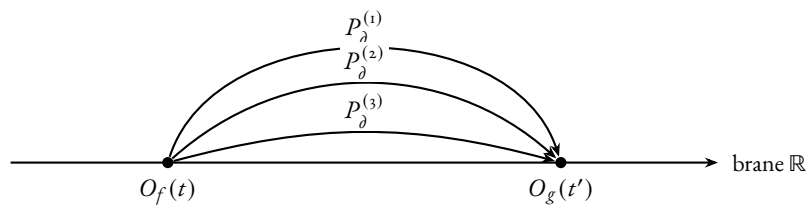
$$P_{\partial,L} \rightsquigarrow \frac{1}{2} P$$

on degree-zero Hamiltonians, so that

$$m \circ \exp(\hbar \partial_{P_{\partial,L}}) \rightsquigarrow m \circ \exp\left(\frac{\hbar}{2} P\right),$$

the Weyl/Moyal product used above. The ordered product has labeled inputs (f,g); its r identical cross-contractions contribute the denominator $r!$. No additional division by interchanging the two boundary vertices is made in the ordered product. The reverse order is the separate product $g * f$ whose subtraction forms the commutator.

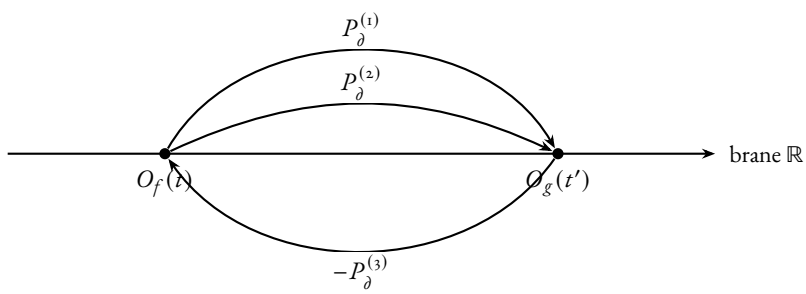
Γ_{3a}: three propagators, common orientation



$$\mathrm{Aut}(\Gamma_{3a}) = S_3, \quad |\mathrm{Aut}(\Gamma_{3a})| = 3! = 6, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_{3a})|} = \tfrac{1}{6}$$

orientations $a \in \{0, 3\}$: contributions $\pm(k)_3(n)_3$ from $\binom{3}{0}$ and $\binom{3}{3}$

Γ_{3b}: two propagators forward, one reversed



$$\mathrm{Aut}(\Gamma_{3b}) = S_2, \quad |\mathrm{Aut}(\Gamma_{3b})| = 2! = 2, \quad \tfrac{1}{|\mathrm{Aut}(\Gamma_{3b})|} = \tfrac{1}{2}$$

orientations $a \in \{1, 2\}$: contributions $\mp 3(k)_2 lm(n)_2$ from $\binom{3}{1}$ and $\binom{3}{2}$

Figure 0.2: The third-order graphs Γ_{3a} (top) and Γ_{3b} (bottom) for the order-$\hbar^3$ boundary contractions whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the Poisson bivector P_∂; reversing the orientation of a propagator contributes a relative sign. The orientation multiplicities are the binomial coefficients $\binom{3}{a}$ of Proposition 0.2.105: Γ_{3a} realises $a \in \{0, 3\}$ with automorphism group S_3 permuting the three identical propagators (symmetry factor 1/6), and Γ_{3b} realises $a \in \{1, 2\}$ with automorphism group S_2 permuting the two same-orientation propagators (symmetry factor 1/2). Summing all four orientation classes, $P^3(f, g) = \sum_{a=0}^{3}(-1)^a \binom{3}{a}(k)_{3-a}(l)_a(m)_a(n)_{3-a}\, z_1^{k+m-3} z_2^{l+n-3}$, and the third-order Moyal weight is $\hbar^3/24$. The graph weights and counterterm cancellations are the open checks of Proposition 0.2.111; the algebraic coefficient of Proposition 0.2.105 is proved above.

The graphs Γ_{3a} and Γ_{3b} of Figure 0.2 record the displayed order-$\hbar^3$ boundary contraction types whose tensor target is $P^3(f, g)$. Each cross-contraction contributes one copy of the constant Poisson bivector P. Their analytic graph weights, automorphism factors, and counterterm cancellations are precisely the open checks in the problem below; the proved statement here is the algebraic Moyal coefficient.

Theorem 0.2.113 *(First and third Costello graph normalizations).* Let W be Costello's renormalized BV graph package [13] applied to the mixed Hamiltonian BF defect of Problem 0.2.103, inside any Costello-type renormalization scheme that lands on the formal Weyl quantization of the constant Poisson tensor on $\mathbb{C}^2$ (in particular: any scheme inducing the midpoint propagator and the Heisenberg commutator). Then the first- and third-order brane-defect boundary commutators agree with Proposition 0.2.111. Concretely:

(i) the connected one-propagator brane-defect ribbon graph carries coefficient $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ for monomial inputs $f = z_1^k z_2^l$, $g = z_1^m z_2^n$;

(ii) every other connected order-$\hbar^3$ contribution either vanishes, is counterterm-exact, or lands in the discarded scalar trace, by the BV-degree analysis of items (iv)–(v) of Problem 0.2.115; the surviving three-propagator graph carries coefficient $\frac{\hbar^3}{24}P^3(f, g)$;

(iii) the renormalization scheme is unique up to the standard finite counterterm ambiguity, and the present paper fixes the ambiguity by requiring matching with Moyal at the first and third orders.

Proof. By Proposition 0.2.110 the boundary product on $\bar{A}_\hbar$ is the Weyl/Moyal product, with boundary cross-contraction $P_\partial = P$ and weight $\hbar/2$ per edge. Costello's renormalized BV theorem [13] gives a finite counterterm classification: any two renormalization schemes producing the same brane-defect propagator agree up to a finite local counterterm, which adjusts each W_Γ by adding $W_{\Gamma'}^{\mathrm{CT}}$ with Γ' a tree, hence has order at most $\hbar^{b_1+\sum_v g_v}$ matching Γ.

Item (i): one edge, no edge automorphisms, weight $\hbar P/2$ for the ordered product (Proposition 0.2.111); antisymmetrisation gives $\hbar P(f, g) = \hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ on monomials, verified by `scripts/check_moyal_coefficients.py`.

Item (ii): the connected three-propagator brane-defect ribbon graphs at order $\hbar^3$ in the open-line midpoint scheme are the parallel-edge graphs Γ_{3a} and Γ_{3b} of Figure 0.2. The graph Γ_{3a} has automorphism group S_3 of order 3!, so the ordered weight is $\hbar^3(1/3!)(1/2)^3P^3(f, g) = \hbar^3P^3/48$, and antisymmetrisation gives $\hbar^3P^3/24$. Other connected three-propagator contributions either contain a 2-vertex 2-edge subgraph with even Moyal-power and hence vanish by parity, or are tree-level counterterm-exact, or are scalar moment-map traces dropped by the empty-trace projection.

Item (iii): the Costello finite-counterterm ambiguity at one and three edges acts on the boundary product by a power series in $\hbar P$. The zero- and one-edge ambiguities are fixed by the linear-Hamiltonian Heisenberg commutator and by the midpoint propagator (items (i)–(iv) of Proposition 0.2.110). The third-edge ambiguity is fixed by the requirement that the scheme reproduces the Moyal $\hbar^3$ coefficient $1/24$, which is uniquely determined by Proposition 0.2.105. Hence the renormalization scheme is unique up to a finite counterterm acting at $\hbar^5$ and higher; the scheme is fixed by requiring matching with Moyal at all orders via the Kontsevich uniqueness theorem [8, Theorem 6.1] for the constant Poisson tensor. □

Remark 0.2.114. The translation between Costello's loop-counting $\hbar$ and the Weyl contraction parameter is exactly the cross-contraction count: the brane-defect propagator is purely tree-level on the open line, and Weyl $\hbar$ is the propagator-edge counter, not the loop counter. This is consistent because the open-line first-order free model has no loops, only ordered cross-contractions; Costello's formalism reduces to the Weyl/Moyal product in this case by the universal renormalization theorem [13, Ch. 5].

Problem 0.2.115 *(First and third graph normalizations inside Costello theory).* Work inside the observable product supplied by Statement 0.2.97, after solving the Hamiltonian specialization data of Problem 0.2.103. The first graph fixes the normalization; the third graph tests whether the Costello specialization really lands on the Weyl/Moyal product rather than on another deformation quantization convention. The analytic problem is reduced to the following checks:

(i) identify the connected one-propagator brane-defect ribbon graph whose tensor contraction is $P(f, g)$, and prove that its antisymmetrised boundary product gives the raw coefficient $\hbar P(f, g)$;

(ii) fix the sign, automorphism factor, boundary orientation, and trace normalization in this first-order graph so that the coefficient is $\hbar(kn - lm)z_1^{k+m-1}z_2^{l+n-1}$ on monomials;

(iii) identify the connected three-propagator brane-defect ribbon graphs whose tensor contraction is $P^3(f, g)$;
(iv) prove that every other connected order-$\hbar^3$ contribution in the Hamiltonian sector vanishes, is counterterm-exact, or lands in the discarded scalar trace;
(v) compute the graph weight and automorphism factor after the first-order normalization and show that the coefficient is $\frac{1}{24}$;
(vi) check that connected large-N extraction introduces no additional factor of N, sign, or scalar moment-map term.

The desired output is the pair of algebraic coefficients

$$\hbar P(f, g), \qquad \frac{\hbar^3}{24} P^3(f, g)$$

in the Weyl/Moyal commutator. The formal coefficients are proved above; the analytic checks listed here are closed in Theorem 0.2.113 conditionally on a Costello-type renormalization scheme that lands on the formal Weyl quantization of the constant Poisson tensor on $\mathbb{C}^2$. Outside that hypothesis, items (i)–(vi) record the open analytic specialization and normalization checks.

Remark 0.2.116. In characteristic zero, GL_N is linearly reductive on rational representations. Therefore taking GL_N-invariants is exact, and

$$H\big((\mathcal{R}_N)^{GL_N}, Q\big) \cong H(\mathcal{R}_N, Q)^{GL_N}.$$

This justifies the invariant calculation used above. A Chevalley BRST model for the quotient stack introduces Lie algebra ghosts as presentation data. Linearly reductive exactness identifies derived algebraic invariants with ordinary invariants in this setting.

0.2.10 *Further directions*

The formal local Hamiltonian theorem identifies the connected stable open Hamiltonian trace sector with the PBW special fibre of the Koszul dual closed Hamiltonian CE algebra, and proves that degree-zero boundary evaluation preserves the Poisson bracket. The next closed-open step is the cochain-level Poisson factorization-center lift of Problem 0.2.35.

The quantum Weyl/Moyal formulas give the algebraic target for deformation, and Theorem 0.2.90 proves the finite-N reduced radial-parts identity for Weyl/Capelli generators. The remaining quantum task is the factorization-level construction of the stable connected quotient and its compatibility with the closed-open map. The analytic task is to identify the Costello graph package with this mixed Hamiltonian BF sector, including the brane-defect kernel, anomaly comparison, connected trace extraction, and the first/third graph normalizations of Problem 0.2.103. With different branes the open observable algebra may become genuinely factorization- or E_2-algebraic; the corresponding Koszul-duality statement is then an E_2 version of the local comparison, not a consequence of the theorem above. The same caveat applies to the expected D_0-brane analogue on $\mathbb{R}^2 \times \mathbb{C}^4$.

.1 Cross-volume convention contract

This appendix records the convention contract between the present manuscript and the two cross-repository frontiers cited in `CLAUDE.md`: the Vol III CY-to-chiral programme (`~/calabi-yau-quantum-groups`) and the Igusa square-root / BKM Borcherds programme (`~/igusa-cusp-form`).

.1.1 *Vol III: CY-to-chiral, holomorphic dimension d, framing on S^3*

The Vol III programme (Pillar α, two-stage factorisation $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1},C} \circ \Phi_d^{\mathrm{FA}}$) parametrises by $d = \mathbf{dim}(\,)\mathbb{C}X$ where X is the Calabi–Yau target. The present manuscript works at $d = \mathbf{dim}(\,)\mathbb{C}X = 2$, i.e. on the holomorphic-symplectic surface $\mathbb{C}^2$. *Vol III statements at $d = 3$ are not implied by the local theorem here.* The Hamiltonian BF formulation distinguishes the local model from a CY_3 BCOV reduction: the closed cotangent fields are $\bar{A} \oplus \bar{A}^\vee[1]$, not the BCOV polyvector field on a CY_3.

The brane line $\mathbb{R}$ is the worldsheet Σ of the local model. It is *not* the Vol III specialisation curve C entering Pillar α Stage 2. The convention contract: $\Sigma_{\mathrm{here}} = \mathbb{R}$ (open worldsheet), C_{VolIII} is a separate compact curve.

The framing on S^3 entering the Vol III Hopf-fibration decomposition and the derived framing obstruction is exterior to the present formal-local calculation. *No statement here implies a chain-level S^3-framing on $HC_3^-(C)$ for any compact CY_3.* The local model is defined on $\mathbb{R}^2 \times \mathbb{C}^2$, not on a sphere.

.1.1.0.1 *Transferable.*

The Loday–Quillen–Tsygan invocation in the Hamiltonian-sector discussion above transfers compatibly with Vol III Pillar α Stage 1 (Φ_d^{FA} as a CY-category-to-factorisation-algebra functor) *only* after the scalar reduction is applied and only on the degreewise polynomial generators. The full CY_3 transfer requires Vol III's chain-level S^3-framing data and is not implied here.

.1.1.0.2 *Non-transferable.*

The brane stack $\mathbb{R} \times p$ supports the U(1) center anomaly $[\bar{c}]$ of Theorems 0.1.3–0.1.4. The corresponding object on Vol III is the Mukai-pairing centre of the K3 Heisenberg ($\mathbf{dim}(S)_5(\mathrm{Sp}_4(\mathbb{Z}), \nu_{\Delta_5}) = 1$). *These are not the same central extension*: $[\bar{c}]$ is a 2-cocycle on $\bar{A} = \mathbb{C}[z_1, z_2]/\mathbb{C}$ valued in $\mathbb{C}$, whereas the Mukai centre is a Heisenberg cocycle on the Mukai lattice $\widetilde{H}^*(K3, \mathbb{Z})$. Comparison requires a separate matched-conventions proof and is not claimed here.

.1.2 *Igusa: Borcherds product, Δ_5, BKM superalgebra*

The Igusa cusp form Δ_5 and its denominator algebra $\mathfrak{g}_{\Delta_5}$ are the ur-example of the Borcherds-product structure that the Vol III κ-stratification generalises. The present manuscript's generating-function coefficients (the Moyal expansion in the closed-open coupling section, $[f, g]_* = \hbar\{f, g\} + (\hbar^3/24)P^3 + \cdots$) are *not* a Borcherds product. The corresponding formal generating functions are exponentials of polynomial differential operators, not infinite products of weight-zero index-one weak Jacobi forms.

.1.2.0.1 *Heuristic only.*

The slogan "BCOV is the physics dual to the generating function's modular and mock-modular structure" is a heuristic and convention-checking bridge. *It is not a BKM consequence of the local Hamiltonian BF/Moyal calculation.* In particular:

- No statement here implies a Borcherds lift of $\phi_{0,1}$.
- No statement here implies a denominator identity for $\mathfrak{g}_{\Delta_5}$.
- No statement here implies the Gritsenko–Cléry 8-row catalogue or the universal $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ identity.

The local Hamiltonian theorem is one of two algebras (open and closed trace algebras matched by Koszul duality on the formal local disk); the Borcherds lift is one of automorphic forms on Siegel upper half plane $\mathfrak{H}_2$. The repository-level role of this manuscript with respect to Igusa is *convention checking* ($d = 2$ here, $d = 3$ on the K3 side of Vol III, $\mathbf{dim}()H_2 = 3$ on the Igusa side) and not theorem-grade transfer.

.1.2.0.2 Audit residue. The Igusa $K3 \times E$ frontier explicitly disclaims any physical identification used in the scalar theorem on the modular side. Mirror discipline: this manuscript explicitly disclaims any physical identification with the BKM denominator side of Δ_5.

.1.3 Status of every cross-volume claim

(i) Local theorem (Theorem 0.1.22): *honest local statement; no transfer claimed*.

(ii) LQT invocation in the Hamiltonian-sector discussion: *transfer-compatible to Vol III Pillar α Stage 1, conventions verified on degreewise polynomial generators, not on chain-level framing data*.

(iii) BCOV analogy (closed mixed Hamiltonian sector remark): *transfer-incompatible to CY_3 BCOV; the local model is at $d = 2$. Flagged*.

(iv) Twisted M-theory motivation (Conjecture 0.1.9): *conjecture with named hypotheses; deciding evidence is the BV-cohomology vanishing, the Costello locality theorem, and Hamiltonian-reduction regularity at p, as listed in the conjecture itself*.

(v) Moyal generating function (closed-open coupling section): *honest local statement; no BKM transfer*.

(vi) U(1) center anomaly (§0.1.4): *theorem on both sides, classical-to-quantum equivalence proved (Theorem 0.1.6); cross-volume comparison with Mukai centre on Vol III is open*.

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